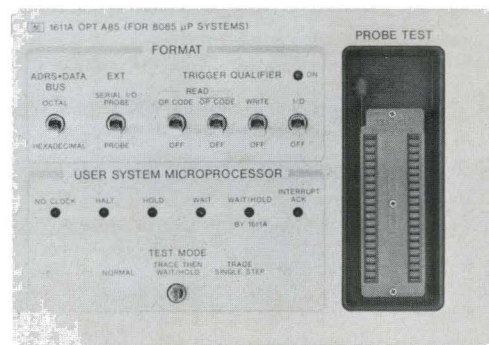


O P E R A T I N G A N D S E R V I C E M A N U A L

1611A OPT A85 (10263A) PERSONALITY MODULE FOR 8085 MICROPROCESSORS

**HEWLETT
PACKARD**



OPERATING AND SERVICE MANUAL
SUPPLEMENT

**MODEL 1611A OPT A85
(10263A)
PERSONALITY MODULE
FOR
(8085 MICROPROCESSORS)**

SERIAL NUMBERS

This manual applies directly to instruments with serial numbers prefixed **1844A**.

With changes described in Section VII, this manual applies to instruments with serial numbers prefixed **1818A** and **1833A**.

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Manual Part Number 10263-90902
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TABLE OF CONTENTS

Section	Page	Section	Page
I GENERAL INFORMATION	1-1	4-8. Memory Display Test	4-2
1-1. Introduction	1-1	4-9. Status Messages for Trace Mode Test.....	4-2
1-3. Description	1-1	4-10. Trigger $\leq$, Trigger Occurrences, Delay, and Trace Triggers Test ..	4-3
1-6. Specifications	1-1	4-11. Trace Triggers End on Disable Test	4-3
1-8. Recommended Test Equipment	1-1	4-12. Pretrigger Test	4-4
II INSTALLATION	2-1	4-13. Count Trigger Test and Stop Test ..	4-4
2-1. Introduction	2-2	4-14. Time Interval Test	4-5
2-4. Model Installation	2-1	4-15. Trigger Qualifier Switch Test.....	4-5
III OPERATION	3-1	4-16. Specification Verification	4-6
3-1. Introduction	3-1	4-18. Test Signal connections	4-7
3-3. Panel Features	3-1	4-19. Initial Setup	4-7
3-5. Probes	3-1	4-20. Clock Test	4-8
3-10. SETTING TRACE SPECIFICATIONS	3-1	4-21. Data Lines Test	4-9
3-15. Triggering	3-2	4-22. Address Lines AD ₀ -AD ₇ Test.....	4-11
3-16. Trace Measurement Execution.....	3-2	4-23. Address Lines A ₈ -A ₁₅ Test	4-12
3-17. Trace	3-2	4-24. Data Bus Status - S ₀ ; S ₁ , and IO/M Test.....	4-14
3-31. Trace Triggers Measurement Execution	3-4	4-25. External Probe Test	4-15
3-32. Trace Trigger	3-4	4-26. Rear-panel Output Test.....	4-17
3-33. Trace Triggers End on Disable	3-4		
3-47. Count Triggers Measurement Execution	3-6	V ADJUSTMENTS.....	5-1
3-48. Count Triggers.....	3-6		
3-50. Count Triggers Set-up.....	3-6	VI REPLACEABLE PARTS	6-1
3-56. Time Interval Measurement Execution	3-6	6-1. Introduction	6-1
3-57. Time Interval	3-6	6-3. Abbreviations.....	6-1
3-65. Operator's Checks.....	3-7	6-5. Replaceable Parts List	6-1
3-66. Self-test	3-7	6-7. Ordering Information	6-1
3-67. Probe Test	3-7	6-10. Direct Mail Order Systems.....	6-1
3-68. Status Messages	3-8		
IV PERFORMANCE TESTS	4-1	VII MANUAL CHANGES.....	7-1
4-1. Introduction	4-1	7-1. Introduction	7-1
4-3. Performance Test Record	4-1	7-3. Manual Changes	7-1
4-5. Operation Verification	4-1	7-6. Manual Change Instructions	7-1
4-7. Self Test	4-1		
		VIII SERVICE	8-1
		8-1. Introduction	8-1

LIST OF ILLUSTRATIONS

Figure	Title	Page	Figure	Title	Page
2-1.	Model 1611A Assembly Locations	2-2	4-3.	Data Lines Test	4-10
2-2.	Personality Panel A11 Installation.....	2-2	4-4.	Address Lines A ₀ -A ₇ Test	4-12
3-1.	Keyboard and Rear-panel Features.....	3-0	4-5.	Address Lines A ₈ -A ₁₅ Test.....	4-13
3-2.	Personality Panel Controls and Indicators	3-0	4-6.	Data Bus Status Test	4-15
3-3.	Turn-on Display	3-1	4-7.	External Probe Test	4-16
3-4.	Display Window	3-2	4-8.	Rear-panel Outputs Test Setup	4-17
3-5.	After Trigger and Before Trigger Displays	3-3	7-1.	Replacement for Figure 8-2, Service Sheet 2, (Sheet 1 of 3)	7-3
3-6.	Absolute Model Display	3-3	7-2.	Replacement for Figure 8-6, Service Sheet 4, (Sheet 1 of 2)	7-6
3-7.	Mnemonic Mode Display	3-3	8-1.	Service Sheet 1, Microprocesor Probe A13 ..	8-1
3-8.	Trace Display	3-4	8-2.	Service Sheet 2, Personality Board A9 (Sheet 1 of 3)	8-4
3-9.	Trace Trigs Display	3-5	8-3.	Service Sheet 3, A85 Personality Board A11 Parts Identification (Sheet 1 of 2)...	8-8
3-10.	Count triggers Display	3-6	8-4.	ROM Test Troubleshooting Tree	8-11
3-11.	Time Internal Display	3-7	8-5.	Service Sheet 4, ROM Board A10 (Sheet 1 of 2)	8-12
3-12.	Probe Test Display	3-7			
4-1.	Specification Verification Test Setup	4-8			
4-2.	Clock Test Waveforms	4-9			

LIST OF TABLES

Table	Title	Page	Table	Title	Page
1-1.	Model 1611A Specifications (Model 10263A Installed)	1-1	4-4.	External Data Field	4-17
1-2.	Recommended Test Equipment	1-2	6-1.	Reference Designators and Abbreviations	6-2
3-1.	A85 Memory Transaction Displays	3-3	6-2.	Replaceable Parts	6-3
4-1.	Opcode/Data Field Display	4-11	7-1.	Manual Changes	7-1
4-2.	Address Field Display	4-12	7-2.	Replaceable Parts	7-2
4-3.	Address Display Field (A ₈ through A ₁₅)	4-14	8-1.	Signature Analysis Procedure No. 1	8-10

SECTION I

GENERAL INFORMATION

1-1. INTRODUCTION.

1-2. This manual supplement contains: (1) information to install and test the Model 10263A; (2) operating information for the Model 1611A with Opt A85 installed; (3) service information that complements service data in Section VIII of the 1611A Operating and Service Manual.

1-3. DESCRIPTION.

1-4. The Hewlett-Packard Model 1611A Option A85 consists of an A85 Personality Module installed in a 1611A standard mainframe for use with microprocessor based systems which use the 8085 (or equivalent) microprocessor. Specifications are listed in table 1-1.

1-5. The Model 10263A is available separately as an accessory to provide the Model 1611A Logic State Analyzer with Option A85 capability.

1-6. SPECIFICATIONS.

1-7. Specifications for the 1611A with the Model 10263A installed are given in table 1-1.

1-8. RECOMMENDED TEST EQUIPMENT.

1-9. Equipment required to test the Model 10263A Option A85 and verify its operation is listed in table 1-2. Other equipment may be substituted if it meets or exceeds the critical specifications listed in the table.

Table 1-1. Model 1611A Specifications (Model 10263A Installed)

CLOCK, DATA, ADDRESS, STATUS, AND CONTROL INPUTS

CLOCK RATE: 300 kHz to 5 MHz.

THRESHOLD: 2 V to 5.5 V, logic 1 (high); -0.5 V to 0.8 V logic 0 (low).

INPUT CURRENT

Clock and Control: approx 50 μ A max with $V_{in} = 2.7$ V; -0.4 mA max with $V_{in} = 0.5$ V.

Data, Address, and Status: approx 20 μ A max with $V_{in} = 2.7$ V; 0.2 mA max with $V_{in} = 0.4$ V.

INPUT CAPACITANCE: approx 25 pF including capacitance of 30.5 cm (12 in.) cable; approx 15 pF with 7.6 cm (3 in.) cable.

SETUP TIME

D0 - D7: 80 ns min relative to rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$.

A0 - A7: 50 ns min relative to falling edge of ALE.

A8 - A15: 345 ns min relative to rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$.

S0, S1, IO/M: 5 ns min relative to falling edge of ALE.

HOLD TIME

D0 - D7: zero relative to rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$.

A0 - A7: 50 ns min relative to falling edge of ALE.

A8 - A15: 60 ns min relative to rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$.

S0 - S1, IO/M: 200 ns min relative to falling edge of ALE.

READY OUTPUT: TTL compatible open-collector output capable of sinking at least 8 mA when active.

EXTERNAL PROBE INPUTS

CURRENT: approx 50 μ A logic 0 or logic 1.

CAPACITANCE: approx 25 pF at probe tip.

THRESHOLD: 2.4 V to 5.5 V, logic 1 (high); -0.8 V to 0.8 V logic 0 (low).

SETUP TIME: 250 ns min relative to rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$.

HOLD TIME: zero, relative to rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$.

OUTPUTS

LOW: <0.4 V into 50 Ω .

HIGH: >2.0 V into 50 Ω (nominally 3.9 V into an open circuit).

TRIGGER: duration, approx 75 ns (RZ format), delay, approx 350 ns after the rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$ during the cycle that defines a valid trigger.

TRACE POINT ($\sqcap$): provides a positive edge approx 350 ns after the rising edge of $\overline{RD}$, $\overline{WR}$, or $\overline{INTA}$ that defines the specific valid trigger to be displayed. If the 1611A is set so that the trigger word is not displayed the Trace Point output occurs for the cycle that defines the valid word immediately preceding the first displayed word.

TRACE POINT ($\sqcap$): complement of Trace Point ($\sqcap$).

MICROPROCESSOR COMPATIBILITY

INTEL: 8085, 8085A, 8085A-2.

AMD: AM9085.

SIEMENS: 8085.

NEC: μ PD8085.

Table 1-2. Recommended Test Equipment

Instrument	Critical Specification	Recommended Model	Use*
Pulse Generators (3)	10 V output into 50 ohms, External trigger, 0 to +2.5V DC offset, 0 to 1.4 μ s adjustable delay	HP 8013B	P
Digital Voltmeter	± 1000 Vdc range, 0.1% accuracy	HP 3465A	A,T
Oscilloscope	50 MHz, BW min, 3 channel display	HP 1740A	P,T
Logic State Analyzer	Pattern recognition and state display	HP 1600A	T
Logic Pulser	Pulse logic circuit	HP 10526T	T
Logic Probe	Monitor digital IC's	HP 10525T	T
50 Ω Feedthroughs (3)	50 Ω feedthrough termination	HP 10100C	P
BNC-to-Alligator Clip Adapter (3)		HP Part No. 8120-1292	P
Current Tracer	1 mA to 1A sensitivity	HP 547A	T
Signature Analyzer		HP 5004A	T
*P=Performance Test; A=Adjustment; T=Troubleshooting			

SECTION II

INSTALLATION

2-1. INTRODUCTION.

2-2. This section contains information and instructions for installing the Model 1611A Option A85 Personality Module.

2-3. Figures 2-1 and 2-2 are provided to facilitate component identification.

2-4. MODULE INSTALLATION.

2-5. The 10263A consists of two printed circuit boards, an A85 Personality Panel, and the dedicated A85 microprocessor probe. To install the 10263A in an HP 1611A, proceed as follows:

a. Set LINE switch to off position and disconnect power cord.

b. Disconnect Microprocessor Probe A13 and External Probe A12 from rear panel.

c. Loosen captive screw on rear of 1611A top cover. Remove top cover and disconnect ribbon cable A11W2 from A9 (see figure 2-1).

d. Remove boards A5, A6, A7, A8, A9, and A10.

e. Disconnect ribbon cable A11W1 from connector A1P2 on Main Board A1 (see figure 2-2).

f. Remove screw that secures ground lug to A11 Board.

g. Remove two mounting screws that secure the A11 Board to Keyboard Support MP27 and remove A11 from instrument.

h. Place Option A85 Personality Panel A11 in position and secure it with mounting screws removed in step g.

i. Secure ground lug to A11 as shown in figure 2-2.



Be sure that lug does not short to CRT mounting bracket.

j. Connect A11W1 into Main Board Connector A1P2 as shown in figure 2-2. Dot on main board indicates pin 1 of connector.

k. Install A10 Option A85 ROM Board and A9 Option A85 Personality Board onto Main Board A1 as shown in figure 2-1.

l. Connect A11W2 from Personality Panel to connector A9P3 on A9 board. Location of connector pin 1 is indicated by dot on board.

m. Reinstall A5, A6, A7, and A8 boards onto Main Board A1 as shown in figure 2-1.

n. Connect new Option A85 microprocessor probe A13 to A9 board through appropriate slot in rear panel and install two retaining screws.

o. Reconnect external probe A12 to A10 board through appropriate slot in rear panel and install two retaining screws.

p. Replace top cover of 1611A.

q. Reconnect power cord.

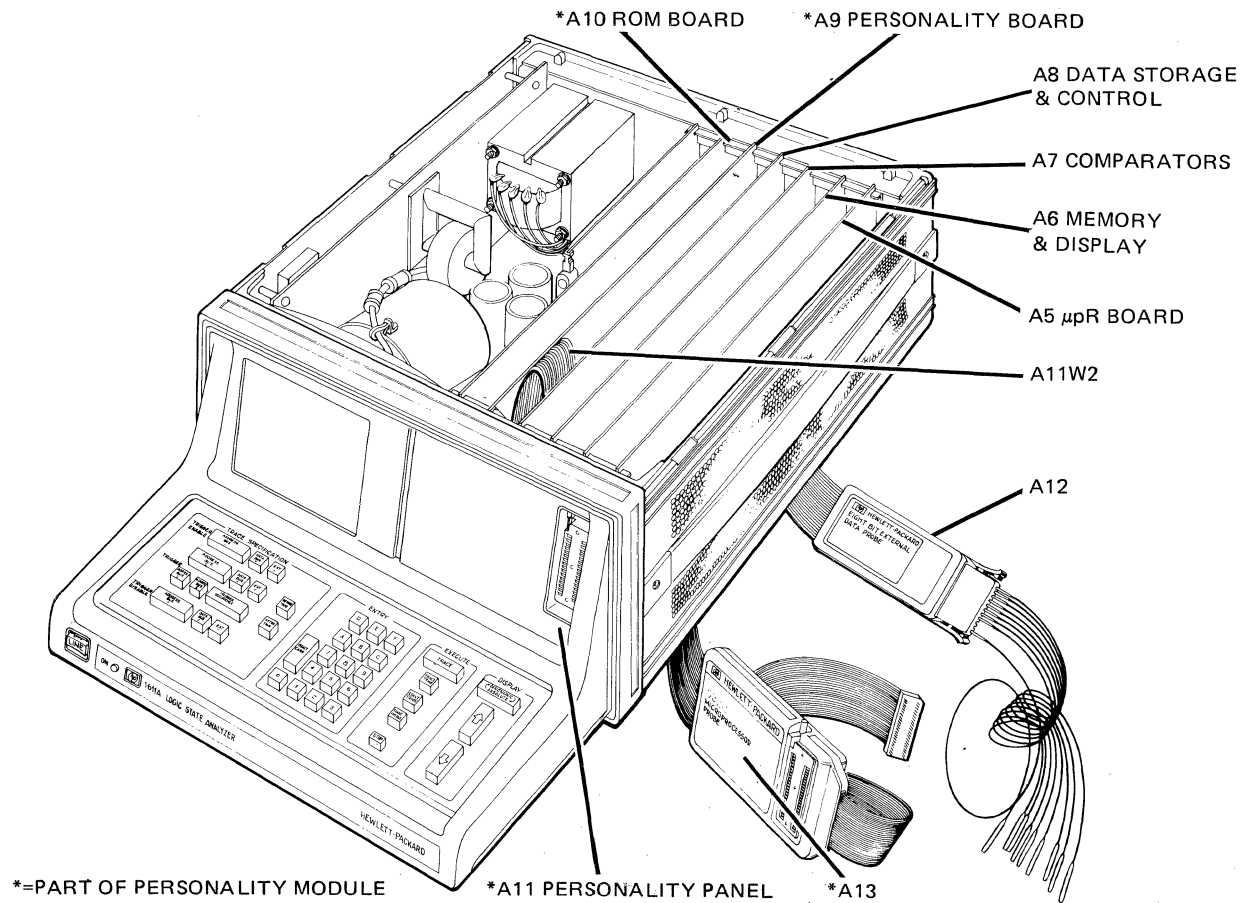


Figure 2-1. Model 1611A Assembly Locations

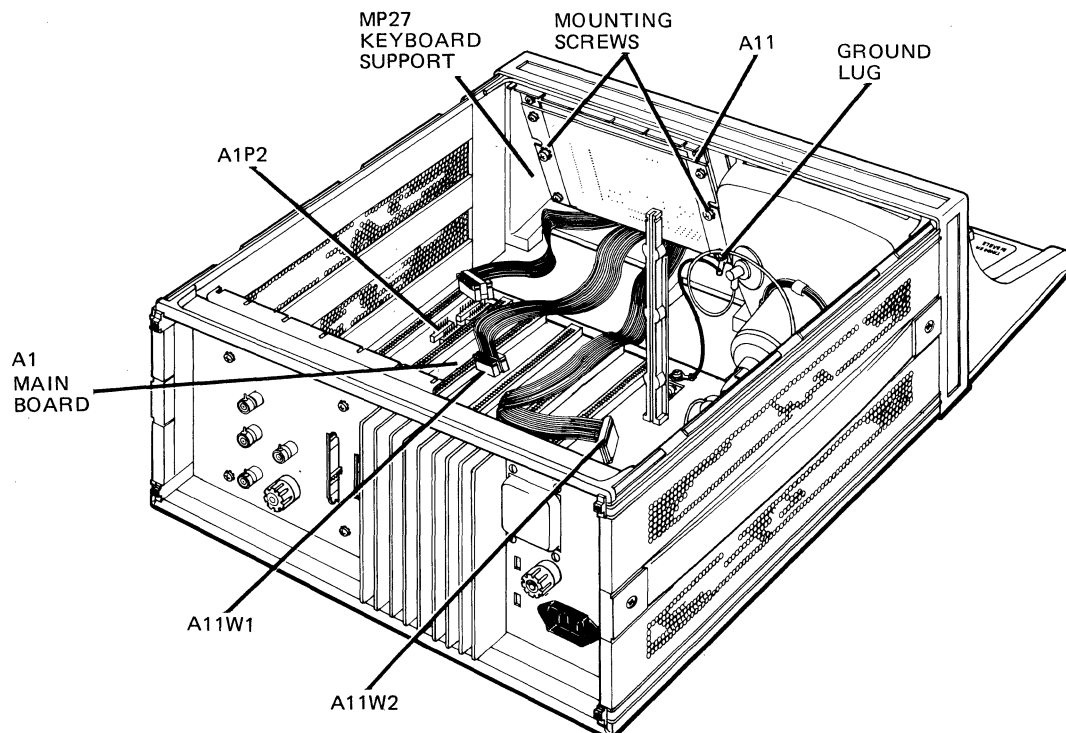
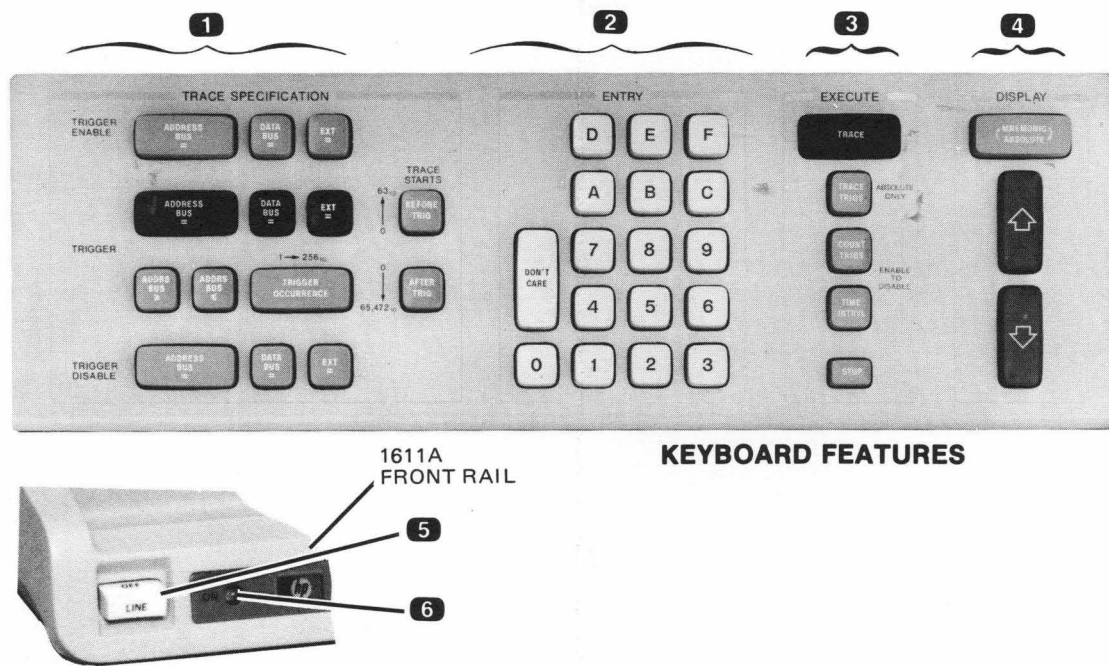


Figure 2-2. Personality Panel A11 Installation



KEYBOARD FEATURES

1 TRACE SPECIFICATION GROUP. Controls for setting up trigger conditions.

TRIGGER Keys.

ADDRESS BUS =. Opens field for entry of desired program or memory address trigger word. Entry is made with up to 4 hex or 6 octal characters and is automatically right-justified with leading zeros assumed.

DATA BUS =. Permits selection of 8-bit trigger word on data bus. Entries may be made with up to 2 hex or 3 octal characters and are automatically right-justified with leading zeros assumed.

EXT =. Permits trigger selection on 8 lines from External Probe A12. Entry is made in binary and is automatically right-justified with leading don't cares assumed.

ADRS BUS $\geq$ and ADRS BUS $\leq$. These keys permit triggering to be limited to activity $\geq$ or $\leq$ selected address. Entry is made with up to 4 hex or 6 octal characters and is automatically right-justified with leading zeros assumed.

TRIGGER OCCURRENCE. Used to determine the number of trigger word occurrences before the trace starts. Entries of a decimal number $1 \leq n \leq 256$ are allowed. Trigger Occurrence is used only in TRACE mode. (If no entry is made, field assumes a value of 1.)

TRIGGER ENABLE and TRIGGER DISABLE Keys. Allow trigger recognition only after Enable and before Disable specifications are met.

ADDRESS BUS =. Controls trigger enable (or disable) recognition on program or memory address word. Entry is made with up to 4 hex

or 6 octal characters and is automatically right-justified with leading zeros assumed.

DATA BUS =. Controls selection of an 8-bit trigger enable (or disable) word on data bus. Entries may be made with up to 2 hex or 3 octal characters and are automatically right-justified with leading zeros assumed.

EXT =. Controls trigger enable (or disable) recognition on 8 lines from External Probe A12. Entry is made in binary and is automatically right-justified with leading don't cares assumed.

TRACE STARTS Keys. (TRACE mode only)

BEFORE TRIG. Selects number of memory transactions that trace starts before trigger (negative time or pretrigger). Entries are made with decimal number $n < 63$.

AFTER TRIG. Selects number of memory transactions that trace starts after trigger (delay). Entries may be made with decimal number $n \leq 65472$. If no entry is made via Before or After Trig, field assumes value of 0.

2 ENTRY GROUP. Alphanumeric keys and DON'T CARE key for entering data into open specification field. Keeping DON'T CARE key depressed approximately 3 seconds clears all TRACE SPECIFICATION entries. If no open data field at the time, the message NO OPEN DATA FIELD will flash until the clear is completed.

3 EXECUTE GROUP. Instructs 1611A to perform a specific task or measurement.

TRACE. This mode provides for capture of 64 consecutive memory transactions in order to

Figure 3-1.
Keyboard and Rear Panel Features
3-0

provide a display of microprocessor program execution.

TRACE TRIGGERS. In this mode, 1611A captures and displays only those memory transactions that match the preselected trigger conditions.

COUNT TRIGS. In this mode, 1611A counts trigger occurrences between Trig Enable and Trig Disable words.

TIME INTERVAL. This mode produces read-out of elapsed time between two states defined by Trigger Enable and Trigger Disable. Timing measurement is made using an internal 1 MHz clock; accuracy is $0.1\% \pm 1 \mu\text{s}$.

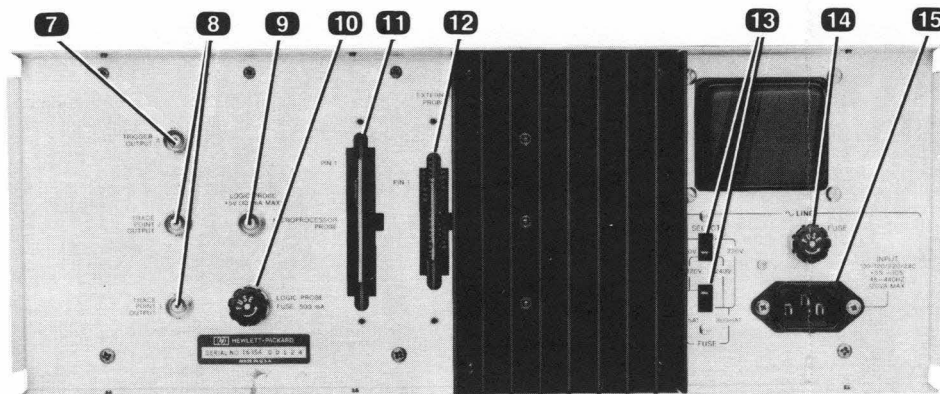
STOP. Halts any execute function in progress; display of activity of μP system under test immediately prior to STOP command will result and STOPPED message will be displayed.

- 4 DISPLAY GROUP.** Used to control format of data displayed; keys do not change data stored in memory.

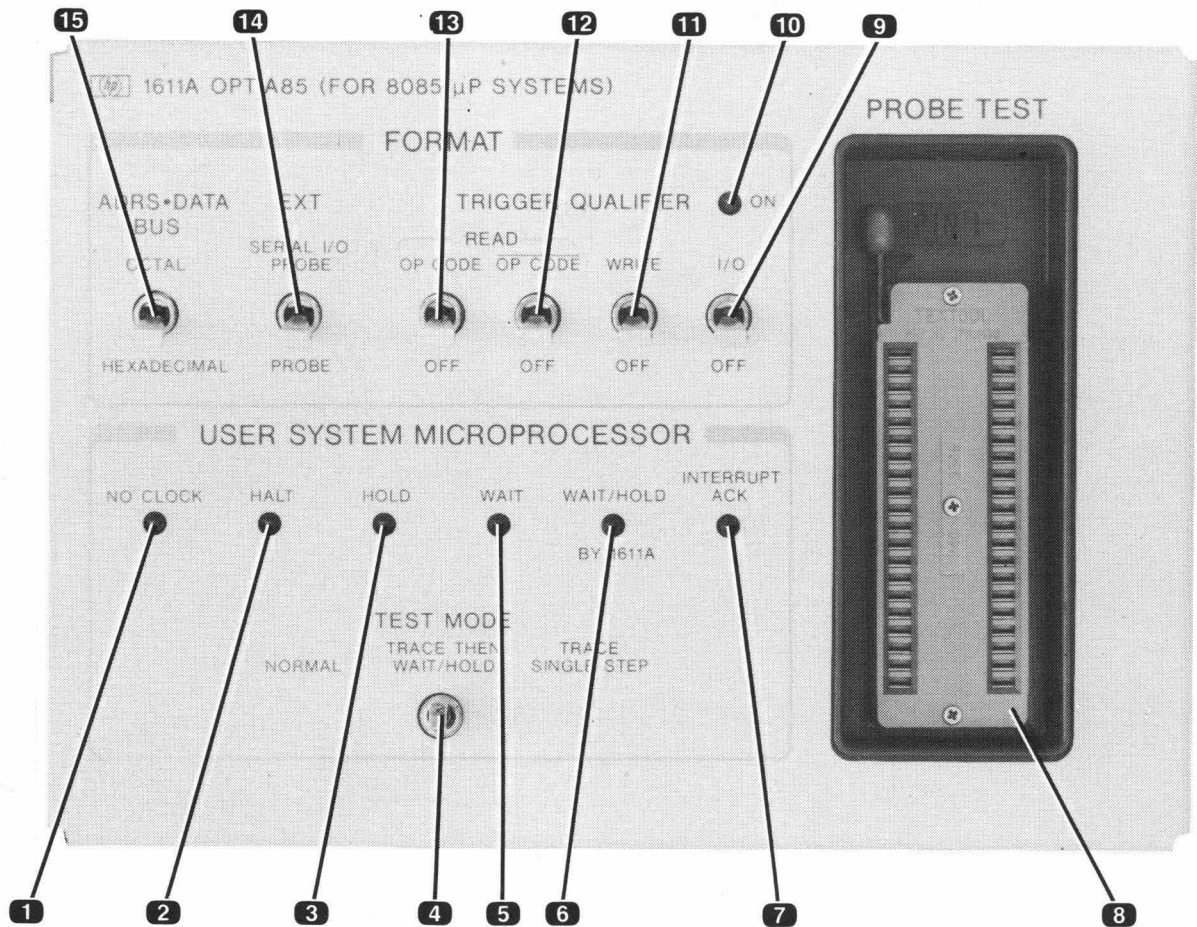
MNEMONIC/ABSOLUTE. Determines if data is displayed as captured (ABSOLUTE) or is to be inverse assembled into mnemonic statements with operands (MNEMONIC). In ABSOLUTE, each line of display represents one memory transaction which includes an address, data word, description of operation (e.g., op code, read, write), and 8 external bits. In MNEMONIC, the data displayed often requires more than one memory transaction per line. The 1611A data display often contains less than 64 lines in MNEMONIC mode although data content is the same as in the 64-line ABSOLUTE display.

ROLL $\uparrow$ $\downarrow$. Used to roll 16-line display window through 64-byte memory.

- 5 LINE.** Line power switch.
- 6 ON.** Indicator lights when line power is on.
- 7 TRIGGER OUTPUT $\square$ $\square$.** Provides $\sim 75 \text{ ns}$, RZ, TTL output on each occurrence of selected trigger word (useful for triggering test equipment).
- 8 TRACE POINT OUTPUT ($\square$) ($\square$).** These outputs are levels which change state when a trace is initiated; they return to previous state when a trigger word is recognized or trace point is reached after delay. This provides a transition for triggering and a pulse for gating, or causing breakpoints with external logic.
- 9 LOGIC PROBE.** +5 V power connector for logic probes requiring less than 100 mA.
- 10 LOGIC PROBE FUSE.** 500 mA fuse for Logic Probe Power.
- 11 MICROPROCESSOR PROBE.** Microprocessor Probe connector.
- 12 EXTERNAL PROBE.** External Probe connector.
- 13 LINE SELECTOR.** Slide switches for selecting 100-, 120-, 220-, or 240-Vac line operation.
- 14 FUSE.** Use 1.0 AT time-delay fuse for 100- 120-, 220-, or 240-Vac operation.
- 15 POWER INPUT.** Power cable connector.



REAR PANEL FEATURES



9 TRIGGER QUALIFIER. Indicates that Trace Specification (including TRIGGER, ENABLE, and DISABLE) must occur during a specific type of machine cycle before 1611A trigger will occur. When trigger requirements are satisfied via Trigger Qualifiers, a TRACE measurement proceeds normally (Before or After Trigger specification and data acquisition are not affected). If all switches are in OFF position, then the 1611A will trigger on ANY machine cycle that meets TRACE SPECIFICATION requirements.

10 I/O. Allows 1611A trigger to occur on Input/Output machine cycles.

11 WRITE. Allows 1611A trigger to occur on Memory Write machine cycles.

12 READ OPCODE. Allows 1611A trigger to occur on Memory Read machine cycles other than Op-Codes.

13 READ OP CODE. Allows 1611A trigger to occur on Op-Code Fetch machine cycles.

NOTE

Any combination of QUALIFIER switches may be used to allow the 1611A trigger to occur on more than one type of machine cycle.

14 SERIAL PROBE. In SERIAL selects bits 6 and 7 of the EXTERNAL display to be S0D and S1D lines of the 8085 while the other 5 bits are from the external probe. In PROBE all bits displayed are from the external probe.

15 OCTAL/HEXADECIMAL. Selects octal or hexadecimal format for data display and Trace Specifications. Binary and decimal numbers are unaffected. Flashing?(s) displayed in Trace Specification field(s) when the code entered is not complete enough for exact conversion to the other base (specification contains "don't care(s)" in least significant digit(s)).

Figure 3-2. Personality Panel Controls and Indicators

SECTION III

OPERATION

3-1. INTRODUCTION.

3-2. This section provides information, explains functions of 1611A controls, connectors, and indicators, and describes measurement capabilities and operating characteristics of the 1611A with a Model 10263A Option A85 installed.

3-3. PANEL FEATURES.

3-4. Keyboard and rear-panel features are shown in figure 3-1; Personality Panel A11 is shown and described in figure 3-2.

3-5. **PROBES.** The 1611A uses two probes to acquire data from the system under test. Primary Probe A13 is dedicated to the microprocessor and is not interchangeable with other 1611A options. It gathers signals from the microprocessor Address and Data busses along with the clock and control signals. Connection is made from the probe pod to the microprocessor socket in the system under test through a ribbon cable and connector. The pod contains a socket for the microprocessor. If the microprocessor cannot be removed from the system, the connector cable and plug can be replaced with a ribbon cable terminated with a dual in-line clip connector which provides a direct connection from the pod to the microprocessor. When the clip is used, the socket on the probe pod is left empty.

3-6. The HOLD-READY switch on Probe A13 allows the 1611A to halt the μ P system under test two ways. Either the μ P Hold input or the μ P Ready input can be used to halt the μ P under test in TRACE THEN WAIT or TRACE SINGLE STEP mode.

3-7. When Probe A13 switch is in HOLD position, the 1611A can halt the μ P under test only if the 8085 CPU is in Probe A13 socket and Probe A13 is connected to the user system. Halting is not possible if Probe A13 is connected to the 8085 CPU by the dual-in-line clip connector.

3-8. When Probe A13 switch is in READY position, the μ P system under test can be halted either with 8085 CPU in Probe A13 socket, or with Probe A13 connected to the 8085 CPU by the dual-in-line clip connector. The ready output of Probe A13 is an open-collector TTL circuit that can sink at least 8 mA when active.

3-9. The external probe A12 is secondary, with no clock input, and is not required for instrument operation. It has 8 uncommitted input lines to accept data coincident with data bus inputs on the primary probe. Input impedance is approx 1 M Ω shunted by ≤ 25 pF at the probe tip.

3-10. SETTING TRACE SPECIFICATIONS.

3-11. At instrument turn on, the display should be as shown in figure 3-3. The display is divided into two parts by a dashed horizontal line. Above this line, all pertinent trace specifications are displayed as they are entered. The trigger ADDRESS bus field is open (inverse video field to right of TRIGGER). A blinking cursor in the lower left corner of the field indicates where a character entered by means of the ENTRY key group will be located.

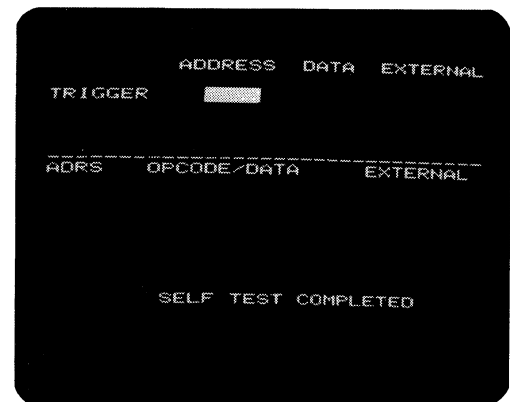


Figure 3-3. Turn-on Display

3-12. The first step in making a measurement is to set up the trace specification. Pressing the key for the desired specification in the TRACE SPECIFICATION key group opens the data field. This field is now ready to accept data from the ENTRY key group for ADDRESS (Octal or Hex), DATA (Octal or Hex), or EXTERNAL (Binary only) specifications.

3-13. To enter data in a field, key in the desired value. If the pressed key is an illegal character for the selected specification field (such as an alpha character for a decimal field), an ILLEGAL CHARACTER message is flashed on the display. This message is cleared when the ENTRY key for an acceptable character is pressed, or when a new TRACE SPECIFICATION, EXECUTE, or DISPLAY field key is selected. All entries are automatically right-justified, and any EXTERNAL entries automatically assume leading don't cares, while ADDRESS and DATA entries automatically assume leading zeros.

3-14. An existing trace specification can be changed or deleted in the following manner. To change a trace specification, press the TRACE SPECIFICATION key for that field, then enter the new data. To delete a trace specification, press

the TRACE SPECIFICATION key for that field, then press DONT CARE in the ENTRY key group. All TRACE SPECIFICATION FIELDS CAN BE "cleared" by pressing DONT CARE key for approximately 3 seconds.

3-15. TRIGGERING. The TRIGGER keys ADDRESS BUS =, DATA BUS =, and EXT =, can be used in any combination to specify a 1611A Trigger word. ADDRESS BUS = key allows 1611A Trigger on a desired μ P program or memory address word. DATA BUS = key allows 1611A to Trigger on a desired μ data bus word. EXT = key allows 1611A to Trigger on a desired external word. When more than one key is used, all requirements must be satisfied. The TRIGGER ENABLE field allows the 1611A to recognize only those Triggers which occur before the Disable specification has been detected. When used together, the Enable and Disable fields define a window, in which the 1611A is allowed to search for specified Triggers. The ADDR BUS $\geq$ key allows any address bus value which is greater than or equal to the value specified (from the value specified up to $FFFF_{16}$ or 17777_8) to be recognized as a Trigger. The ADDR BUS $\leq$ key allows any address bus value specified (from the value specified down 0000_{16} or 00000_8) to be recognized as a Trigger. These two keys may be used together to define ranges over which any address bus activity will generate a trigger. For example, if a smaller number is entered in $\geq$ field than is entered in the $\leq$ field, then only the address bus activity within this window will be recognized. If a larger number is entered in $\geq$ field than is entered in the $\leq$ field, then the address bus activity which occurs outside of this window will be recognized and displayed. Trigger requirements, including Enable and Disable, can be restricted to particular μ P machine cycles with the TRIGGER QUALIFIER switches on Personality Panel A11.

3-16. TRACE MEASUREMENT EXECUTION.

3-17. TRACE. This is the most common mode of data acquisition. Trace mode allows capture of up to 64 consecutive memory transactions from the microprocessor system under test. After a completed Trace measurement run, the 1611A high-speed memory contains up to 64 memory transactions. The memory transactions are viewed through a 16-line movable window (figure 3-4). Thus, in Trace mode the 1611A can monitor and display the program execution of the microprocessor system under test.

3-18. The starting point for data acquisition is selected via the TRIGGER, TRIGGER OCCURRENCE, and BEFORE or AFTER TRIG Keys. The TRIGGER OCCURRENCE requirement will allow the 1611A to recognize the Trigger only on the nth occurrence (with n being any desired value from 1 to 256; if no entry is made, field assumes a value of 1). The BEFORE TRIG Key can be used to start the trace up to 63 words before the Trigger. This permits viewing of events leading up to the Trigger (negative-time or pretrigger). The AFTER TRIG Key can be used to delay start of the trace up to 65 472 transactions after the Trigger event (figure 3-5). When AFTER TRIG delay is used, the data captured in high-speed memory will not contain the Trigger word.

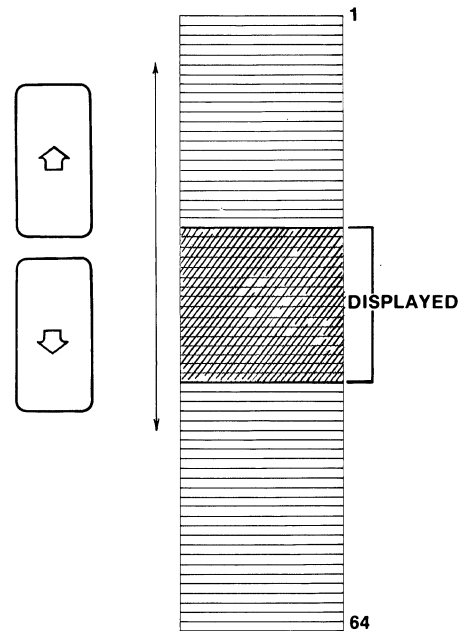


Figure 3-4. Display Window

3-19. Trace display may be viewed in ABSOLUTE or MNEMONIC mode. In both modes the program or memory address is displayed in the ADDR column. In ABSOLUTE (figure 3-6), the machine language op-code or data and the type of μ P machine cycle (e.g., READ, OPCODE, WRITE) are displayed in the OPCODE/DATA column (table 3-1). In MNEMONIC, the assembly language mnemonic and 8 or 16 bit operand (if the operand exists) are displayed in the OPCODE/DATA column (figure 3-7). If an invalid op-code is detected, then *** will be flashing in the OPCODE/DATA column when in MNEMONIC mode. If *'s are displayed NOT flashing, then the operand was not captured by high-speed memory. The * also occurs when 8085 does not take the jump on a conditional jump. Single-byte operand displays are ** (Hex) or *** (Octal). Two-byte operand displays are **** (Hex) or ***** (Octal). Hexadecimal or octal format can be selected and interchanged in both display modes.

3-20. Trace measurements may be made in NORMAL, TRACE THEN HALT, or TRACE SINGLE STEP mode. These modes are selected by the TEST MODE switch on Personality Panel A11. In NORMAL, the 1611A captures and displays data while the microprocessor system under test operates without interruption of program flow. In TRACE THEN HALT, the 1611A places the μ P system under test in a Halt state when the data acquisition cycle is completed. In TRACE SINGLE STEP, the 1611A Halts the μ P system under test after each instruction cycle. The SINGLE STEP mode is functional only in TRACE measurements. During each TRACE execution only one μ P instruction cycle is processed and displayed.

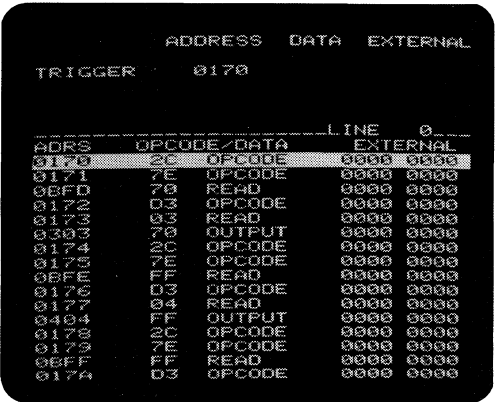
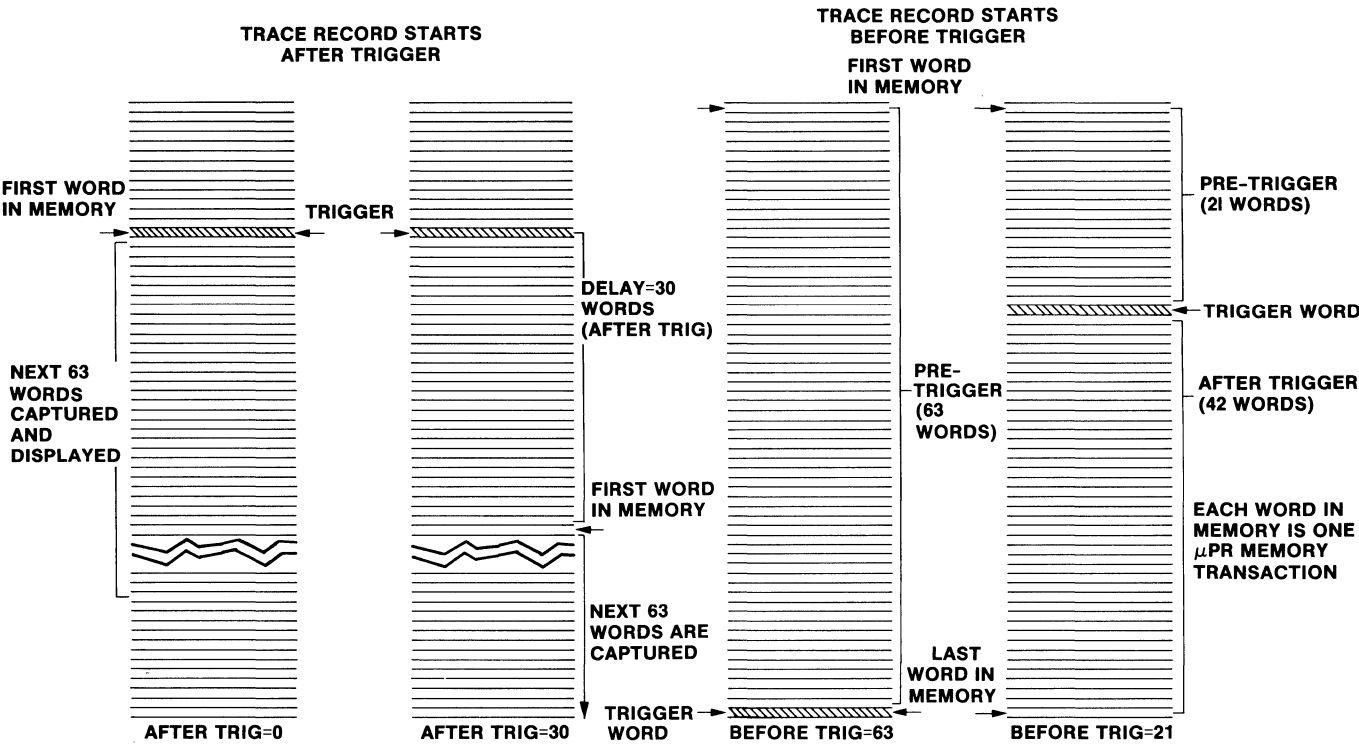


Figure 3-6. Absolute Mode Display

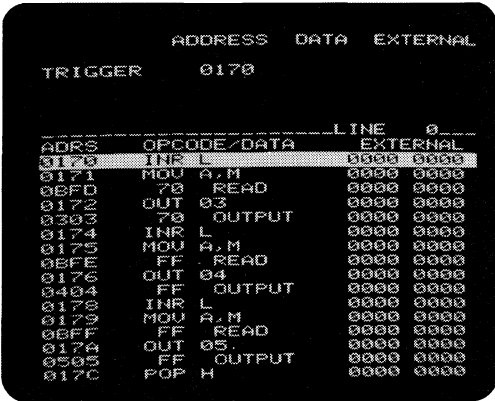


Figure 3-7. Mnemonic Mode Display

Table 3-1. A85 Memory Transaction Displays

OPCODE	The contents of the data bus is an op-code.
WRITE	The contents of the data bus is information being written to memory.
READ	The contents of the data bus is being read from memory.
INPUT	The contents of the data bus was read from an I/O Port.
OUTPUT	The contents of the data bus is data to be written to an I/O Port.

3-21. A TRACE measurement is initiated each time the TRACE key is pressed. Repeated TRACE runs may be made with continuous-run execution. Continuous execution is initiated by pressing TRACE key approximately 2 seconds. The message CONTINUOUS will be displayed. Pressing the TRACE key again restores regular operation. Continuous runs may be stopped by pressing STOP key or aborted by pressing any other key.

3-22. **Trace Setup.** Enter TRIGGER requirements. Any desired combination of Trigger Enable, Trigger Disable, Trigger Occurrence, and Before or After Trigger requirements may be entered. Ranges of address bus values which will be recognized as Triggers may also be entered

with ADDRS BUS $\geq$ or $\leq$ keys. Triggering (including Enable and Disable) may be restricted to certain types of μ P under test machine cycles with the TRIGGER QUALIFIER switches on Personality Panel A11. Press TRACE key.

3-23. Display in TRACE.

3-24. Before the first Trigger is detected, two messages may be displayed: (1) WAITING FOR ENABLE flashes until the Enable specification (if entered) is detected; (2) WAITING FOR TRIGGER flashes until the Trigger specification is detected (after the Enable, if specified, has been detected).

3-25. WAITING FOR ENABLE and WAITING FOR TRIGGER can be alternating. This happens when both Enable and Disable are specified and no Trigger is found in the window. The Enable is found, no Trigger is found, the Disable is found, and the 1611A looks for another Enable.

3-26. After the first Trigger is detected, but before the TRACE measurement is complete, two messages may be displayed: (1) TRIGR OCCUR = shows how many Triggers are detected from the first Trigger until the Trigger Occurrence specification is met; (2) DELAYING is shown from the time Trigger Occurrence specification is met until the measurement is complete.

3-27. Any data shown below the dashed line will not change until the measurement is complete or until data acquisition is stopped by the STOP key. When a measurement is completed, the Trigger word is shown in an inverse video field, if no AFTER TRIGGER DELAY is specified (figure 3-8). The message LINE 0 also appears. The DISPLAY ROLL keys may be used to move the 16-line display window through the 64 lines (0 to 63) stored in high-speed memory. The LINE message will vary from 0 up to 48. The LINE number shown is the number of the top line in the 16-line display (48 + 15 = 63).

3-28. If an incomplete TRACE measurement is stopped by the STOP key, then all data acquired by the high-speed memory during measurement will be displayed (memory dump). If the memory has no new data, then the data display section will be blank. The message STOPPED is shown.

3-29. When a TRACE measurement is incomplete and any key other than an EXECUTE key is pressed, the measurement run is aborted. The display remains as it was before the measurement. Any attempt to change the display with DISPLAY group keys will result in the message RUN ABORTED, NO DISPLAY CHANGE.

3-30. Partial displays (less than 64 memory transactions stored for display) can occur in Trace mode when using a BEFORE TRIGGER specification. The minimum number of lines that will be displayed in ABSOLUTE mode is 64 minus the BEFORE TRIG specification. Internally, the 1611A completes a measurement

ADDRESS	OPCODE/DATA	EXTERNAL
0290	MOV A.M	0000 0000
02C0	78 READ	0000 0000
029E	NOP	0000 0000
029F	STAX B	0000 0000
02FD	70 WRITE	0000 0000
02A0	NOP	0000 0000
02A1	POP D	0000 0000
02BC	F3 READ	0000 0000
02ED	0E READ	0000 0000
02A2	INR E	0000 0000
02A3	INR C	0000 0000
02A4	JNZ 02BE	0000 0000
02BE	LMI H, 02C0	0000 0000
0291	LDAX D	0000 0000
02F4	10	0000 0000
0292	NOP	0000 0000

Figure 3-8. Trace Display

by counting the number of memory transactions after the Trigger. Because of this, the 1611A may not display the number of memory transaction BEFORE TRIG that have been specified. If a BEFORE TRIG entry of 60 is made, then the 1611A will terminate the measurement 3 memory transactions after the Trigger. This allows for up to 60 memory transactions before the Trigger, the Trigger, and 3 memory transactions after the Trigger to be captured in high-speed memory. If, after the Trace measurement was started, the system under test executed only 15 memory transactions before the Trigger occurred, then the 1611A would display only 19 memory transactions. The ROLL keys can be used to view the 15 transactions before the Trigger, the Trigger, and 3 transactions after the Trigger. The display will not roll up any further once the last memory transaction captured by the 1611A is in the bottom line of the display.

3-31. TRACE TRIGGERS MEASUREMENT EXECUTION.

3-32. **TRACE TRIGGERS.** In this mode the 1611A captures and displays only memory transactions that match the Trigger specification. For example, the operator may want to see only "Write" operations from a block of memory to verify that proper data was entered. This is easily done by using the Write TRIGGER QUALIFIER switch on Personality Panel A11. The run automatically terminates when 64 Triggers have been found. The run can be terminated before 64 Triggers have been found by using the STOP key.

3-33. **TRACE TRIGGERS End on DISABLE.** In this mode the TRACE TRIGGERS measurement ends only when Trigger Disable is detected. The 1611A will display up to 64 Triggers that occurred immediately prior to detection of the Disable.

NOTE

Models 1611A with Serial Prefix 1723A and above (Opt A85 installed) have this capability. If Trace Triggers end on Disable mode is desired in Models 1611A with Serial Prefix below 1723A, the old A8 board (01611-66508) must be changed to a modified A8 board (01611-66535).

To enter this mode: (1) Specify a BEFORE TRIG (PRE-TRIGR) of 63; (2) Specify a TRIGGER DISABLE; (3) Press TRACE TRIGS.

3-34. Trace Triggers and Trace Triggers End on Disable displays may be viewed in ABSOLUTE or MNEMONIC mode. In both modes the program or memory address of the Trigger is displayed in the ADRS column. In ABSOLUTE, the machine language op-code or data and the type of μ P machine cycle (e.g., READ, OPCODE, WRITE) are displayed in the OPCODE/DATA column. In MNEMONIC, the assembly language mnemonic and 8 or 16 bit operand (if the operand exists) are displayed in the OPCODE/DATA column. If the operand is not captured by high-speed memory, then *** is shown in the appropriate line of OPCODE/DATA column. Hexadecimal or octal format can be selected and interchanged in both display modes.

3-35. Trace Triggers or Trace Triggers End on Disable measurements can be made in NORMAL or TRACE THEN HALT modes (selected by TEST MODE switch on Personality Panel A11). In NORMAL, the 1611A captures and displays Triggers while the μ P system under test operates without interruption of program flow. In TRACE THEN HALT, the 1611A places the μ P system under test in a Halt state when the data acquisition cycle is completed.

3-36. A TRACE TRIGGERS or TRACE TRIGGERS End on DISABLE measurement run is initiated each time TRACE TRIGS key is pressed. Repeated runs may be made with Continuous-run execution. Continuous execution is initiated by pressing TRACE TRIGS key approximately 2 seconds. The message CONTINUOUS will be displayed. Pressing the TRACE TRIGS key again restores regular operation. Continuous runs may be stopped by pressing STOP key or aborted by pressing any other key.

3-37. **Trace Triggers Set-up.** Enter TRIGGER requirements. Trigger Enable and/or Disable requirements may be entered if desired. Ranges of address bus values which will be recognized as Triggers may also be entered with ADDRS BUS $\geq$ or $\leq$ keys. Triggering (including Enable and Disable) may be restricted to certain types of μ P under test machine cycles with the TRIGGER QUALIFIER switches on Personality Panel A11. Press TRACE TRIGS key.

3-38. Display in TRACE TRIGGERS.

3-39. Before the first Trigger is detected two messages may be displayed. (1) WAITING FOR ENABLE flashes until the Enable specification (if entered) is detected. (2) WAITING FOR TRIGGER flashes until the Trigger specification is detected (after the Enable, if specified, has been detected).

3-40. WAITING FOR ENABLE and WAITING FOR TRIGGER can be alternating. This happens when both Enable and Disable are specified and no Trigger is found in the window. The Enable is found, no Trigger is

found, the Disable is found, and the 1611A looks for another Enable.

3-41. During TRACE TRIGGERS End on DISABLE measurements, the message TRIGR OCCUR=0 flashes (instead of WAITING FOR TRIGGER) until the first Trigger is detected.

3-42. After the first Trigger is detected, but before the TRACE TRIGGERS measurement is complete, the message TRIGR OCCUR = may appear. This shows how many Triggers are detected from the first Trigger until high-speed memory is full (64 Triggers Stored).

3-43. In Trace Triggers End on Disable measurements, the message TRIGR OCCUR = >64 can appear for two reasons: (1) When 64 Triggers are detected before the Disable is detected; (2) The 1611A does not contain the TRACE TRIGGERS End on DISABLE modification (1611A with Serial Numbers Prefix below 1723A).

3-44. Any data shown below the dashed line will not change until the measurement is complete or until data acquisition is stopped by the STOP key. When a measurement is completed, the message TRACE TRIGGERS appears. Triggers are shown in the data display (figure 3-9). The message LINE 0 also appears. The DISPLAY ROLL keys may be used to move the 16-line display window through the 64 lines (0 to 63) stored in high-speed memory. The line message will vary from 0 to 48. The LINE number shown is the number of the top line in the 16-line display ($48 + 15 = 63$). TRACE TRIGGERS End on DISABLE measurements can result in partial displays. This is because less than 16 Triggers were detected before Disable.

ADDRESS	DATA	EXTERNAL
TRIGGER	0BFF	
TRACE TRIGGERS	-----	LINE 0
ADRS	OPCODE/DATA	EXTERNAL
0BFD	70 READ	0000 0000
0BFE	FF READ	0000 0000
0BFF	FF READ	0000 0000
0BF0	15 READ	0000 0000
0BFA	7E WRITE	0000 0000
0BF1	16 READ	0000 0000
0BFB	F9 WRITE	0000 0000
0BF2	00 READ	0000 0000
0BFC	00 WRITE	0000 0000
0BF3	00 READ	0000 0000
0BFD	70 WRITE	0000 0000
0BF4	10 READ	0000 0000
0BFE	FF WRITE	0000 0000
0BFF	10 READ	0000 0000
0BFF	FF WRITE	0000 0000
0BFA	7E READ	0000 0000

Figure 3-9. Trace Trigs Display

3-45. If an incomplete TRACE TRIGGERS or TRACE TRIGGERS End on DISABLE measurement is stopped by the STOP key, then all data acquired by the high-speed memory during the measurement will be displayed (memory dump). If the memory has no new data, then the data display section will be blank. The message STOPPED is shown.

3-46. When a TRACE TRIGGERS or TRACE TRIGGERS End on DISABLE measurement is incomplete and any key other than an EXECUTE key is pressed, the measurement is aborted. The display remains as it

was before the measurement. Any attempt to change the display with DISPLAY Group Keys will result in the message RUN ABORTED, NO DISPLAY CHANGE.

3-47. COUNT TRIGGERS MEASUREMENT EXECUTION.

3-48. COUNT TRIGGERS. In this mode the 1611A counts Triggers detected between (and including) the Trigger Enable and Trigger Disable words. Count Triggers measurements can be performed in two ways: (1) A regular measurement run is executed each time COUNT TRIGS key is pressed; (2) Repeated COUNT TRIGGERS runs are made with Continuous-run execution. Count Triggers measurements are made with TEST MODE switch on Personality Panel A11 in the NORMAL position.

3-49. A COUNT TRIGGERS measurement run is initiated each time COUNT TRIGS key is pressed. Repeated COUNT TRIGGERS runs may be made with Continuous-run execution (initiated by pressing COUNT TRIGS key approximately 2 seconds). The message CONTINUOUS will be displayed. Pressing COUNT TRIGS keys again restores regular operation. Continuous runs may be stopped by pressing STOP key or aborted by pressing any other key.

3-50. Count Triggers Set-up. Enter Trigger ENABLE, TRIGGER, and, Trigger DISABLE requirements. Ranges of address bus values which will be recognized as Triggers may also be entered with ADDR BUS $\geq$ or $\leq$ keys. Triggering (including Enable and Disable) may be restricted to certain types of μ P under test machine cycles with the TRIGGER QUALIFIER switches on Personality Panel A11. Press COUNT TRIGS Key.

3-51. Display in COUNT TRIGGERS.

3-52. Before Enable is detected, the message WAITING FOR ENABLE flashes. The bottom line of the data display will show COUNT = 0 TRIGGERS. This line shows the incomplete result of a count in progress. After the Enable is detected, but before the DISABLE is detected, the message COUNTING may flash and the bottom line COUNT = shows how many Triggers have been detected. If the cumulative count exceeds $2^{24} - 1$ (approximately 16.78×10^6) Triggers, then the message COUNTER OVERFLOW flashes. This means that the Disable was not detected before the counter limit was exceeded.

3-53. When Disable is detected the measurement is completed (figure 3-10). The bottom line of the data display COUNT = disappears. The top line of the data display COUNT = shows how many Triggers were detected between (and including) the Enable and Disable. MAX = and MIN = show the same number of Triggers detected as COUNT=. This is because both the maximum and minimum results of the first COUNT TRIGS run are the same as the completed measurement result.

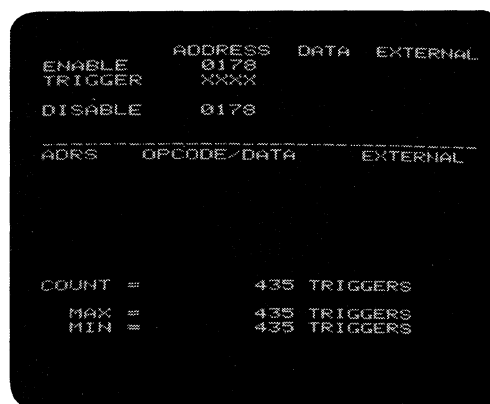


Figure 3-10. Count Triggers Display

3-54. In COUNT TRIGGERS Continuous-run execution, the bottom line COUNT = will often be flashing. This is due to updating of the incomplete result of a count in progress. The top line COUNT = shows the result of the last measurement completed. MAX = shows the maximum number of Triggers detected by any of the repeated executions of Count Triggers. MIN = shows the minimum number of Triggers detected by any of the repeated executions of Count Triggers.

3-55. Any Count Triggers measurement in progress may be stopped with the STOP key or aborted with any other key. When the STOP key is pressed while a run is in progress, the message STOPPED appears.

3-56. TIME INTERVAL MEASUREMENT EXECUTION.

3-57. TIME INTERVAL. This mode produces a readout of elapsed time between the occurrence of two states defined by Trigger Enable and Trigger Disable. The timing measurement is made using an internal 1 MHz reference oscillator. Accuracy is within $0.1\% \pm 1\mu s$. Verifying the execution time of wait loops is one application of the time interval feature. It is also useful for measuring time required for the execution of a subroutine when a critical timing interface is involved with the system hardware or an external device. Time Interval measurements can be performed in two ways: (1) A regular measurement run is executed each time TIME INTVL key is pressed; (2) Repeated TIME INTERVAL runs are made with Continuous-run execution. Time Interval measurements are made with TEST MODE switch on Personality Panel A11 in the NORMAL position.

3-58. A TIME INTERVAL measurement run is initiated each time TIME INTVL key is pressed. Repeated TIME INTERVAL runs may be made with Continuous-run execution. This is initiated by pressing TIME INTVL key approximately 2 seconds. The message CONTINUOUS will be displayed. Pressing TIME INTVL key again restores regular operation. Continuous runs may be stopped with STOP key or aborted by pressing any other key.

3-59. Time Interval Set-up. Enter Trigger ENABLE and Trigger DISABLE requirements. No TRIGGER entry is

required. The Enable and Disable may both be restricted to certain types of μ P under test machine cycles with the TRIGGER QUALIFIER switches on Personality Panel A11. Press TIME INTVL key.

3-60. Display in TIME INTERVAL.

3-61. Before the Enable is detected, the message WAITING FOR ENABLE flashes. The bottom line of the data display will show TIME = 0 MICROSECONDS. This line shows the incomplete result of a count in progress. After the Enable is detected, but before the DISABLE is detected, the message COUNTING may flash and the bottom line TIME = shows the elapsed time in microseconds. If the cumulative count exceeds $2^{24}-1$ (approximately 16 seconds), then the message COUNTER OVERFLOW flashes. This means that the Disable was not detected before the counter limit was exceeded.

3-62. When the Disable is detected the measurement is completed (figure 3-11). The bottom line of the data display TIME = disappears. The top line of the data display TIME = shows elapsed time in microseconds between occurrence of the Enable and the Disable. MAX = and MIN = will show the same elapsed time as TIME =. This is because both maximum and minimum results of the first TIME INTVL run are the same as the completed measurement result.

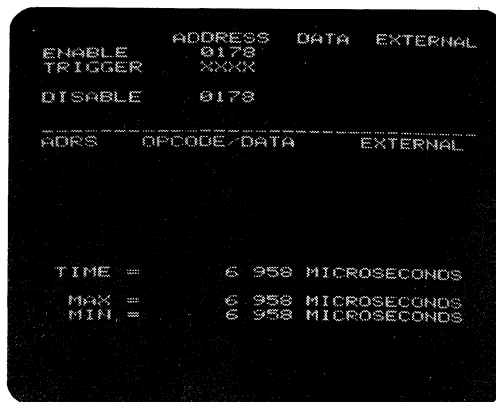


Figure 3-11. Time Interval Display

3-63. In TIME INTERVAL Continuous-run execution, the bottom line TIME = will often be flashing. This is due to updating of the incomplete result of a count in progress. The top line TIME = shows the result of the last measurement completed. MAX = shows maximum elapsed time of any repeated executions of Time Interval. MIN = shows minimum elapsed time of any repeated executions of Time Interval.

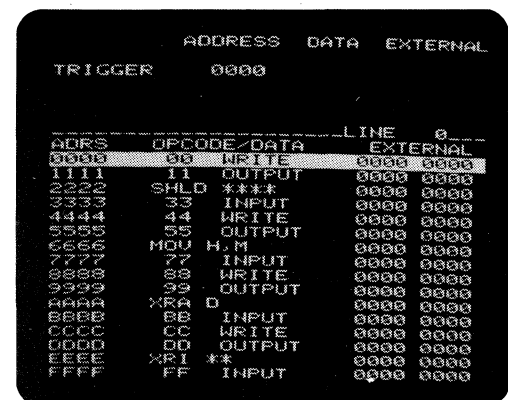
3-64. All TRIGGER field entries are ignored by the 1611A in Time Interval mode. Any Time Interval measurement in progress may be stopped with the STOP key or aborted with any other key.

3-65. OPERATOR'S CHECKS.

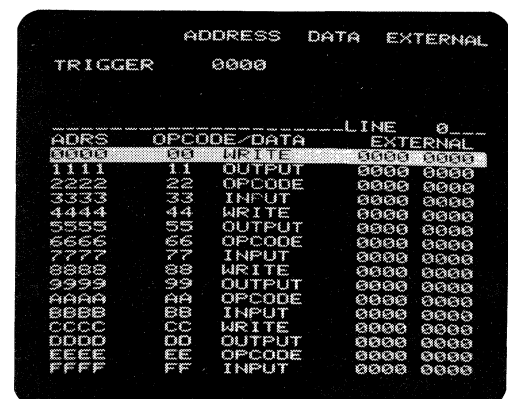
3-66. SELF-TEST. When power is applied to the 1611A, the internal microprocessor automatically performs an

internal self-test on the 1611A. A check sum is performed to verify contents of the ROM. Next, a test pattern is written to the RAM (to check for RAM ERROR), then read back and checked for accuracy. Also, some system hardware (most of the circuitry on A7 and A8 boards) is tested at this time. Upon completion of these tests, the message SELF-TEST COMPLETED is written on screen by the microprocessor to give a visual indication that the display circuitry and microprocessor are working properly. If a malfunction is detected, then the messages ERROR IN RAM, ERROR IN ROM, or ERROR IN HARDWARE may be displayed. These give quick indications of where to look when 1611A troubleshooting is necessary.

3-67. PROBE TEST. (Switch on probe must be in HOLD position during probe test.) Functional tests may be made on the 1611A without any external circuits by connecting the microprocessor probe plug into the Personality Panel Probe Test socket. Executing a trace generates a test display (see figure 3-12). Most system functions and controls may then be exercised on this pattern. In addition to generating a test pattern, the PROBE TEST feature permits checkout of Personality Panel indicators by cycling through NO CLOCK, HALT, WAIT, and INTERRUPT ACK lights at a 1-Hz rate. Check WAIT/HOLD BY 1611A light by setting TEST MODE switch to SINGLE STEP. Return TEST MODE switch to NORMAL.



Mnemonic



Absolute

Figure 3-12. Probe Test Display

3-68. STATUS MESSAGES. The following messages are displayed on the 1611A screen to provide an indication of 1611A operating status. (Messages may vary with different 1611A options.)

CONTINUOUS: Appears after an EXECUTE key has been held down approximately 2 seconds, which initiates continuous-run execution. Single run is restored by pressing the same key briefly. Continuous operation is stopped by STOP key, or aborted by pressing any other key.

COUNTER OVERFLOW: Appears in COUNT TRIGS or TIME INTERVAL mode when event counters overflow (Count exceeds $2^{24} - 1$).

COUNTING: Appears in COUNT TRIGS or TIME INTERVAL mode when 1611A is in the process of counting.

DELAYING: Displayed in TRACE mode from the time Trigger Occurrence specification has been met until the measurement is complete.

ERROR IN HARDWARE: Displayed when 1611A self-test has detected a malfunction in its internal hardware (on A7 or A8 boards).

ERROR IN RAM/ROM: Appropriate message appears when self-test display comes on after turn-on if 1611A determines that Random-Access Memory or Read Only Memory is not functioning correctly.

ILLEGAL CHARACTER: Appears if the operator attempts to enter an unacceptable character into an open trace specification field. For example, attempting to enter an alpha character in the BEFORE TRIG field (decimal field) will generate this message.

LINE 0 - 48: Indicates line number of top line in data display in TRACE and TRACE TRIGGERS modes. When LINE 48 is displayed, the next 15 lines on the display complete the 64 lines (0 to 63) stored in memory.

NO OPEN DATA FIELD: Displayed when the operator tries to enter a character from the ENTRY field without pressing appropriate key in the TRACE SPECIFICATION keyboard section to open a data field.

RUN ABORTED, NO DISPLAY CHANGE: Displayed for two conditions:

1. If a DISPLAY key is pressed when a TRACE or TRACE TRIGS run is in progress (WAITING FOR ENABLE, WAITING FOR TRIGGER, or the required number of TRIGR OCCUR has not been met).
2. If a DISPLAY key is pressed after a previous TRACE or TRACE TRIGS measurement run has been aborted by pressing any key other than an EXECUTE key while run is in progress.

SELF-TEST COMPLETED: Indicates satisfactory completion of self-test (unless an error message is also displayed).

SINGLE STEP: Displayed after TRACE is pressed in single-step mode.

SINGLE STEP, NO DISPLAY CHANGE: Displayed when an attempt is made to change the display by using DISPLAY keys after a trace has been executed in TRACE SINGLE STEP test mode.

SINGLE STEP, TRACE ONLY: Displayed when TEST MODE switch is set to SINGLE STEP and TRACE TRIGS, COUNT TRIGS, or TIME INTVL in the keyboard EXECUTE field is pressed. Only TRACE can be executed in SINGLE STEP test mode.

STOPPED: Displayed when STOP key is pressed, indicates a halt of any EXECUTE run in progress.

TRIGR OCCUR =: Displayed for three conditions:

1. During TRACE measurements, from first trigger recognition until number of triggers recognized equals the number selected by trigger occurrence specification.
2. During TRACE TRIGS measurements, from first trigger recognition until display memory is full (64 triggers stored).
3. During TRACE TRIGS End on Disable measurements, TRIGR OCCUR = 0 is displayed until first trigger is found, and TRIGR OCCUR = >64 is displayed if 64 triggers have been found without finding the Disable word, or if the 1611A does not contain the Trace Triggers End on Disable hardware modification (1611A with Serial Number Prefix below 1723A).

NOTE

If Trace Triggers End on Disable mode is desired, the old A8 board (01611-66508) may be changed to A8 board (01611-66535).

WAITING FOR ENABLE: Only displayed when a Trigger Enable has been specified and one of three conditions exists:

1. ENABLE specification has not been met.
2. System under test is running poorly or not at all.
3. This message will alternate with WAITING FOR TRIGGER if a trigger is not found between trigger ENABLE and DISABLE specifications, but the DISABLE is being found so the search for ENABLE is resumed.

WAITING FOR TRIGGER: Displayed for two conditions:

1. During TRACE and TRACE TRIGS measurements after Trigger ENABLE specification has been met, if an ENABLE has been specified, but before the first trigger has been detected.
2. Enable has not been specified and system under test is running poorly or not at all.

SECTION IV

PERFORMANCE TESTS

4-1. INTRODUCTION.

4-2. This section provides two types of performance tests: an Operation Verification and a Specification Verification. The Operation Verification provides approximately 90% assurance that the 1611A Option A85 is functioning properly. The Specification Verification checks electrical parameters of the instrument using the specifications in table 1-1 as the performance standards. Unless the detailed Specification Verification is required, it is recommended that only the Operation Verification be performed. All tests can be performed without access to the interior of the instrument.

4-3. PERFORMANCE TEST RECORD.

4-4. Performance Test results may be recorded in the Performance Test Record at the end of the procedures. The test record lists all tested specifications and their acceptable limits. The results recorded at incoming inspection can be used for comparison in periodic maintenance and troubleshooting and after repairs or adjustments.

PERFORMANCE TESTS/OPERATION VERIFICATION

4-5. OPERATION VERIFICATION.

4-6. To assure that the 1611A Option A85 is operating properly without testing all of the specifications listed in table 1-1, perform the following Operation Verification.

4-7. SELF TEST.

DESCRIPTION:

This test verifies proper operation of the ROM, display RAM, and CRT display circuits. If the ROM/RAM test fails, the test will be repeated and result in a display that indicates the failed circuit. When self test passes, a display is generated to indicate the circuit is operating properly.

PROCEDURE:

- a. Set power LINE switch to on. Verify that CRT displays message SELF TEST COMPLETED.

PERFORMANCE TESTS/OPERATION VERIFICATION

4-8. MEMORY DISPLAY TEST.

DESCRIPTION:

This test verifies that the 1611A storage memory can be loaded and the information stored in memory can be viewed through a 16 line movable window. This test also verifies that information is being sensed correctly by the address and data lines.

PROCEDURE:

- a. Connect μ P probe A13 to front panel PROBE TEST socket.
- b. Set 1611A front panel controls as follows:

ADRS DATA BUS	HEXADECIMAL
EXT	PROBE
READ OP CODE	OFF
READ OP CODE	OFF
WRITE	OFF
I/O	OFF
TEST MODE	NORMAL

- c. Press TRACE. Verify that a list is displayed.
- d. Use ROLL keys to scroll through memory. Verify that LINE 0 through LINE 48 can be displayed. (Note: It may be necessary to use ABSOLUTE/MNEMONIC key to view all 64 lines of memory).

4-9. STATUS MESSAGES FOR TRACE MODE TEST.

DESCRIPTION:

This test verifies that before the first trigger condition is detected, two messages may be displayed: WAITING FOR ENABLE flashes until the Enable specification (if entered) is detected; WAITING FOR TRIGGER flashes until the Trigger specification is detected.

PROCEDURE:

- a. Press TRIGGER ENABLE/ADDRESS BUS = key.
- b. Enter CDEF.
- c. Press TRACE. Verify that WAITING FOR ENABLE message is flashing.
- d. Press TRIGGER ENABLE/ADDRESS BUS = key.
- e. Press DON'T CARE.
- f. Press TRIGGER/ADDRESS BUS = key.
- g. Enter 89AB.
- h. Press TRACE. Verify that WAITING FOR TRIGGER message is flashing.

PERFORMANCE TESTS/OPERATION VERIFICATION

4-10. TRIGGER $\leq$, TRIGGER OCCURRENCES, DELAY, AND TRACE TRIGGERS TEST.

DESCRIPTION:

This test verifies that the 1611A can recognize $\leq$ or $\geq$ trigger specifications, count trigger occurrences, and count delay before the memory is loaded and a display is generated. This test also verifies that in TRACE TRIGGERS, only information matching Trigger conditions are stored in memory and displayed.

PROCEDURE:

- a. Press AD RS BUS $\geq$ key.
- b. Enter 0000.
- c. Press AD RS BUS $\leq$ key.
- d. Enter 1110.
- e. Press TRIGGER OCCURRENCE.
- f. Enter 256.
- g. Press AFTER TRIG key.
- h. Enter 65472.
- i. Press TRACE. WAITING FOR TRIGGER message may flash briefly. Verify that DELAYING message flashes until trace is completed (LINE 0 displayed).
- j. Press TRACE TRIGGERS. Verify that every row of data display shows:

ADRS
0000

OPCODE/DATA
00 WRITE

EXTERNAL
0000 0000

4-11. TRACE TRIGGERS END ON DISABLE TEST.

DESCRIPTION:

This test verifies that the 1611A can execute a TRACE TRIGGERS function and end only when a Triggers Disable condition is detected.

PROCEDURE:

- a. Press and hold DON'T CARE key approximately 3 seconds until all TRACE SPECIFICATIONS are cleared.
 - b. Press TRIGGER/ADDRESS BUS = key.
 - c. Enter 0000.
 - d. Press BEFORE TRIG key.
 - e. Enter 63.
 - f. Press TRIGGER DISABLE/ADDRESS BUS = key.
 - g. Enter ABAB.
 - h. Press TRACE TRIGS.
-

PERFORMANCE TESTS/OPERATION VERIFICATION

- i. Verify that TRIG OCCUR => 64 message flashes.
- j. Press TRIGGER DISABLE/ADDRESS BUS = key.
- k. Enter FFFF.
- l. Press TRACE TRIGS.
- m. Verify that data display shows only one line as follows:

ADRS
0000

OPCODE/DATA
00 WRITE

EXTERNAL
0000 0000

Note: If your 1611A serial number prefix is below 1723A, then TRIG OCCUR = 64 is displayed immediately to indicate that the 1611A does not have Trace Triggers End On Disable capability.

4-12. PRETRIGGER TEST.
DESCRIPTION:

This test verifies that a specified number of memory transactions can be viewed that occur before a selected trigger word.

PROCEDURE:

- a. Clear all TRACE SPECIFICATIONS entered on display by pressing the appropriate field selection keys and then pressing DON'T CARE.
 - b. Press TRIGGER/ADDRESS BUS = key.
 - c. Enter AAAA.
 - d. Press BEFORE TRIGGER key.
 - e. Enter 4.
 - f. Press TRACE. Verify that trigger word AAAA in inverse video is located on the 5th line of the display.
-

4-13. COUNT TRIGGER TEST AND STOP TEST.
DESCRIPTION:

This test verifies that the 1611A can count Triggers detected between (and including) the Trigger Enable and Trigger Disable words. The test also verifies that when STOP key is pressed, operation is halted and STOPPED message is displayed.

PROCEDURE:

- a. Clear all TRACE SPECIFICATIONS entered on display by pressing the appropriate field selection keys and then pressing DON'T CARE.
 - b. Press TRIGGER ENABLE/ADDRESS BUS = key.
 - c. Enter 0000.
-

PERFORMANCE TESTS/OPERATION VERIFICATION

- d. Press TRIGGER DISABLE/ADDRESS BUS = key.
 - e. Enter FFFF.
 - f. Press TRIGGER/ADDRESS BUS = key.
 - g. Enter 4444.
 - h. Press and hold COUNT TRIGS key for approximately 3 seconds. Verify that CONTINUOUS message is displayed and that COUNT = , MAX = , and MIN = , all display 1 TRIGGERS.
 - i. Press STOP and verify that STOPPED message is displayed.
-

4-14. TIME INTERVAL TEST.**DESCRIPTION:**

This test verifies that the 1611A can count the time interval occurring between states determined by Trigger Enable and Trigger Disable specifications. If no Trigger disable condition is found, message COUNTER OVERFLOW occurs after approximately 16 seconds. The test also verifies that if the DON'T CARE key is depressed approximately 3 seconds, all TRACE SPECIFICATION entries are cleared.

PROCEDURE:

- a. Clear all TRACE SPECIFICATIONS entries on the display by pressing and holding DON'T CARE for approximately 3 seconds. Verify that all TRACE SPECIFICATIONS messages are cleared (NO OPEN DATA FIELD message will flash until the clear is complete.)
 - b. Press TRIGGER ENABLE/DATA BUS = key.
 - c. Enter 00.
 - d. Press TRIGGER DISABLE/DATA BUS = key.
 - e. Enter FF.
 - f. Press and hold TIME INTERVAL key for approximately 3 seconds. Verify that TIME = , MAX = , and MIN = all display 62 MICROSECONDS (± 1).
 - g. Press TRIGGER DISABLE/DATA BUS = key.
 - h. Enter AB.
 - i. Press TIME INTVL. Verify that COUNTING, followed by COUNTER OVERFLOW messages are displayed (flashing).
-

4-15. TRIGGER QUALIFIER SWITCH TEST.**DESCRIPTION:**

This test verifies that the TRIGGER QUALIFIER switches can restrict triggering of the 1611A to certain types of machine cycles.

PROCEDURE:

- a. Clear TRACE SPECIFICATIONS on the display by pressing and holding the DON'T CARE key for approximately 3 seconds.
-

PERFORMANCE TESTS/OPERATION VERIFICATION

- b. Press TRACE. Verify that list is displayed.
 - c. Set OCTAL/HEXIDECIMAL switch to each position and verify that displayed data converts to selected base.
 - d. Set TRIGGER QUALIFIER I/O switch to I/O position.
 - e. Press TRACE. Verify that inverse video trigger displays INPUT or OUTPUT in the OPCODE/DATA column. It may be necessary to use ABSOLUTE/MNEMONIC key to obtain this display.
 - f. Return I/O switch to OFF.
 - g. Set WRITE switch to WRITE position.
 - h. Press TRACE. Verify that inverse video trigger displays WRITE in the OPCODE/DATA column.
 - i. Return WRITE switch to OFF.
 - j. Set $\overline{\text{OP CODE}}$ switch to $\overline{\text{OP CODE}}$ position.
 - k. Press TRACE. Verify that WAITING FOR TRIGGER message is flashing.
 - l. Return $\overline{\text{OP CODE}}$ switch to OFF.
 - m. Set OP CODE switch to OP CODE position.
 - n. Press TRACE. Verify that inverse video trigger displays OP CODE in the OP CODE/DATA column. It may be necessary to use ABSOLUTE/MNEMONIC key to obtain this display.
 - o. Return OP CODE switch to OFF.
 - p. With EXT switch set to PROBE, press TRACE. Verify that EXTERNAL data field displays 0000 0000.
 - q. Set EXT switch to SERIAL I/O PROBE.
 - r. Press TRACE. Verify that EXTERNAL data field displays 1100 0000.
 - s. Return EXT switch to PROBE.
 - t. Verify that NO CLOCK, HALT, HOLD, WAIT, and INTERRUPT ACK LEDs light in sequence.
 - u. Set TEST MODE switch to TRACE SINGLE STEP. Verify that WAIT/HOLD BY 1611A LED is on.
 - v. Return TEST MODE switch to NORMAL.
 - w. Check TRIGGER QUALIFIER LED by cycling OP CODE, $\overline{\text{OP CODE}}$, WRITE, and I/O switches on and off. All TRIGGER QUALIFIER switches must be OFF for TRIGGER QUALIFIER LED to be extinguished.
-

4-16. SPECIFICATION VERIFICATION.

- 4-17. The following procedures test the electrical parameters of the 1611A Option A85 using the specifications of table 1-1 as the performance standard. Threshold specifications are tested by applying the minimum voltage that the 1611A will recognize as a logic "1" and the maximum voltage that the 1611A will recognize as a logic "0". Setup time is the minimum time that data must be present at a 1611A probe input before it can be clocked into the 1611A. Hold time is the minimum time that data must be present at a 1611A probe input after it is clocked into the 1611A. When minimum threshold, setup time, and hold time specifications are met, then known data that is clocked into the 1611A can be shown by the CRT (or front-panel LEDs) to verify proper 1611A circuit operation.
-

PERFORMANCE TESTS/OPERATION VERIFICATION

4-18. TEST SIGNAL CONNECTIONS.

No standard adapter is available for connecting test signals to microprocessor probe A13. Therefore, it is recommended that short lengths of small-gauge solid wire be inserted in the probe socket and that test signals be connected to these wires with a BNC-to-alligator clip adapter. The solid wires allow socket clamping action to secure a good connection between the signal source and the probe.

NOTE

Some tests will not give correct results if the wrong 1611A keys are accidentally pressed. If the 1611A does not give a correct indication, cycle LINE power switch off and on, and then repeat the test to verify that the incorrect indication was not caused by pressing the wrong key.

4-19. INITIAL SETUP.

- a. Connect external probe A12 and μ P probe A13 to their respective connectors on 1611A rear panel. Do not connect μ P probe A13 to front-panel test socket.
- b. Set 1611A front-panel controls as follows:

ADRS DATA BUS	HEXADECIMAL
EXT	PROBE
READ OP CODE	OFF
READ $\overline{\text{OP CODE}}$	OFF
WRITE	OFF
I/O	OFF
TEST MODE	NORMAL

NOTE

Perform tests in the sequence given. Test equipment should not be disconnected unless specified in the procedure. Remove μ P probe cable with 40-pin male connector from μ P probe A13 during these tests. Do not specify an ENABLE. Specify TRIGGER = DON'T CARE.

PERFORMANCE TESTS/OPERATION VERIFICATION

4-20. CLOCK TEST.

SPECIFICATION:

Clock rate: 300 kHz to 5 MHz.

DESCRIPTION:

This test verifies that a proper signal applied to the μ P probe A13 CLOCK input will extinguish the front-panel NO CLOCK LED.

EQUIPMENT:

Pulse Generators (3)
 50-Ohm Feedthroughs (3)
 BNC-to-Alligator Clip Adapters (3)
 Oscilloscope

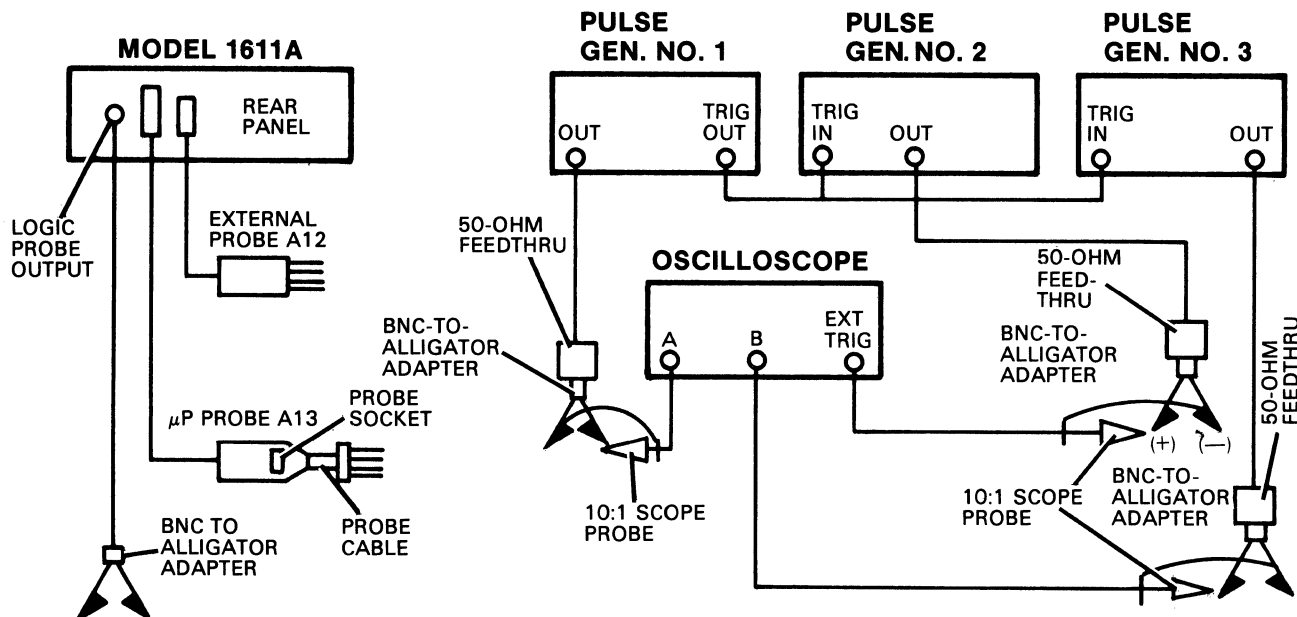


Figure 4-1. Specification Verification Test Setup

PERFORMANCE TESTS/OPERATION VERIFICATION

PROCEDURE:

- a. Connect test equipment as shown in figure 4-1.
- b. Verify that NO CLOCK LED is on.
- c. Connect pulse generator No. 1 between pin 37 (CLK OUT) and pin 20 (GND) of μ P probe A13 socket.
- d. Adjust pulse generator No. 1 for output as shown in figure 4-2a.
- e. Verify that NO CLOCK LED is off.
- f. Adjust pulse generator No. 1 for output as shown in figure 4-2b.
- g. Verify that NO CLOCK is off.
- h. Disconnect pulse generator No. 1 from pin 37 (CLK OUT) of μ P probe.

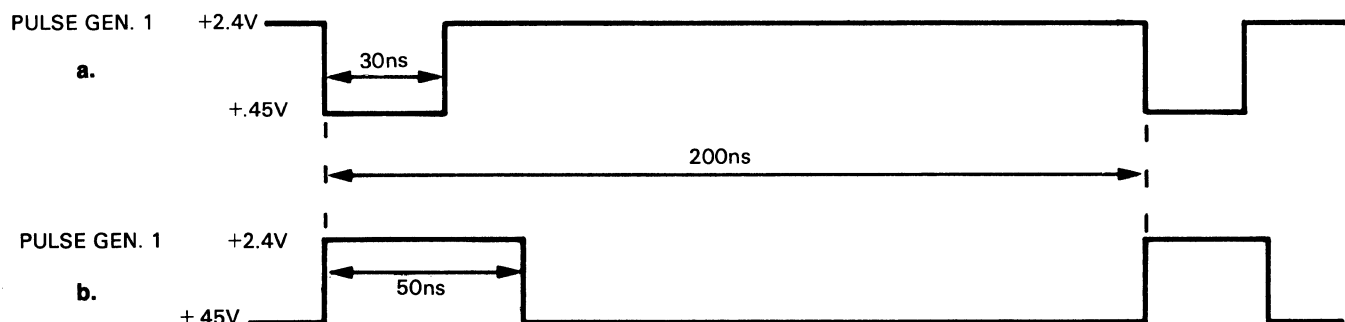


Figure 4-2. CLOCK Test Waveforms

4-21. DATA LINES TEST.**SPECIFICATION:**

- Setup Time: ≥ 80 ns
 Hold Time: ≤ 0 ns relative to positive transition of $\overline{RD}$
 Threshold: Logic 1 (HI) = +2.0 to 5.5V
 Logic 0 (LO) = -0.5 to +0.45V

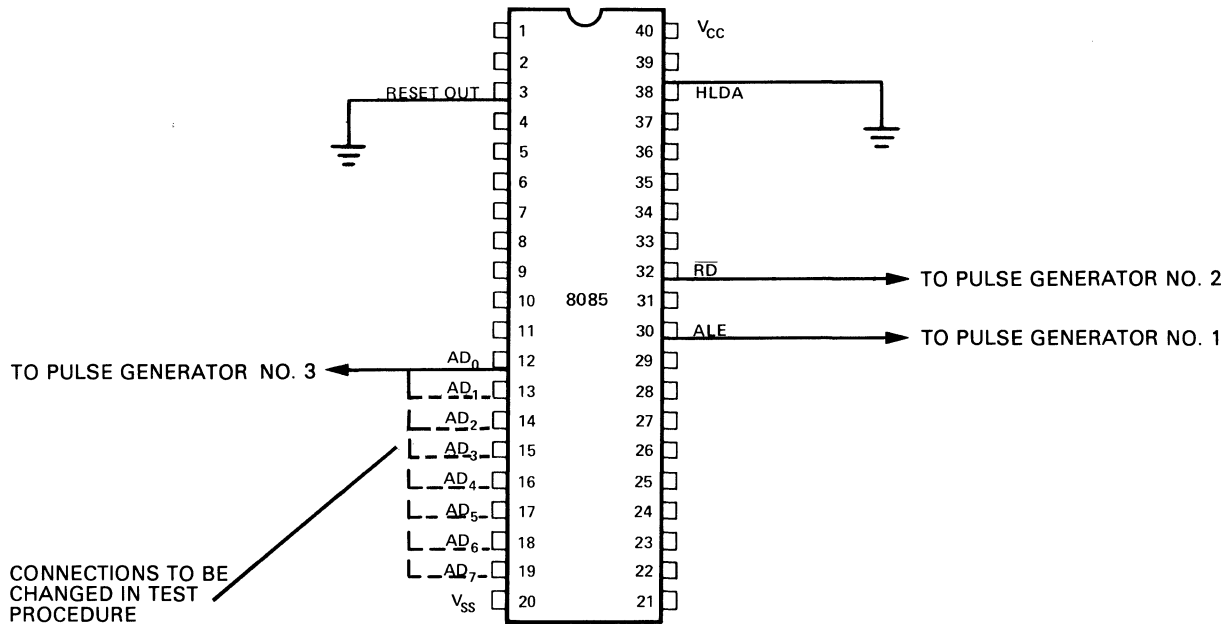
DESCRIPTION:

The HLDA and RESET OUT inputs of μ P probe A13 are connected to ground, enabling data to be clocked into the 1611A. A LO pulse (logic "0") that satisfies minimum threshold, setup time, and hold time is applied to each Data line input and clocked into the 1611A. The CRT display is then checked to verify proper operation of the Data line under test.

- a. Disconnect pulse generator No. 1 from pin 37 of μ P probe A13.
- b. Connect pin 3 (RESET OUT) and pin 38 (HLDA) of μ P probe to ground.
 (Note: This connection will remain the same for all following tests.)
- c. Connect pulse generator No. 1 to pin 30 (ALE) of μ P probe.
 (Note: This connection will remain the same for all following tests.)

PERFORMANCE TESTS/OPERATION VERIFICATION

- d. Connect pulse generator No. 2 to pin 32 ($\overline{RD}$) of μP probe. If dual channel oscilloscope with TRIGGER VIEW is used, monitor pulse generator No. 2 with EXT SYNC INPUT and Select TRIGGER VIEW.
- e. Connect pulse generator No. 3 to pin 12 (AD_0) of μP probe.



- f. Adjust pulse generators for test waveforms as shown in figure 4-3.
- g. Press TRACE key and verify that displayed list OPCODE/DATA field matches that shown in table 4-1.
- h. Disconnect pulse generator No. 3 from pin 12 of μP probe and repeat steps e through g for each of the remaining Data lines (D1 through D7) and verify that the display OPCODE/DATA field is as shown in table 4-1.

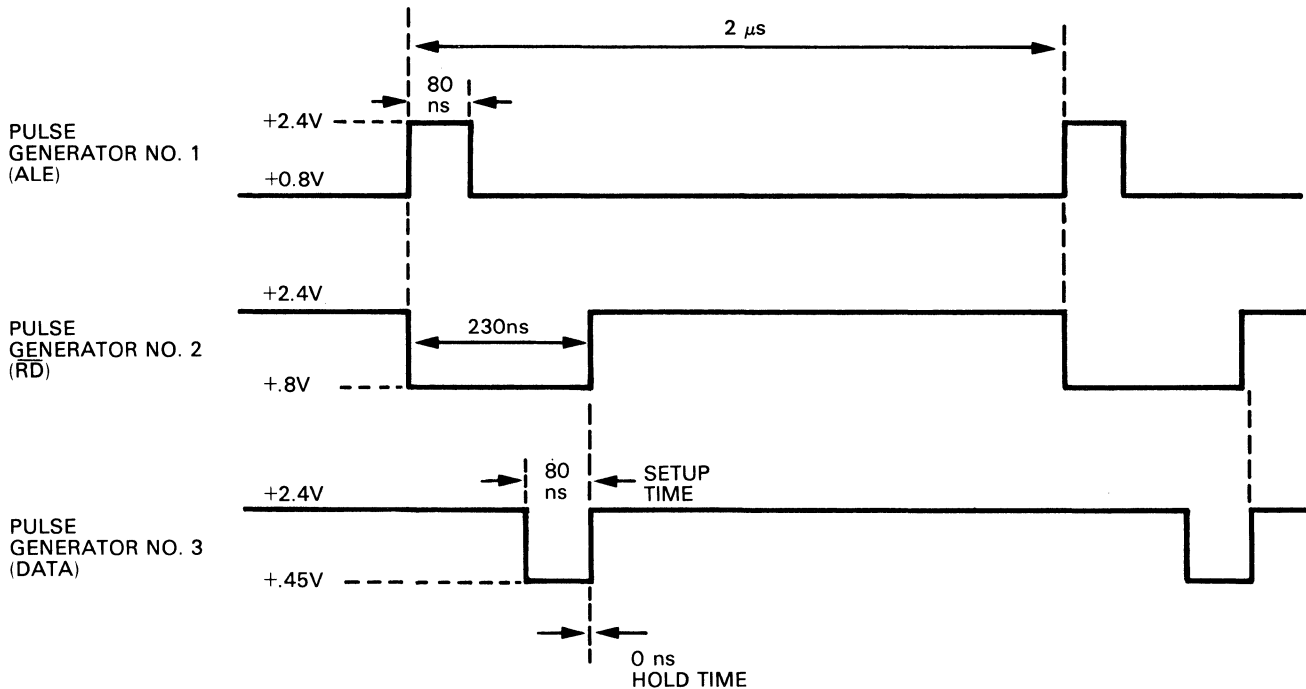


Figure 4-3. Data Lines Test

PERFORMANCE TESTS/OPERATION VERIFICATION

Table 4-1. Opcode/Data Field Display

Data Line	μ P Probe Socket Pin	OPCODE/DATA Field
D0	12	FE INPUT
D1	13	FD INPUT
D2	14	FB INPUT
D3	15	F7 INPUT
D4	16	EF INPUT
D5	17	DF INPUT
D6	18	BF INPUT
D7	19	7F INPUT

4-22. ADDRESS LINES AD₀-AD₇ TEST.

SPECIFICATION:

Setup Time: ≥ 50 ns

Hold Time: ≥ 50 ns relative to following edge of ALE

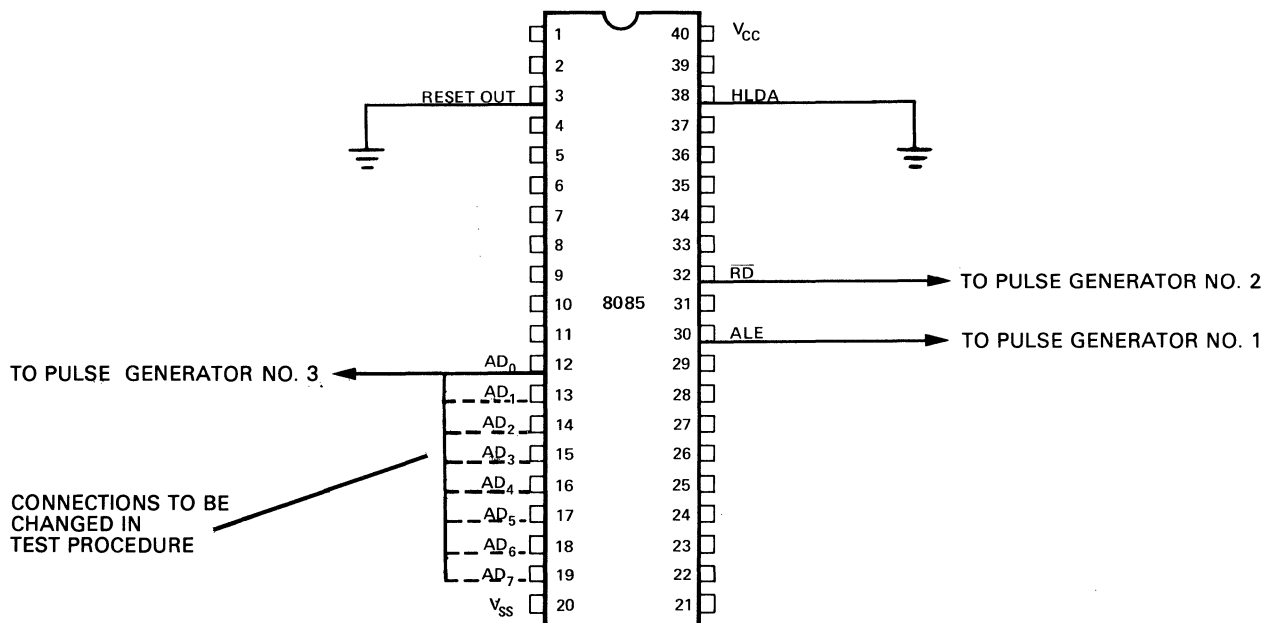
Threshold: Logic 1 (HI) = +2.4 to 5.5V

Logic 0 (LO) = -.5 to .8V

DESCRIPTION:

The lower 8 bits of the memory address bus are multiplexed between Address and Data information. When signal ALE (ADDRESS LATCH ENABLE) is high, information on the memory bus is the lower 8 bits of Address. The HLDA and RESET OUT inputs of μ P probe A13 are connected to ground to enable address information to be clocked into the 1611A. A LO pulse (logic "0") that satisfies minimum threshold, setup time, and hold time is applied to each Address line input and clocked into the 1611A. The CRT display is then checked to verify proper operation of the Address line under test.

- a. Connect pulse generator No. 3 to pin 12 (AD₀) of μ P probe A13.
- b. Adjust pulse generators for waveform shown in figure 4-4.



PERFORMANCE TESTS/OPERATION VERIFICATION

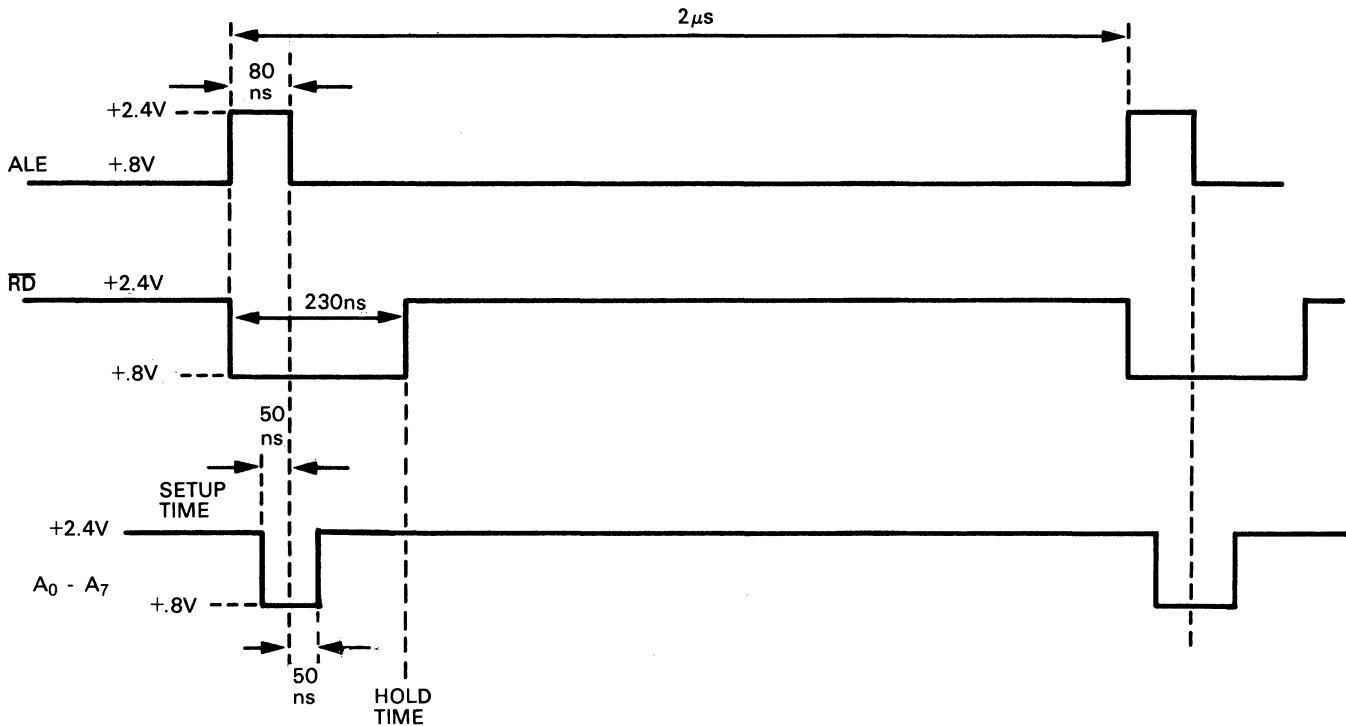


Figure 4-4. Address Lines A₀ - A₇ Test

- c. Press TRACE key and verify that displayed ADRS field matches that shown in table 4-2.
- d. Disconnect pulse generator No. 3 from pin 12 of μP probe and repeat steps a through c for each of the remaining ADDRESS lines (A₀ through A₇) and verify that the display ADRS field is as shown in table 4-2.

Table 4-2. Address Field Display

Address Line	μP Probe Socket Pin	ADRS Field
A0	12	FFFE
A1	13	FFFD
A2	14	FFFB
A3	15	FFF7
A4	16	FFEF
A5	17	FFDF
A6	18	FFBF
A7	19	FF7F

4-23. ADDRESS LINES A₈-A₁₅ TEST.

SPECIFICATION:

- Setup Time: ≥345 ns
- Hold Time: ≥60 ns relative to positive transition on $\overline{RD}$
- Threshold: Logic 1 (HI) = +2.4 to +5.5V
- Logic 0 (LO) = -0.5 to 0.8V

DESCRIPTION:

The HLDA and RESET OUT inputs of μP probe A13 are connected to ground to enable address information to be clocked into the 1611A. A LO pulse (logic “0”) that satisfies minimum threshold, setup time, and hold time is applied to each Address line input and clocked into the 1611A. The CRT display is then checked to verify proper operation of the Address line under test.

PERFORMANCE TESTS/OPERATION VERIFICATION

- a. Connect pulse generator No. 3 to pin 21 (A_8) of μP probe.
- b. Adjust pulse generators for waveforms shown in figure 4-5.

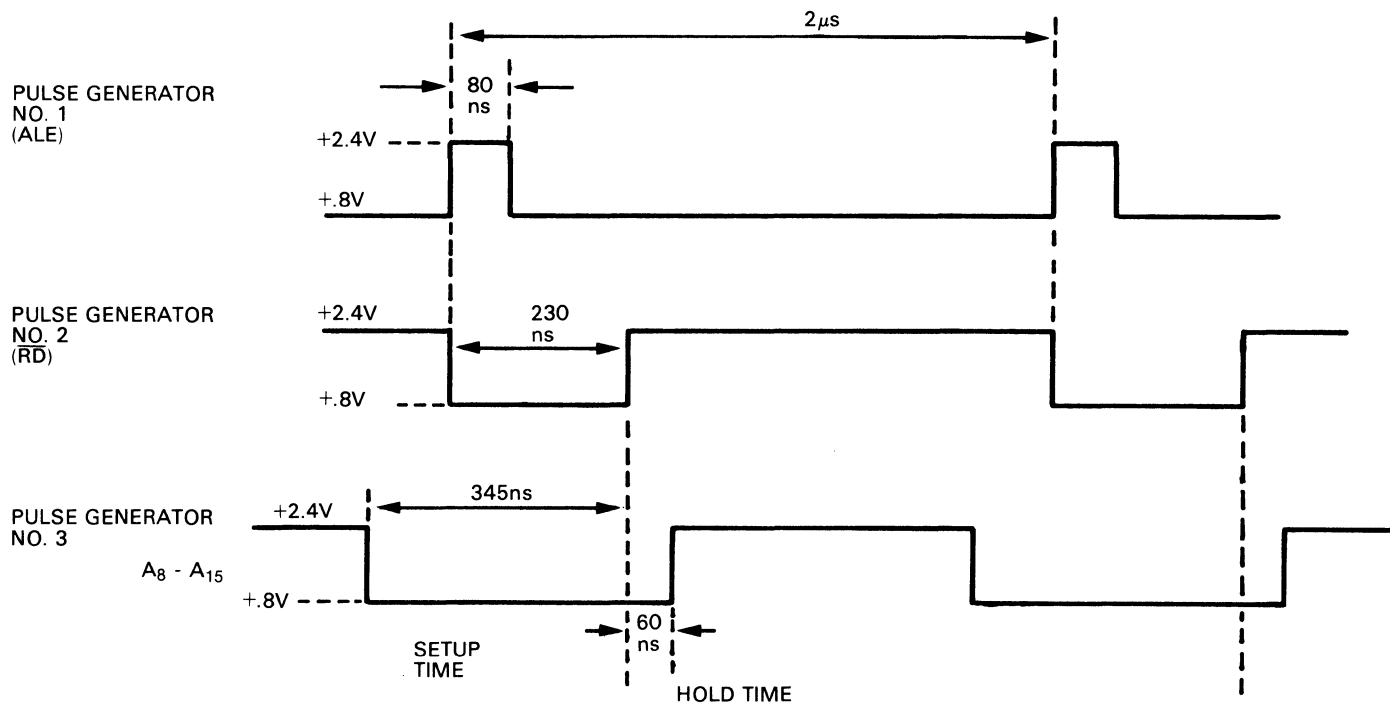
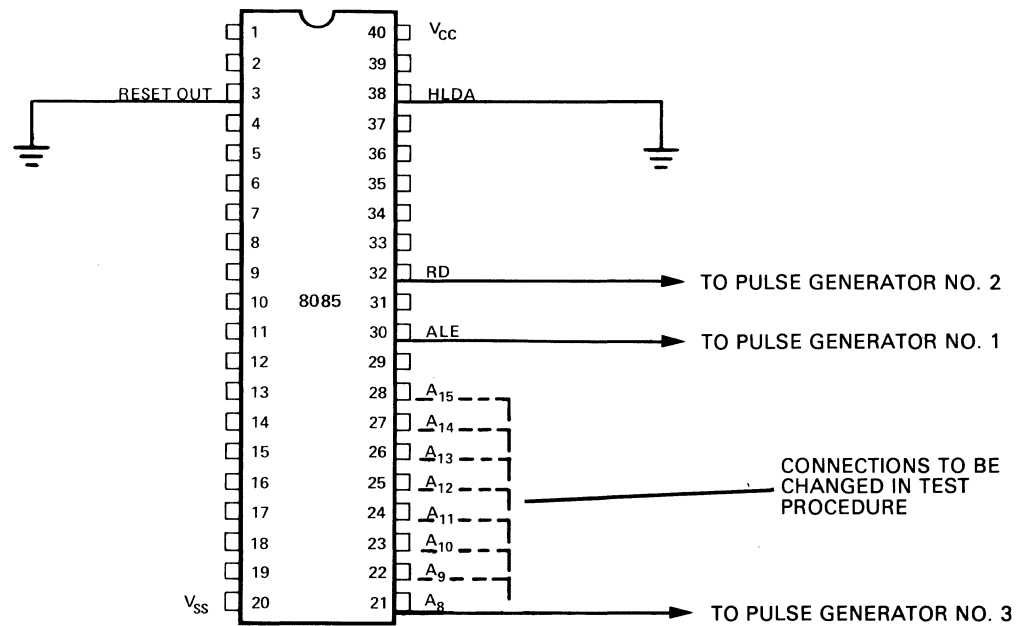


Figure 4-5. Address Lines A_8 - A_{15} Test

PERFORMANCE TESTS/OPERATION VERIFICATION

- c. Press TRACE key and verify that displayed ADRS field matches that shown in table 4-3.
- d. Disconnect pulse generator No. 3 from pin 21 of μ P probe and repeat steps a through c for each of the remaining ADDRESS lines (A_8 through A_{15}) and verify that the display ADRS field is as shown in table 4-3.

Table 4-3. Address Display Field (A_8 through A_{15})

Address Line	μ P Probe Socket Pin	ADRS Field
A8	21	FEFF
A9	22	FDFF
A10	23	FBFF
A11	24	F7FF
A12	25	EFFF
A13	26	DFFF
A14	27	BFFF
A15	28	7FFF

4-24. DATA BUS STATUS - S_0 , S_1 , AND $IO/\overline{M}$ TESTS.**SPECIFICATION:**Setup Time: ≥ 50 nsHold Time: ≥ 200 ns relative to negative edge of ALE

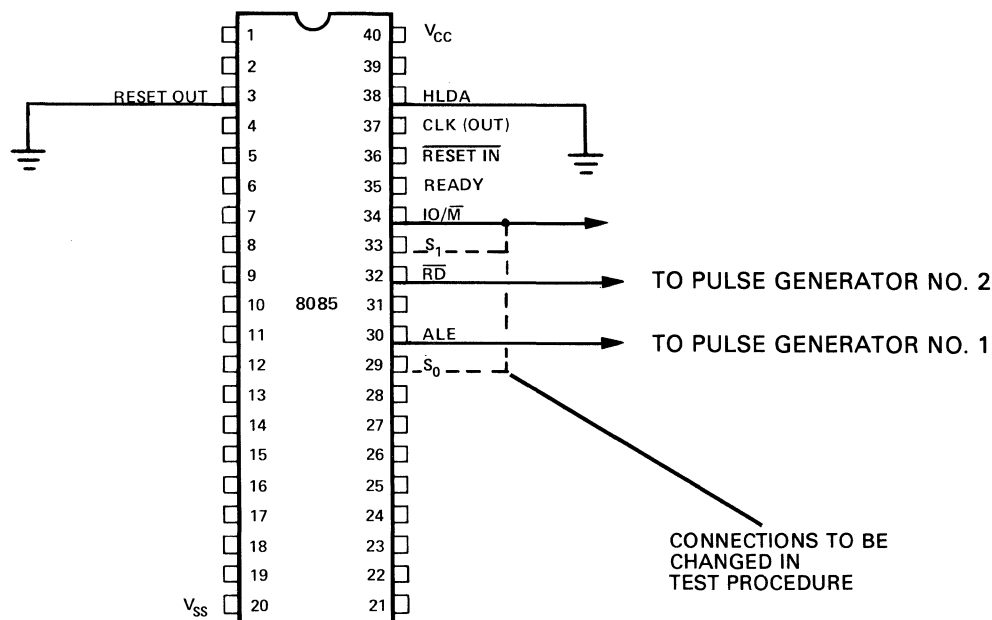
Threshold: Logic 1 (HI) = +2.4 to +5.5V

Logic 0 (LO) = -0.5 to +0.8V

DESCRIPTION:

HLDA and RESET OUT inputs of μ P probe A13 are connected to ground to enable Data bus status information to be clocked into the 1611A. A LO pulse (logic "0") that satisfies minimum threshold, setup time, and hold time is applied to S_0 , S_1 , and $IO/\overline{M}$ and clocked into the 1611A. The CRT display is then checked to verify proper operation of the data bus status line under test.

- a. Connect pulse generator No. 3 to pin 34 ($IO/\overline{M}$) of μ P probe A13.
- b. Adjust pulse generators for test waveforms shown in figure 4-6.



PERFORMANCE TESTS/OPERATION VERIFICATION

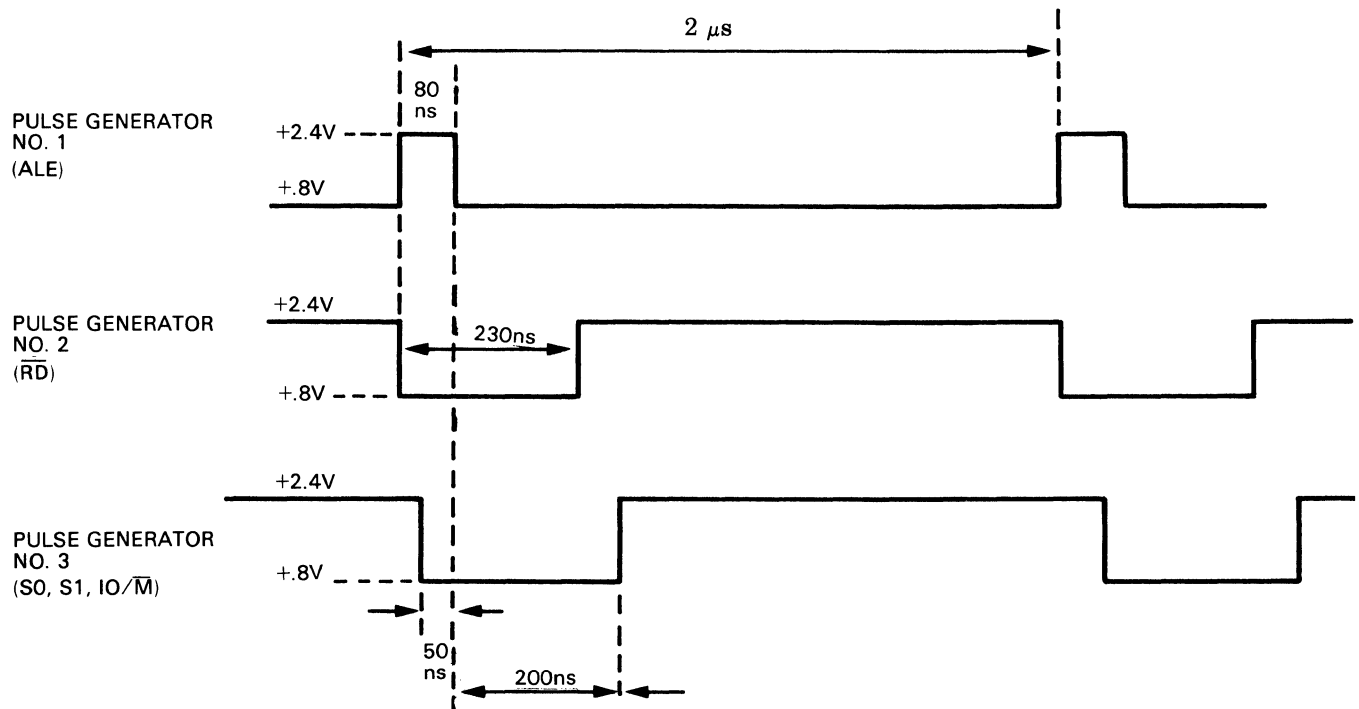


Figure 4-6. Data Bus Status Test

- c. Press TRACE key and verify that displayed OPCODE/DATA field indicates FF OPCODE. Use ABSOLUTE/MNEMONIC key if necessary to obtain this display.
 - d. Connect pulse generator No. 3 to pin 33 (S_1) of μP probe.
 - e. Ground pin 34 ($IO/\overline{M}$) of μP probe.
 - f. Press TRACE key and verify that displayed OPCODE/DATA field indicates FF WRITE.
 - g. Connect pulse generator No. 3 to pin 29 (S_0) of μP probe.
 - h. Press TRACE key and verify that displayed OPCODE/DATA field indicates FF READ.
 - i. Remove ground from pin 34 ($IO/\overline{M}$) of μP probe.
-

4-25. EXTERNAL PROBE TEST.

SPECIFICATION:

Setup Time: ≥ 250 ns

Hold Time: ≥ 0 ns relative to positive transition of $\overline{RD}$, $\overline{WR}$, $\overline{INTA}$

Threshold: Logic 1 (HI) = +2.4 to +5.5V

Logic 0 (LO) = -0.8 to +0.8V

DESCRIPTION:

The HLDA and RESET OUT pins of μP probe A13 are connected to ground to enable External information to be clocked into the 1611A. A HI pulse (logic "1") that satisfies minimum threshold, setup time, and hold time is applied to each External probe line and clocked into the 1611A. The CRT display is then checked to verify proper operation of the External line under test.

PERFORMANCE TESTS/OPERATION VERIFICATION

- Connect pulse generator No. 3 to input 0 of External data probe.
- Adjust pulse generators for test waveforms shown in figure 4-7.

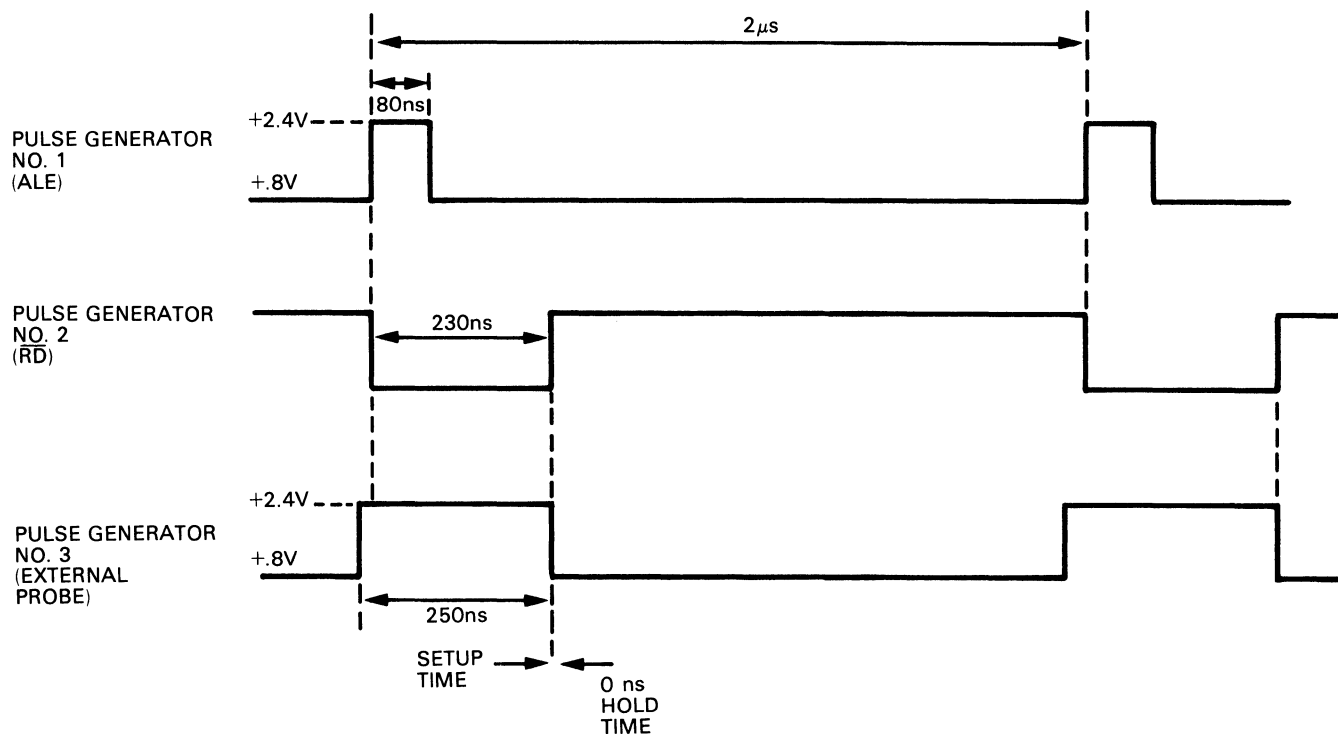
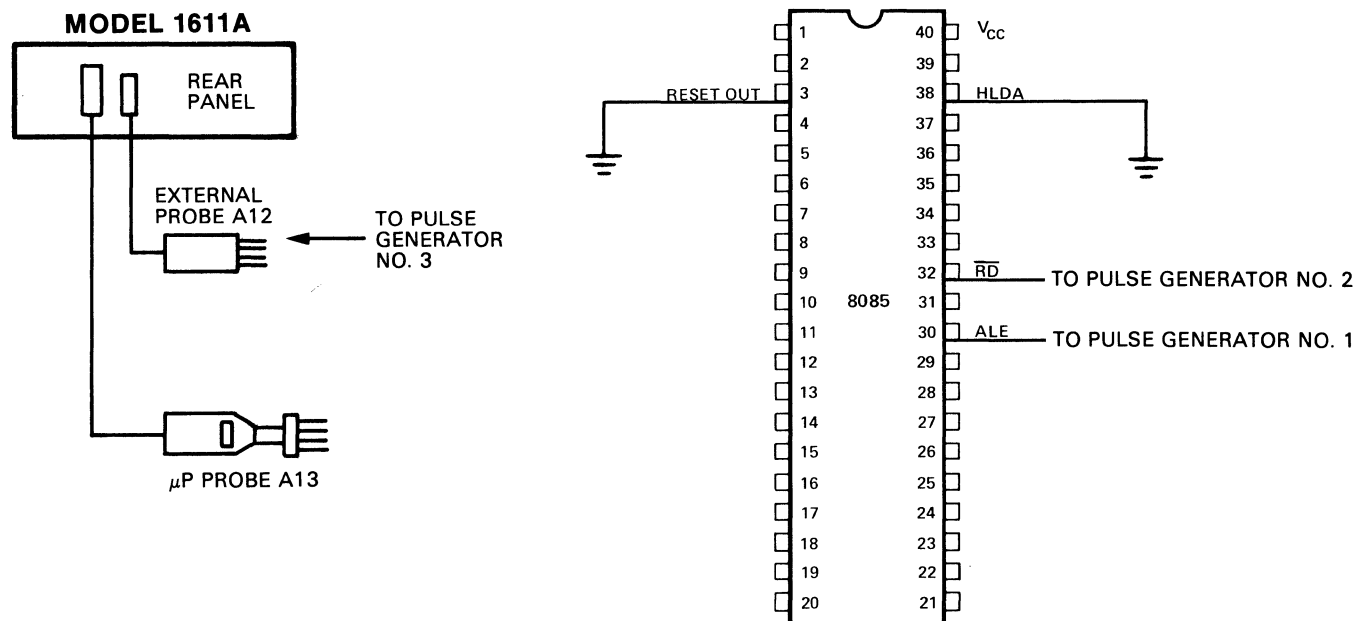


Figure 4-7. External Probe Test

PERFORMANCE TESTS/OPERATION VERIFICATION

- c. Press TRACE key and verify that displayed EXTERNAL field matches that shown in table 4-4.
- d. Repeat steps a through c for each of the remaining External Probe lines (1 through 7) and verify that the displayed External field matches table 4-4.
- e. Disconnect all test equipment and connections from μ P probe A13 and External probe A12.

Table 4-4. External Data Field

EXTERNAL INPUT	EXTERNAL FIELD
0	0000 0001
1	0000 0010
2	0000 0100
3	0000 1000
4	0001 0000
5	0010 0000
6	0100 0000
7	1000 0000

4-26. REAR-PANEL OUTPUT TESTS.**SPECIFICATION:**

Threshold: Logic 1 (HI) >2.0V into 50 ohms

Logic 0 (LO) < +0.4V

DESCRIPTION:

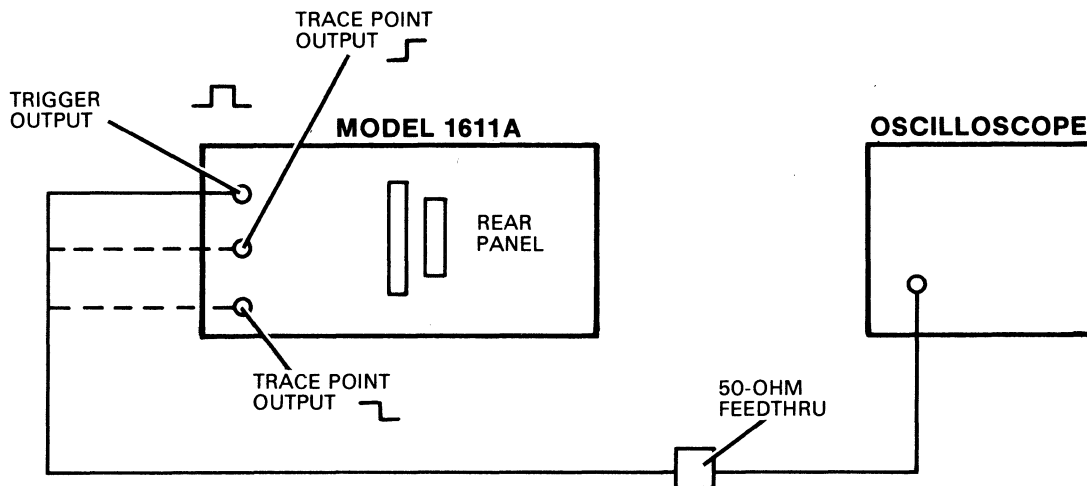
These tests verify that the 1611A produces proper rear-panel outputs.

Trigger Output.

- a. Plug 40-pin male connector from μ P probe cable into 1611A front-panel PROBE TEST socket. (Reconnect probe cable to μ P probe A13 if disconnected.)
- b. Connect TRIGGER OUTPUT to oscilloscope as shown in figure 4-8.
- c. Verify that TRIGGER OUTPUT pulses have maximum amplitude greater than +2.0 V and minimum amplitude less than +0.4 V.

Trace Point $\square$ / $\sqcap$ Outputs.

- a. Connect TRACE POINT $\square$ output from 1611A rear panel to oscilloscope.
- b. While holding TRACE key down, verify presence of pulses with maximum amplitude greater than +2.0 V and minimum amplitude less than +0.4 V.
- c. Repeat steps a and b for TRACE POINT $\sqcap$.

*Figure 4-8. Rear-panel Outputs Test Setup*

PERFORMANCE TEST RECORD

HEWLETT-PACKARD

MODEL 1611A OPTION A85

LOGIC ANALYZER

Tested by _____

Serial No. _____

Date _____

Paragraph Number	Test	Specifications	Passed	Failed
	Clock	$0 \leq +.45 \text{ V}$, $1 \geq +2.4 \text{ V}$ 30 ns min LO, 50 ns min Hi		
	Data Lines D ₀ - D ₇	$0 \leq +0.8 \text{ V}$, $1 \geq +2.0 \text{ V}$ Setup $\geq 80 \text{ ns}$ Hold $\geq 0 \text{ ns}$		
	Address Lines A ₀ A ₇	$0 \leq +.45 \text{ V}$, $1 \geq +2.4 \text{ V}$ Setup $\geq 50 \text{ ns}$ Hold $\geq 50 \text{ ns}$		
	Address Lines A ₈ - A ₁₅	$0 \leq +.45 \text{ V}$, $1 \geq +2.4 \text{ V}$ Setup $\geq 345 \text{ ns}$ Hold $\geq 60 \text{ ns}$		
	Data Bus Status S ₀ , S ₁ , IO/M	$0 \leq +.45 \text{ V}$, $1 \geq +2.4 \text{ V}$ Setup $\geq 50 \text{ ns}$ Hold $\geq 200 \text{ ns}$		
	External Inputs	$0 \leq +.8 \text{ V}$, $1 \geq +2.4 \text{ V}$ Setup $\geq 250 \text{ ns}$ Hold $\geq 0 \text{ ns}$		
	Trigger Output	$0 \leq +0.4 \text{ V}$, $1 \geq +2.0 \text{ V}$		
	Trace Point Outputs	$0 \leq +0.4 \text{ V}$, $1 \geq +2.0 \text{ V}$ into 50 ohms		

SECTION V

ADJUSTMENTS

There are no adjustments on the Model 10263A/1611A Option A85.

SECTION VI

REPLACEABLE PARTS

6-1. INTRODUCTION.

6-2. This section contains information for ordering parts. Table 6-1 lists abbreviations used in the parts list and throughout the manual. Table 6-2 lists all replaceable parts in reference designator order. Table 6-3 contains the names and addresses that correspond to the manufacturers' five-digit code numbers.

6-3. ABBREVIATIONS.

6-4. Table 6-1 lists abbreviations used in the parts list, the schematics and throughout the manual. In some cases, two forms of the abbreviation are used: one all in capital letters, and one partial or no capitals. This occurs because the abbreviations in the parts list are always all capitals. However, in the schematics and other parts of the manual, other abbreviation forms are used with both lowercase and uppercase letters.

6-5. REPLACEABLE PARTS LIST.

6-6. Table 6-2 is the list of replaceable parts and is organized as follows:

- a. Chassis-mounted parts in alphanumerical order by reference designation.
- b. Electrical assemblies and their components in alphanumerical order by reference designation.
- c. Option parts.

The information given for each part consists of the following:

- a. The Hewlett-Packard part number.
- b. The total quantity (Qty) in the instrument.
- c. The description of the part.
- d. A five-digit code that indicates the manufacturer.

- e. The manufacturer's part number.

The total quantity for each part is given only once—at the first appearance of the part number in the list.

6-7. ORDERING INFORMATION.

6-8. To order a part listed in the replaceable parts table, quote the Hewlett-Packard part number, indicate the quantity required, and address the order to the nearest Hewlett-Packard office.

6-9. To order a part that is not listed in the replaceable parts table, include the instrument model number, instrument serial number, the description and function of the part, and the number of parts required. Address the order to the nearest Hewlett-Packard office.

6-10. DIRECT MAIL ORDER SYSTEM.

6-11. Within the USA, Hewlett-Packard can supply parts through a direct mail order system. Advantages of using the system are as follows:

- a. Direct ordering and shipment from the HP Parts Center in Mountain View, California.
- b. No maximum or minimum on any mail order (there is a minimum order amount, for parts ordered through a local HP office when the orders require billing and invoicing).
- c. Prepaid transportation (there is a small handling charge for each order).
- d. No invoices—to provide these advantages, a check or money order must accompany each order.

6-12. Mail-order forms and specific ordering information are available through your local HP office. Addresses and phone numbers are located at the back of this manual.

Table 6-1. Reference Designators and Abbreviations

REFERENCE DESIGNATORS							
A	= assembly	F	= fuse	MP	= mechanical part	U	= integrated circuit
B	= motor	FL	= filter	P	= plug	V	= vacuum, tube, neon bulb, photocell, etc
BT	= battery	IC	= integrated circuit	Q	= transistor	VR	= voltage regulator
C	= capacitor	J	= jack	R	= resistor	W	= cable
CP	= coupler	K	= relay	RT	= thermistor	X	= socket
CR	= diode	L	= inductor	S	= switch	Y	= crystal
DL	= delay line	LS	= loud speaker	T	= transformer	Z	= tuned cavity network
DS	= device signaling (lamp)	M	= meter	TB	= terminal board		
E	= misc electronic part	MK	= microphone	TP	= test point		
ABBREVIATIONS							
A	= amperes	H	= henries	N/O	= normally open	RMO	= rack mount only
AFC	= automatic frequency control	HDW	= hardware	NOM	= nominal	RMS	= root-mean square
AMPL	= amplifier	HEX	= hexagonal	NPO	= negative positive zero (zero temperature coefficient)	RWV	= reverse working voltage
BFO	= beat frequency oscillator	HG	= mercury	NPN	= negative-positive-negative	S-B	= slow-blow
BE CU	= beryllium copper	HR	= hour(s)	NRFR	= not recommended for field replacement	SCR	= screw
BH	= binder head	HZ	= hertz	NSR	= not separately replaceable	SE	= selenium
BP	= bandpass	IF	= intermediate freq	OBD	= order by description	SECT	= section(s)
BRS	= brass	IMPG	= impregnated	OH	= oval head	SEMICON	= semiconductor
BWO	= backward wave oscillator	INCD	= incandescent	OX	= oxide	SI	= silicon
CCW	= counter-clockwise	INCL	= include(s)	P	= peak	SIL	= silver
CER	= ceramic	INS	= insulation(ed)	PC	= printed circuit	SL	= slide
CMO	= cabinet mount only	INT	= internal	PF	= picofarads= 10-12 farads	SPG	= spring
COEF	= coefficient	K	= kilo=1000	PH BRZ	= phosphor bronze	SPL	= special
COM	= common	LH	= left hand	PHL	= philips	SST	= stainless steel
COMP	= composition	LIN	= linear taper	PIV	= peak inverse voltage	SR	= split ring
COMPL	= complete	LK WASH	= lock washer	PNP	= positive-negative-positive	STL	= steel
CONN	= connector	LOG	= logarithmic taper	P/O	= part of	TA	= tantalum
CP	= cadmium plate	LPF	= low pass filter	POLY	= polystyrene	TD	= time delay
CRT	= cathode-ray tube	M	= milli=10-3	PORC	= porcelain	TGL	= toggle
CW	= clockwise	MEG	= meg=106	POS	= position(s)	THD	= thread
DEPC	= deposited carbon	MET FLM	= metal film	POT	= potentiometer	TI	= titanium
DR	= drive	MET OX	= metallic oxide	PP	= peak-to-peak	TOL	= tolerance
ELECT	= electrolytic	MFR	= manufacturer	PT	= point	TRIM	= trimmer
ENCAP	= encapsulated	MHZ	= mega hertz	PWV	= peak working voltage	TWT	= traveling wave tube
EXT	= external	MINAT	= miniature	RECT	= rectifier	U	= micro=10-6
F	= farads	MOM	= momentary	RF	= radio frequency	VAR	= variable
FH	= flat head	MOS	= metal oxide substrate	RH	= round head or right hand	VDCW	= dc working volts
FIL H	= fillister head	MTG	= mounting			W/	= with
FXD	= fixed	MY	= "mylar"			W	= watts
G	= giga (109)	N	= nano (10-9)			WIV	= working inverse voltage
GE	= germanium	N/C	= normally closed			WW	= wirewound
GL	= glass	NE	= neon			W/O	= without
GRD	= ground(ed)	NI PL	= nickel plate				

Table 6-2. Replaceable Parts

Reference Designation	HP Part Number	Qty	Description	Mfr Code	Mfr Part Number
A9	01611-66566	1	BOARD ASSEMBLY, OPTION A85 PERSONALITY	28480	01611-66566
A10	01611-66571	1	BOARD ASSEMBLY, OPTION A85 ROM	28480	01611-66571
A11	01611-68710	1	ASSEMBLY, PERSONALITY PANEL, OPTION A85	28480	01611-68710
A13	01611-62112	1	ASSEMBLY, A85 MPR PROBE	28480	01611-62112
MP1	1540-0325	1	CASE, CRYG HANDLE	28480	1540-0325
A9	01611-66554		BOARD ASSEMBLY, OPTION A85 PERSONALITY	28480	01611-66554
A9C1	0160-2198	1	CAPACITOR-FXD 20PF $\pm 5\%$	28480	0160-2198
A9C2	0160-2055	30	CAPACITOR-FXD .01UF $+80 -20\%$	04200	C023F101F103ZS22-CDH
THRU C6					
A9C7	0180-0291	1	CAPACITOR-FXD 1UF $\pm 10\%$ 35VDC TA	56289	150D105X9035A2
A9C8	0160-2055		CAPACITOR-FXD .01UF $+80 -20\%$	04200	C023F101F103ZS22-CDH
A9C10	0140-0198	2	CAPACITOR-FXD 200PF $\pm 5\%$ 300 WVDC	04522	DM15F201J0300WVICR
A9C11	0160-3449	1	CAPACITOR-FXD 2000PF $\pm 10\%$ 250 WVDC CER	28480	0160-3449
A9C12	0140-0205	1	CAPACITOR-FXD 62PF $\pm 5\%$ 300 WVDC	04522	DM15E620J0300WVICR
A9C13	0140-0198		CAPACITOR-FXD 200PF $\pm 5\%$ 300 WVDC		
A9C14 THRU C29	0160-2055		CAPACITOR-FXD .01UF $+80 -20\%$	04200	C023F101F103ZS22-CDH
A9C30 & C31	0160-2306	2	CAPACITOR-FXD 27PF $\pm 5\%$ 300 WVDC	28480	0160-2306
A9R19 & R20	0757-0415	2	RESISTOR 475 1% .125W F TC=0+ -100	01121	CC
A9R1	0757-0438	4	RESISTOR 5.11K 1% .125 W F TC=0+ -100	24546	C4-1/8-TO-5111-F
A9R2			NOT ASSIGNED		
A9R3	0757-0910	6	RESISTOR 270 2% .125 W F TC=0+ -100	02273	CEA-993
THRU R6					
A9R7	0757-0438		RESISTOR 5.11K 1% .125W F TC=0+ -100	24546	C4-1/8-TO-511-F
THRU R9					
A9R10	0757-0442	7	RESISTOR 10K 1% .125W	02273	CEA-93
THRU R13					
A9R14	0757-0910		RESISTOR 270 2% .125W F TC=0+ -100	02273	CEA-993
A9R15	0757-0910		RESISTOR 270 2% .125W F TC=0+ -100	02273	CEA-993
A9R16	0757-0442		RESISTOR 10K 1% .125W	02273	CEA-93
THRU R18					
A9TP1	0360-0535	7	TERMINAL, TEST POINT	28480	OBD
THRU TP7					
A9U1	1820-1464	1	IC-DIGITAL SN74393N TTL DUAL	01295	SN74B93N
A9U2	1820-1423	1	IC-DIGITAL SN74LS123N TTL LS DUAL	01295	SN74LS123N
A9U3	1820-1191	3	IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U4	1820-1191		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U5	1820-2169	2	IC-DIGITAL	01295	SN74L04N
A9U6			NOT ASSIGNED		
A9U7			NOT ASSIGNED		
A9U8	1816-1248	2	IC-DIGITAL SN74LS74N TTL LS DUAL	28480	1816-1248
A9U9	1818-0707	10	IC-DIGITAL SN74LS175N TTL LS QUAD	28480	1818-0707
A9U10	1818-0711	1	IC-DIGITAL SN74S15N TTL S	28480	1818-0711
A9U11	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U12	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U13	1820-2169		IC-DIGITAL	01295	SN74L04N
A9U14	1820-2195	1	IC-DIGITAL	28480	1820-2195
A9U15	1820-1191		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U16	1810-0283	5	NETWORK-RES 16-PIN-DIP .1-PIN SPCG 270	28480	1810-0283
A9U17	1820-1207	1	IC-DIGITAL SN74LS30N TTL LS	01295	SN74LS30N
A9U18	1820-0077	1	IC-DIGITAL SN7474N TTL DUAL	01295	SN7474N
A9U19	1820-1199		IC-DIGITAL	01295	SN74LS04
A9U20	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U21	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U22	1820-1782	1	IC-DIGITAL	01295	SN74S00N
A9U23	1820-0681	1	IC-DIGITAL SN74S00N TTL S QUAD	01295	SN74S00N
A9U24	1820-0661	1	IC-DIGITAL SN7432N TTL QUAD	01295	SN7432N
A9U25	1820-1112		IC-DIGITAL SN74LS74N TTL LS DUAL	01295	SN74LS74N
A9U26	1820-0579	1	IC-DIGITAL SN74123N TTL DUAL	01295	SN74123N
A9U27	1820-1144	1	IC-DIGITAL SN74LS02N TTL LS QUAD	01295	SN74LS02N
A9U28 THRU U30	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U31	1820-1449	1	IC-DIGITAL SN74S32N TTL S QUAD	01295	SN74S32N
A9U32	1820-1367	1	IC-DIGITAL SN74S08N TTL S QUAD	01295	SN74S08N
A9U33	1820-1322	1	IC-DIGITAL SN74S02N	01295	SN74S02N
A10	01611-66571		BOARD ASSEMBLY, OPTION A85 ROM	28480	01611-66571
A10C1	0180-0228	1	CAPACITOR-FXD 22UF $\pm 100\%$ 15VDC TA	56289	150D226X9015B2
A10C2	0160-3443	1	CAPACITOR-FXD .1UF $+80 -20\%$ 5VDC CER	02813	CY20C104Z2
A10C3 THRU C6	0160-2055		CAPACITOR-FXD .01UF $+80 -20\%$	04200	C023F101F103ZS22-CDH
A10C7			NOT ASSIGNED		

Table 6-2. Replaceable Parts (Cont'd)

Reference Designation	HP Part Number	Qty	Description	Mfr Code	Mfr Part Number
A10U1	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A10U2	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A10U3	1820-1244	1	IC-DIGITAL SN74LS153N TTL LS	01295	SN74LS153N
A10U4	1820-1208	1	IC-DIGITAL SN74LS32N TTL LS QUAD	01295	SN74LS32N
A10U5	1820-1917	1	IC-DIGITAL SN74LS240N TTL LS OCTL	01295	SN74LS240N
			NOT ASSIGNED		
			NOT ASSIGNED		
A10U8	1816-1248	1	IC-8K PROM	28480	1816-1248
A10U9	1816-1114	1	IC-8K PROM	28480	1816-1114
A10U10	1818-0711	1	IC-8K PROM	28480	1818-0711
A10U11	1820-1216	1	IC-DIGITAL SN74LS138N TTL LS	01295	SN74LS138N
A10U12	1810-0055	1	NETWORK-RES 10K+ -5%	04200	203C90
A10XU8 THRU XU10	1200-0622	3	SOCKET-IC-24-PIN DIP	28480	1200-0622
A11	01611-68710		ASSEMBLY, PERSONALITY PANEL, OPTION A85		
			NOTE		
			BOARD ASSEMBLY A11A1 IS NOT SOLD SEPARATELY. TO OBTAIN A REPLACEMENT, THE ENTIRE PERSONALITY PANEL (HP PART NO. 01611-68710) MUST BE ORDERED.		
A11H1	2200-0103	4	SCREW-MACH 4-40 .25-IN. LG PAN-HD-POZI	28480	2200-0103
A11MP1	01611-00213	1	PANEL, A85 PERSONALITY	28480	01611-00213
A11MP2	5040-0565	1	BEZEL, CONNECTOR	28480	5040-0565
A11W1	01611-61605	1	CABLE ASSEMBLY, PERSONALITY	28480	01611-61605
A11W2	01611-61620	1	CABLE ASSEMBLY, A85 PERSONALITY PANEL	28480	01611-61620
A11A1DS1 THRU DS6	1990-0486	6	LED-VISIBLE LUM-INT=1MCD 1F=20MA-MAX	28480	1990-0486
A11A1DS7	1990-0487	1	LED-VISIBLE LUM-INT=1MCD 1F=20MA MAX	28480	1990-0487
A11A1J1	1200-0623	2	SOCKET-40-PIN FEM ZIF	28480	1200-0623
A11A1R1	0684-1021	1	RESISTOR 1K 10% .25W FC TC= -400+600	01121	CB1021
A11A1R2	0757-0280	5	RESISTOR 1K 1% .125W F TC=0+ -100	24546	C4-1/8-TO-1001F
A11A1R3	0757-0409	1	RESISTOR 274 1% .125W F TC=0+ -100	02995	MF4C-1
A11A1S1	3101-2119	1	SWITCH-TGL SUBMIN DPDT NS 2A 250 VAC	28480	3101-2119
A11A1S2	3101-2118	2	SWITCH-TGL SUBMIN SPDT NS 2A 250 VAC	28480	3101-2118
A11A1S3	3101-2118		SWITCH-TGL SUBMIN SPDT NS 2A 250 VAC	28480	3101-2118
A11A1S4	3101-2214	4	SWITCH-TGL SUBMIN DPDT SN 2A 250 VAC	28480	3101-2214
THRU S7 A13	01611-62112	1	ASSEMBLY, A85 MICROPROCESSOR PROBE	28480	01611-62112
			NOTE		
			BOARD ASSEMBLY A13A1 IS NOT SOLD SEPARATELY. TO OBTAIN A REPLACEMENT ASSEMBLY, THE ENTIRE PROBE (HP PART NO. 01611-62112) MUST BE ORDERED.		
A13H1	0624-0306	3	SCREW-TPG 2-28 5-IN-LG PAN-HD-POZI SIL	28480	0624-0306
A13H2	0624-0390	2	SCREW-TPG 2-28 .625-IN-LOG PAN-HD-POZI SIL	28480	0624-0390
A13H3	2200-0111	2	SCREW-MACH 4-40 .5-IN-LG PAN-HD-POZI	28480	0200-0111
A13MP1	5040-8126	1	PROBE POD-BOTTOM	28480	5040-8126
A13MP2	01611-24102	1	PROBE POD-TOP	28480	01611-24102
A13MP3	7120-7178	1	LABEL 8085 PROBE	28480	7120-7178
A13MP4	1540-0440	1	CASE, VINYL	28480	1540-0440
A13W1A	01611-61610	1	CABLE ASSEMBLY, DIP LONG	28480	01611-61610
A13W1B	01611-61612	1	CABLE ASSEMBLY, DIP SHORT (NOT SUPPLIED WITH A13; ORDER SEPARATELY)	28480	01611-61612
A13W1C	01611-61609	1	CABLE ASSEMBLY, TEST CLIP (NOT SUPPLIED WITH A13; ORDER SEPARATELY)	28480	01611-61609
A13A1C1 THRU C3	0160-2055		CAPACITOR-FXD .01UF +80-20%	04200	C023F101F103ZS22-CDH
A13A1C4	0160-3508	1	CAPACITOR-FXD 1UF +80-20% 50WVDC CER	28480	0160-3508
A13A1C5	0180-0228	1	CAPACITOR-FXD 22UF+ -10% 15VDC TA	56289	150D226X9015B2
A13A1J1	1200-0623		SOCKET, 40-PIN FEM ZIF	28480	1200-0623
A13A1P1	1251-3782	1	CONNECTOR 40-PIN M RECTANGULAR	76381	3432-1002
A13A1R1 THRU R4	0757-0280		RESISTOR 1K 1% .125W FTC=0+ -100	24546	C4-1/8-TO-1001F
A13A1R5	0757-0394	1	RESISTOR 51.1 1% .125W F TC=0! -100	24546	C4-1/8-TO-51R1-F
A13A1S1	3101-0973	1	SWITCH-SLIDE MINI NS .5A 125VAC	04876	GF126-0018
A13A1U1	1820-1624	1	IC-DIGITAL SN74LS241N TTL OCTL	01295	SN74LS241N
A13A1U2 THRU U4	1820-1918	3	IC-DIGITAL SN74LS241N TTL LS OCTL	01295	SN74LS241N
A13A1U5	1820-1198	1	IC-DIGITAL SN74LS03N TTL LS QUAD	01295	SN74LS03N
A13A1U6 THRU U9	1810-0283		NETWORK-RES 16-PIN-DIP .1PIN SPCG 270 2%	28480	1810-0283

SECTION VII

MANUAL CHANGES

7-1. INTRODUCTION.

7-2. This section contains information required to back-date or update this manual for a specific instrument.

7-3. MANUAL CHANGES.

7-4. This manual applies directly to the instrument having the serial prefix shown on the manual title page. If the serial prefix of your instrument is not the same as the one on the title page, find your serial prefix in table 7-1 and make the changes to the manual that are listed for that serial prefix. When making changes listed in table 7-1, make the change with the highest number first. Example: if backdating changes 1, 2, and 3 are required for your serial prefix, do change 3 first, then change 2, and finally change 1.

7-5. If the serial prefix of your instrument is not listed either on the title page or in table 7-1, refer to an enclosed MANUAL CHANGES sheet for updating information. Also, if a MANUAL CHANGES sheet is supplied, make all indicated ERRATA corrections.

Table 7-1. Manual changes

Serial Prefix	Make Changes
1818A 1833A	2 and 1 2

7-6. MANUAL CHANGE INSTRUCTIONS.

CHANGE 1

Table 6-2,

Replace page 6-2 with table 7-2.

Figure 7-1, Service Sheet 2 (Sheet 2 of 3),

In upper left-hand corner of page (between pins 32 and 33) delete pins and C7.

Figure 7-2, Service Sheet 2 (Sheet 3 of 3), ICs ON THIS SCHEMATIC table;

Add: REF DES U6, HP PART NO. 1820-0686, MFR PART NO. 74S11N.

Figure 8-2, Service Sheet 2 (Sheet 2 of 3),

Replace with figure 7-1.

Figure 8-2, Service Sheet 2 (Sheet 3 of 3),

Replace with figure 7-1.

Figure 8-2, Service Sheet 2, A9 Component Location, (Sheet 1 of 2),

Replace with figure 7-1.

Figure 8-5, Service Sheet 4, A10 Component Location, (Sheet 1 of 2),

Replace with figure 7-2.

CHANGE 2

Table 6-2,

A10: Change HP Part No. and Mfr Part No. to 01611-66555.

Add: A10C7, HP Part No. 0160-2055, CD 9, Qty 1, CAPACITOR-FXD .01 UF +80-20%, Mfr Code 04200, Mfr Part No. CD23F101F103ZS22-CDH.

Add: A10U6, HP Part No. 1816-1248, CD 2, Qty 1, IC-8K PROM, Mfr Code 28480, Mfr Part No. 1816-1248.

Add: A10U7, HP Part No. 1816-1108, CD 3, Qty 1, IC-8K PROM, Mfr Code 28480, Mfr Part No. 1816-1108.

A10U8: Change HP Part No. and Mfr Part No. to 1816-1109.

A10U9: Change HP Part No. and Mfr Part No. to 1816-1109.

A10U10: Change HP Part No. and Mfr Part No. to 1816-1112.

A10XU8 THRU A10X10: Change to A10XU6 THRU A10XU10 and change Qty from 3 to 5.

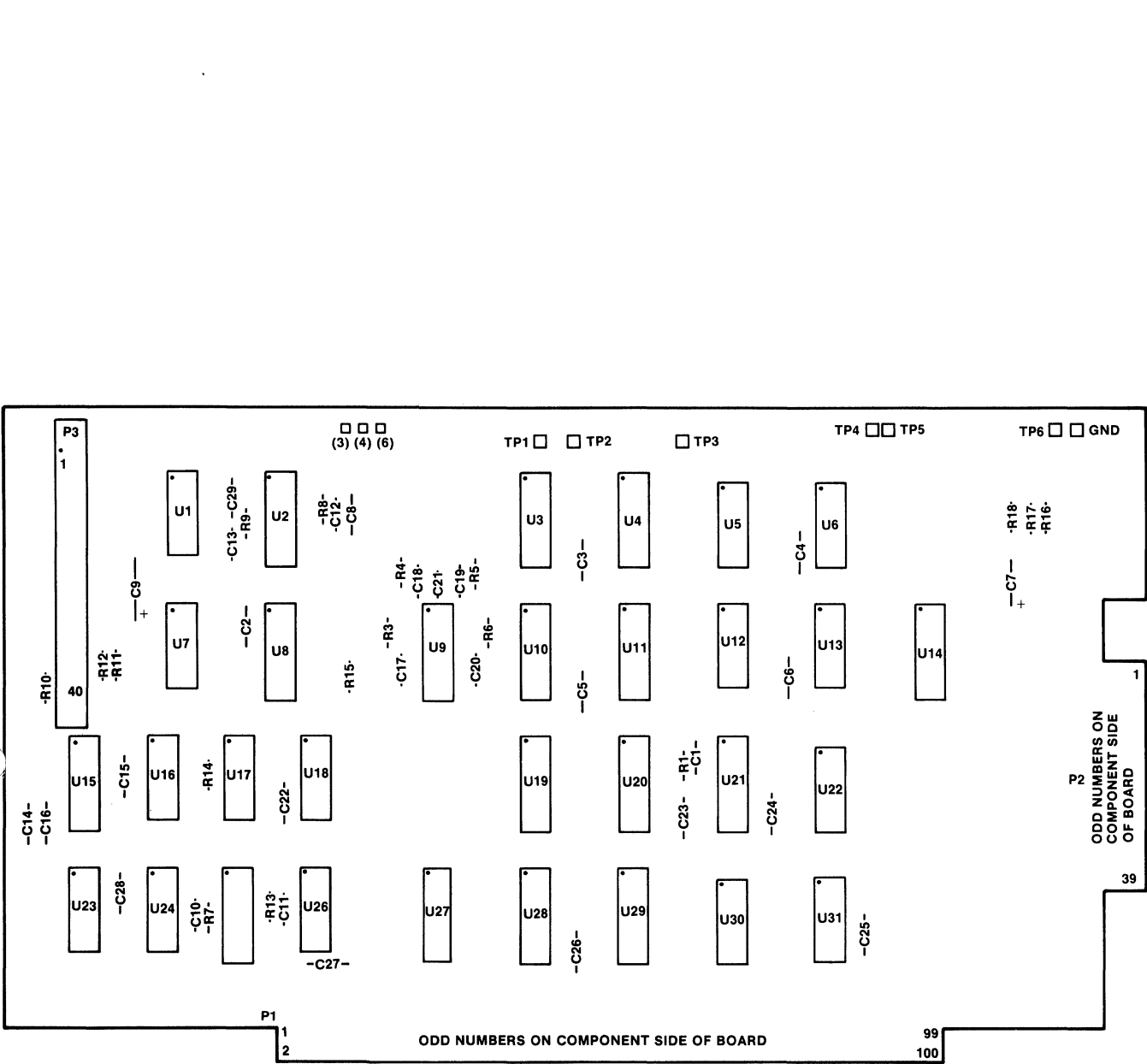
Figure 8-5, Service Sheet 4, (Sheet 2 of 2),

Replace with figure 7-2.

Page 8-10, Delete Signature Analysis Procedure No. 1 and figure 8-4, ROM Test Troubleshooting Tree.

Table 7-2. Replaceable Parts

Reference Designation	HP Part Number	Qty	Description	Mfr Code	Mfr Part Number
A9	01611-66554	1	BOARD ASSEMBLY, OPTION A85 PERSONALITY	28480	01611-66554
A10	01611-66555	1	BOARD ASSEMBLY, OPTION A85 ROM	28480	01611-66555
A11	01611-68710	1	ASSEMBLY, PERSONALITY PANEL, OPTION A85	28480	01611-68710
A13	01611-62112	1	ASSEMBLY, A85 MPR PROBE	28480	01611-62112
MP1	1540-0325	1	CASE, CRYG HANDLE	28480	1540-0325
A9	01611-66554		BOARD ASSEMBLY, OPTION A85 PERSONALITY	28480	01611-66554
A9C1	0160-2198	1	CAPACITOR-FXD 20PF $\pm 5\%$	28480	0160-2198
A9C2	0160-2055	30	CAPACITOR-FXD .01UF +80 -20%	04200	C023F101F103ZS22-CDH
THRU C6					
A9C7	0180-0291	1	CAPACITOR-FXD 1UF +10% 35VDC TA	56289	150D105X9035A2
A9C8	0160-2055		CAPACITOR-FXD .01UF +80-20%	04200	C023F101F103ZS22-CDH
A9C10	0140-0198	2	CAPACITOR-FXD 200PF + -5% 300 WVDC	04522	DM15F201J0300WVICR
A9C11	0160-3449	1	CAPACITOR-FXD 2000PF + -10% 250 WVDC CER	28480	0160-3449
A9C12	0140-0205	1	CAPACITOR-FXD 62PF + -5% 300 WVDC	04522	DM15E620J0300WVICR
A9C13	0140-0198		CAPACITOR-FXD 200PF +—5% 300 WVDC		
A9C14 THRU C29	0160-2055		CAPACITOR-FXD .01UF +80-20%	04200	C023F101F103ZS22-CHD
A9C30 & C31	0160-2306	2	CAPACITOR-FXD 27PF +—5% 300 WVDC	28480	0160-2306
A9R19 & R20	0757-0415	2	RESISTOR 475 1% .125W F TC=0+—100	01121	CC
A9R1	0757-0438	4	RESISTOR 5.11K 1% .125 W F TC=0+ -100	24546	C4-1/8-TO-5111-F
A9R2			NOT ASSIGNED		
A9R3	0757-0910	6	RESISTOR 270 2% .125 W F TC=0+ -100	02273	CEA-993
THRU R6					
A9R7	0757-0438		RESISTOR 5.11K 1% .125W F TC=0+ -100	24546	C4-1/8-TO-511-F
THRU R9					
A9R10	0757-0442	7	RESISTOR 10K 1% .125W	02273	CEA-93
THRU R13					
A9R14	0757-0910		RESISTOR 270 2% .125W F TC=0+ -100	02273	CEA-993
A9R15	0757-0910		RESISTOR 270 2% .125W F TC=0+ -100	02273	CEA-993
A9R16	0757-0442		RESISTOR 10K 1% .125W	02273	CEA-93
THRU R18					
A9TP1	0360-0535	7	TERMINAL, TEST POINT	28480	OBD
THRU TP7					
A9U1	1820-1464	1	IC-DIGITAL SN74393N TTL DUAL	01295	SN74B93N
A9U2	1820-1423	1	IC-DIGITAL SN74LS123N TTL LS DUAL	01295	SN74LS123N
A9U3	1820-1191	3	IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U4	1820-1191		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U5	1820-2169	2	IC-DIGITAL	01295	SN74L04N
A9U6	1820-0686	1	IC-DIGITAL SN74S11N TTL S TPL	01295	SN74S11N
A9U7	1820-1112	2	IC-DIGITAL SN74LS74N TTL LS DUAL	01295	SN74LS74N
A9U8	1820-1195	10	IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U9	1820-1319	1	IC-DIGITAL SN74S15N TTL S	01295	SN74S15N
A9U10	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U11	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U12	1820-2169		IC-DIGITAL	01295	SN74L04N
A9U13					
A9U14	1820-1191		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U15	1810-0283	5	NETWORK-RES 16-PIN-DIP .1-PIN SPCG 270	28480	1810-0283
A9U16	1820-1207	1	IC-DIGITAL SN74LS30N TTL LS	01295	SN74LS30N
A9U17	1820-0077	1	IC-DIGITAL SN7474N TTL DUAL	01295	SN7474N
A9U18	1820-1199		IC-DIGITAL	01295	SN74LS04
A9U19	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U20	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
A9U21	1820-1782	1	IC-DIGITAL	01295	SN74S00N
A9U22	1820-0681	1	IC-DIGITAL SN74S00N TTL S QUAD	01295	SN74S00N
A9U23	1820-0661	1	IC-DIGITAL SN7432N TTL QUAD	01295	SN7432N
A9U24	1820-1112		IC-DIGITAL SN74LS74N TTL LS DUAL	01295	SN74LS74N
A9U25	1820-0579	1	IC-DIGITAL SN74123N TTL DUAL	01295	SN74123N
A9U26	1820-1144	1	IC-DIGITAL SN74LS02N TTL LS QUAD	01295	SN74LS02N
A9U27	1820-1195		IC-DIGITAL SN74LS175N TTL LS QUAD	01295	SN74LS175N
THRU 29					
A9U30	1820-1449	1	IC-DIGITAL SN74S32N TTL S QUAD	01295	SN74S32N
A9U31	1820-1367	1	IC-DIGITAL SN74S08N TTL S QUAD	01295	SN74S08N
A10	01611-66555		BOARD ASSEMBLY, OPTION A85 ROM	28480	01611-66555
A10C1	0180-0228	1	CAPACITOR-FXD 22UF + -100% 15VDC TA	56289	150D226X9015B2
A10C2	0160-3443	1	CAPACITOR-FXD .1UF +80-20% 5VDC CER	02813	CY20C104Z2
A10C3 THRU C7	0160-2055		CAPACITOR-FXD .01UF +80-20%	04200	C023F101F103ZS22-CDH



A9 Component Location (01611-66554)

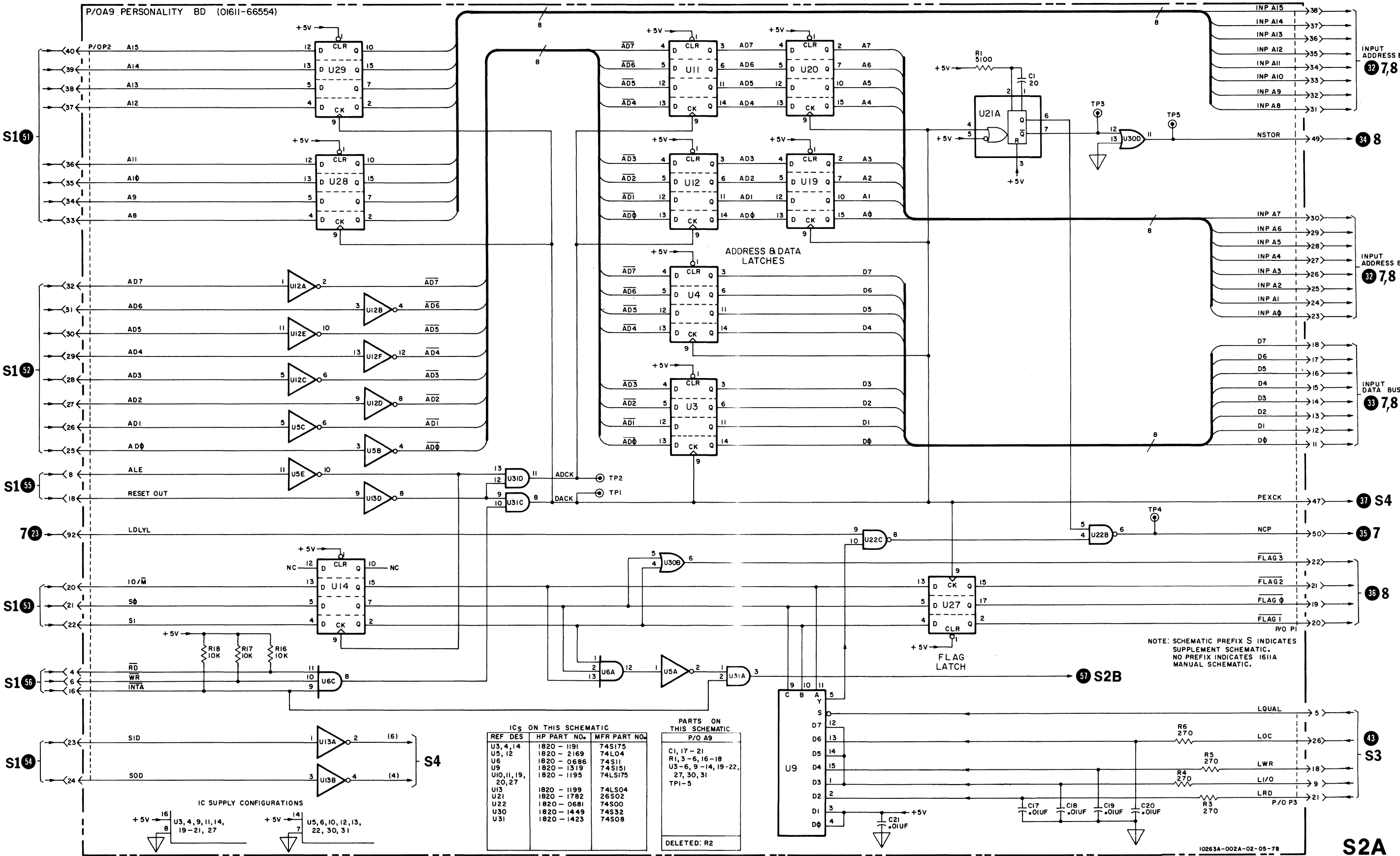


Figure 7-1. Replacement for Figure 8-2, Service Sheet 2, (Sheet 1 of 3)

Figure 7-1. Replacement for Figure 8-2, Service Sheet 2, (Sheet 2 of 3) 7-3/(7-4 blank)

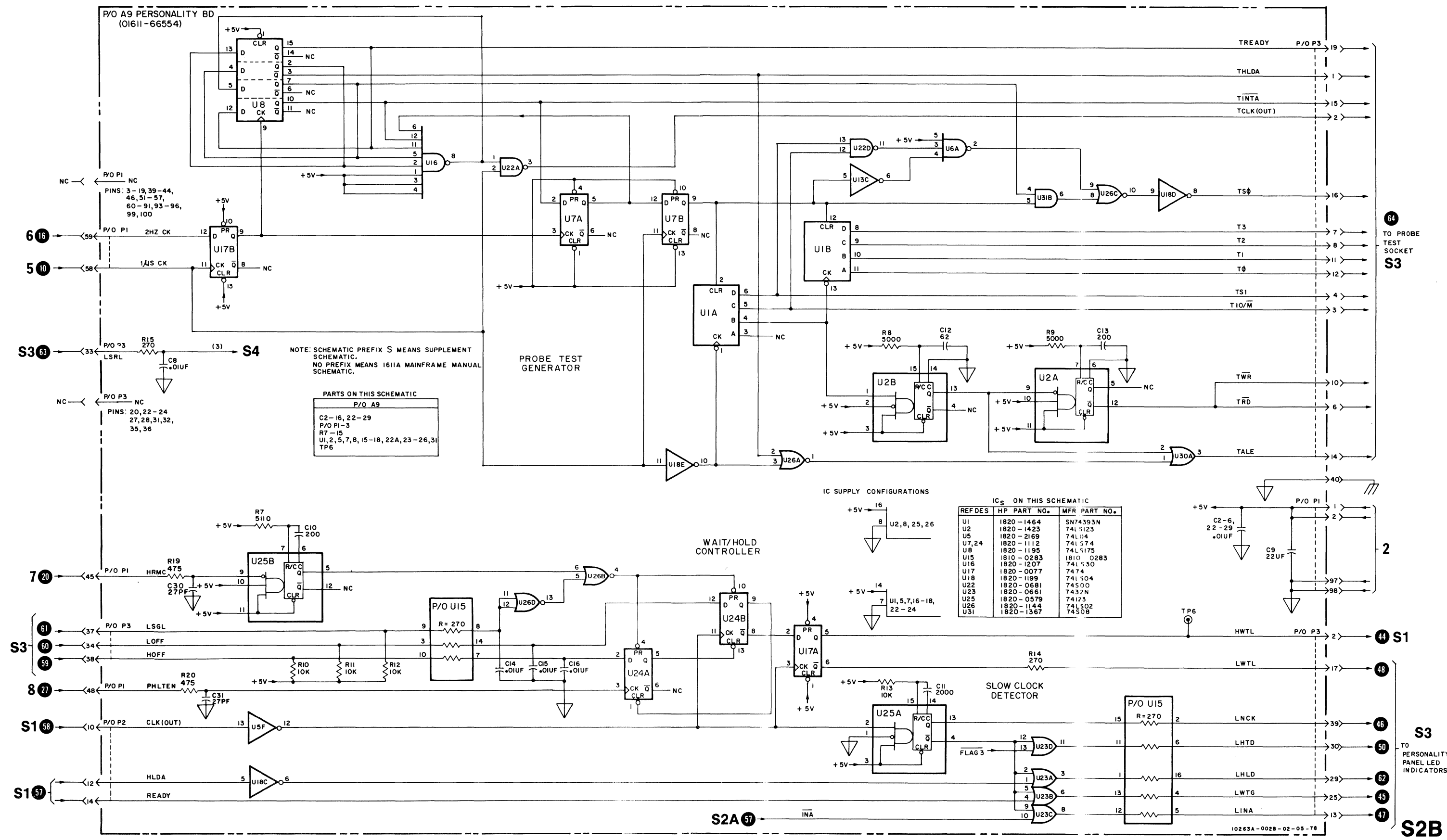
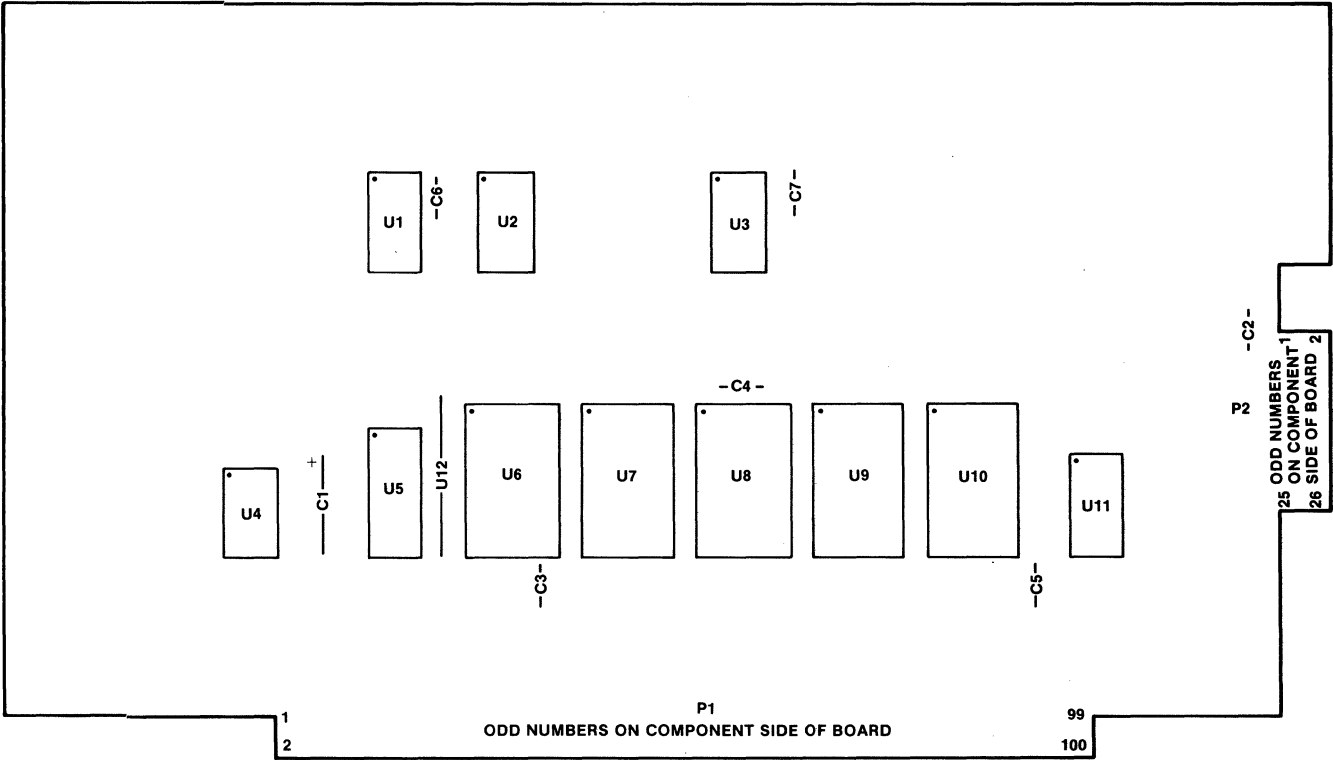


Figure 7-1.
Replacement for Figure 8-2, Service Sheet 2, (Sheet 3 of 3)
7-5



A10 Component Location (01611-66555)

Figure 7-2. Replacement for Figure 8-6, Service Sheet 4, (Sheet 1 of 2)

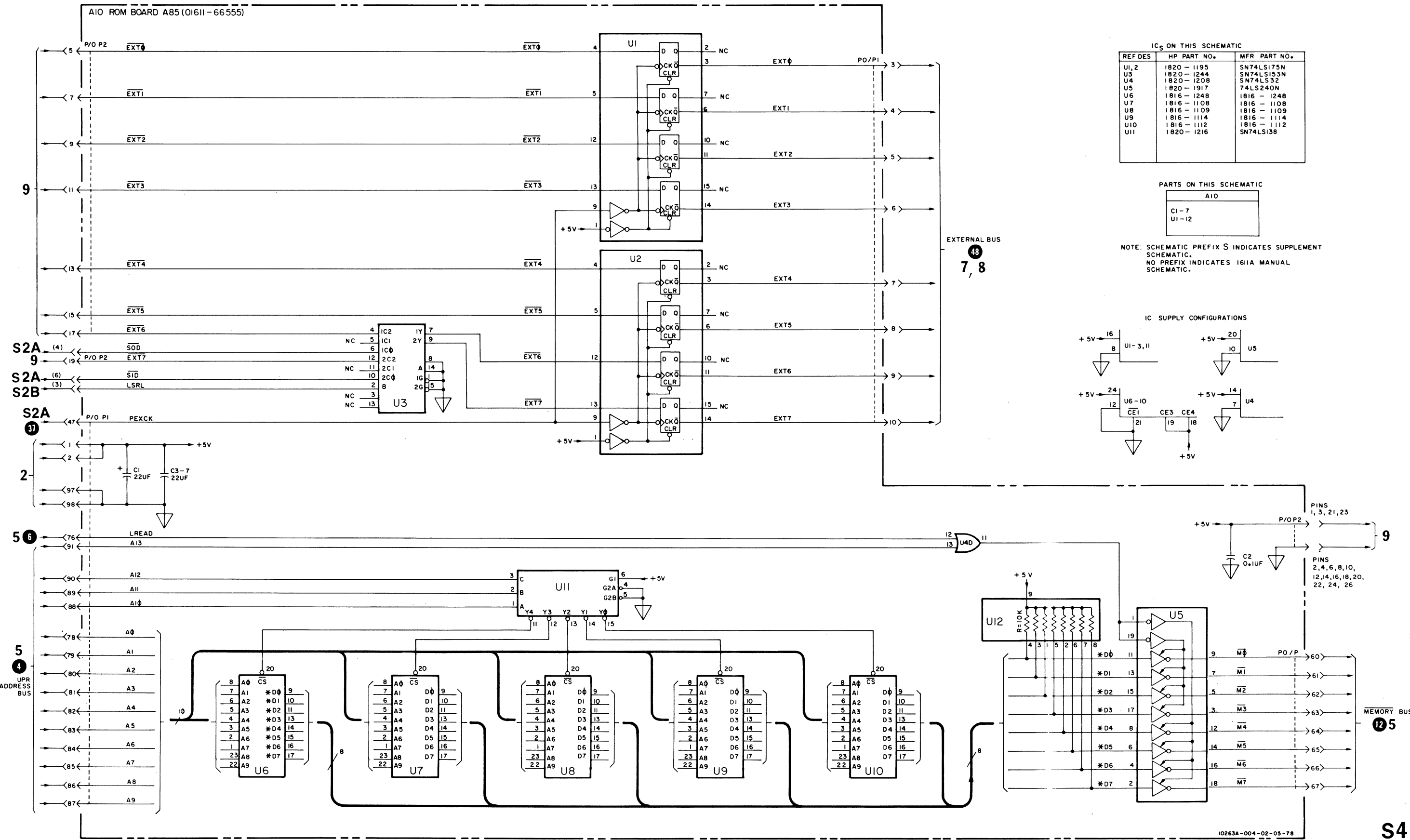


Figure 7-2.
Replacement for Figure 8-6, Service Sheet 4, (Sheet 2 of 2)
7-7/(7-8 blank)

SECTION VIII**SERVICE****8-1. INTRODUCTION.**

8-2. This section contains schematic diagrams for the Model 10263A/1611A Option A85 as well as signature analysis troubleshooting procedure for Service Sheet 4.

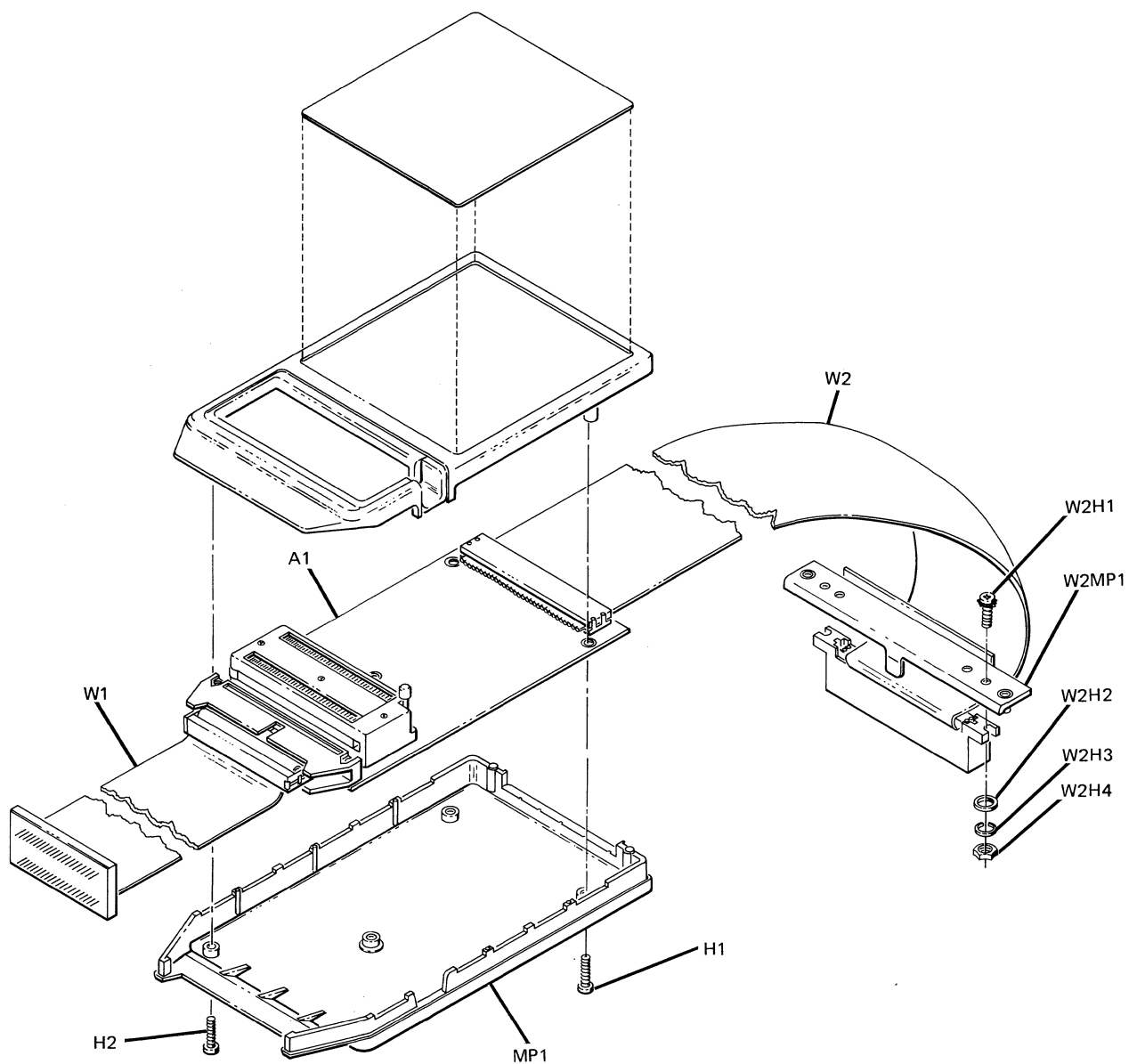
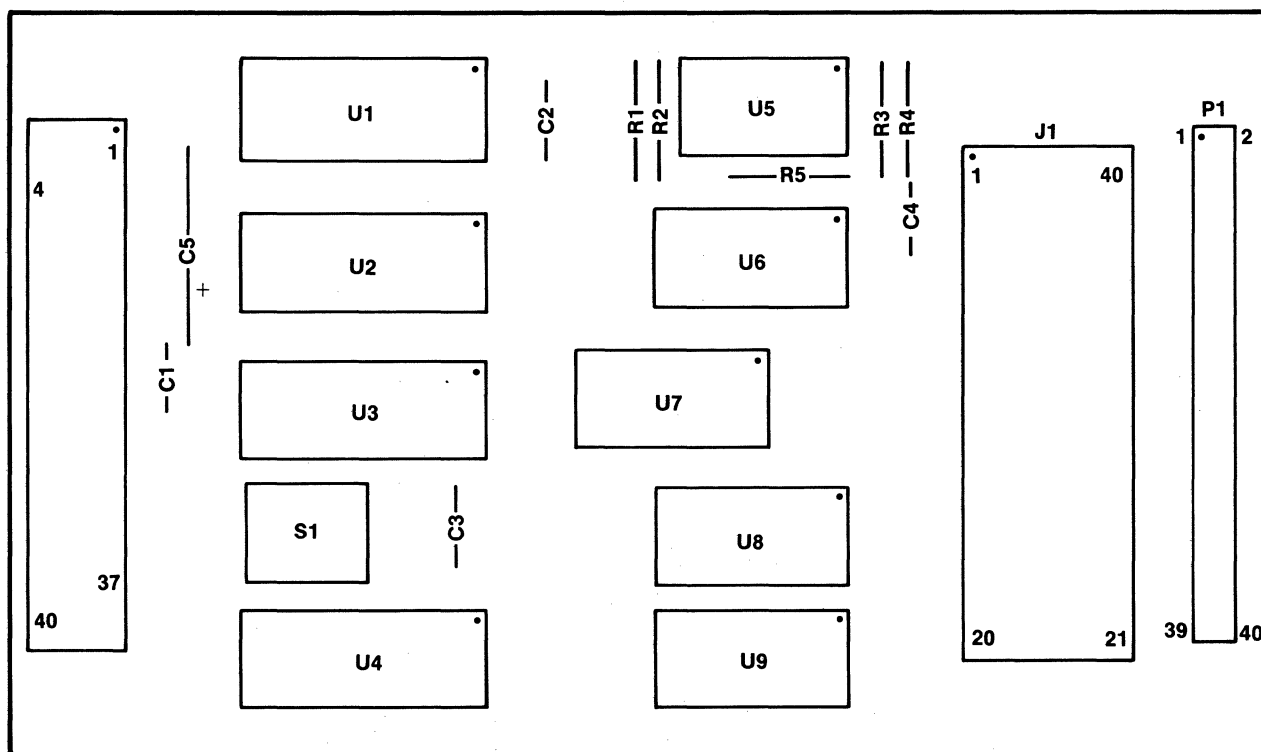


Figure 8-1. Service Sheet 1, Microprocessor Probe A13 (Sheet 1 of 3)



A13A1 Component Location (01611-62112)

Figure 8-1. Service Sheet 1, Microprocessor Probe A13 (Sheet 2 of 3)

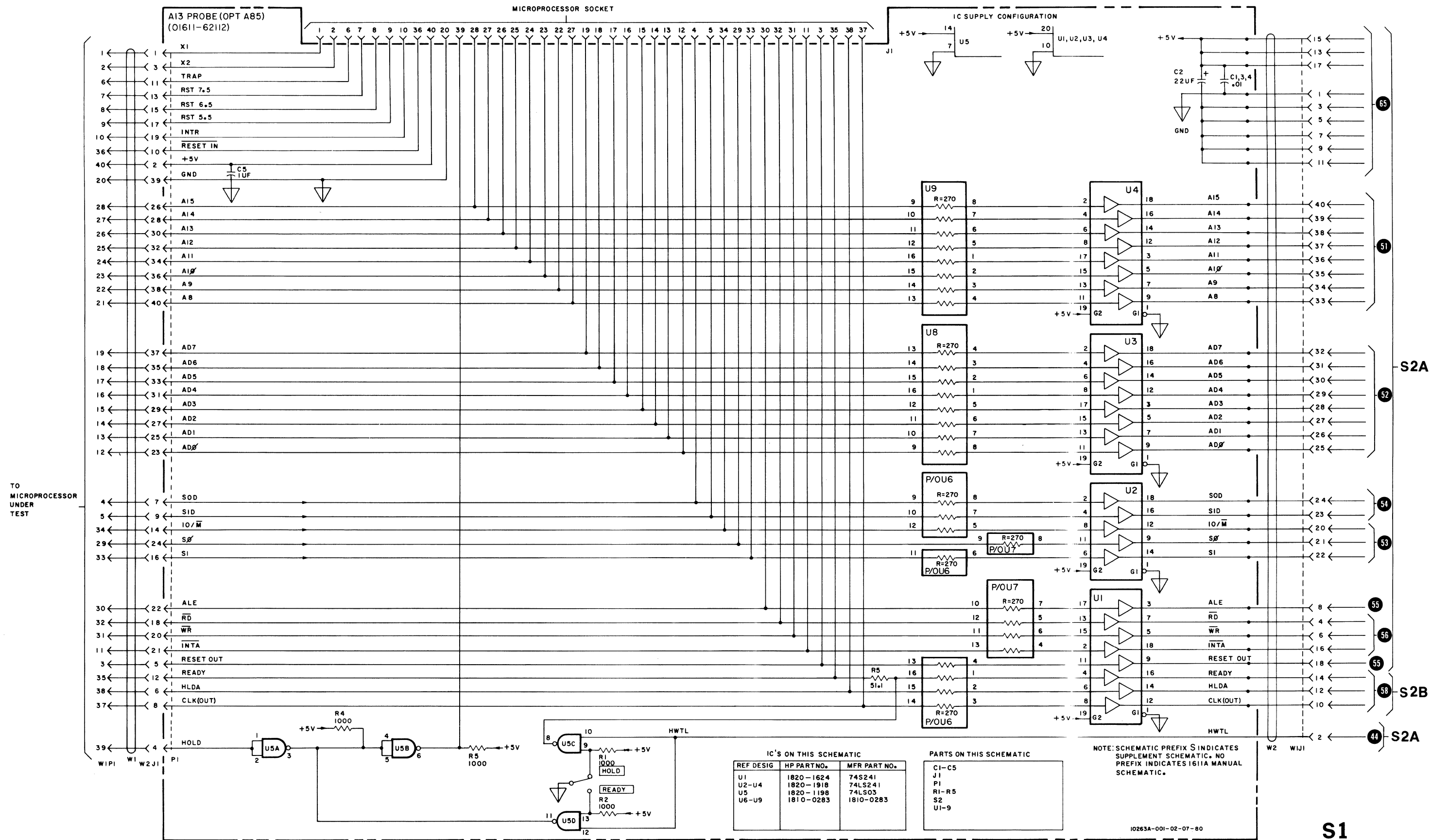
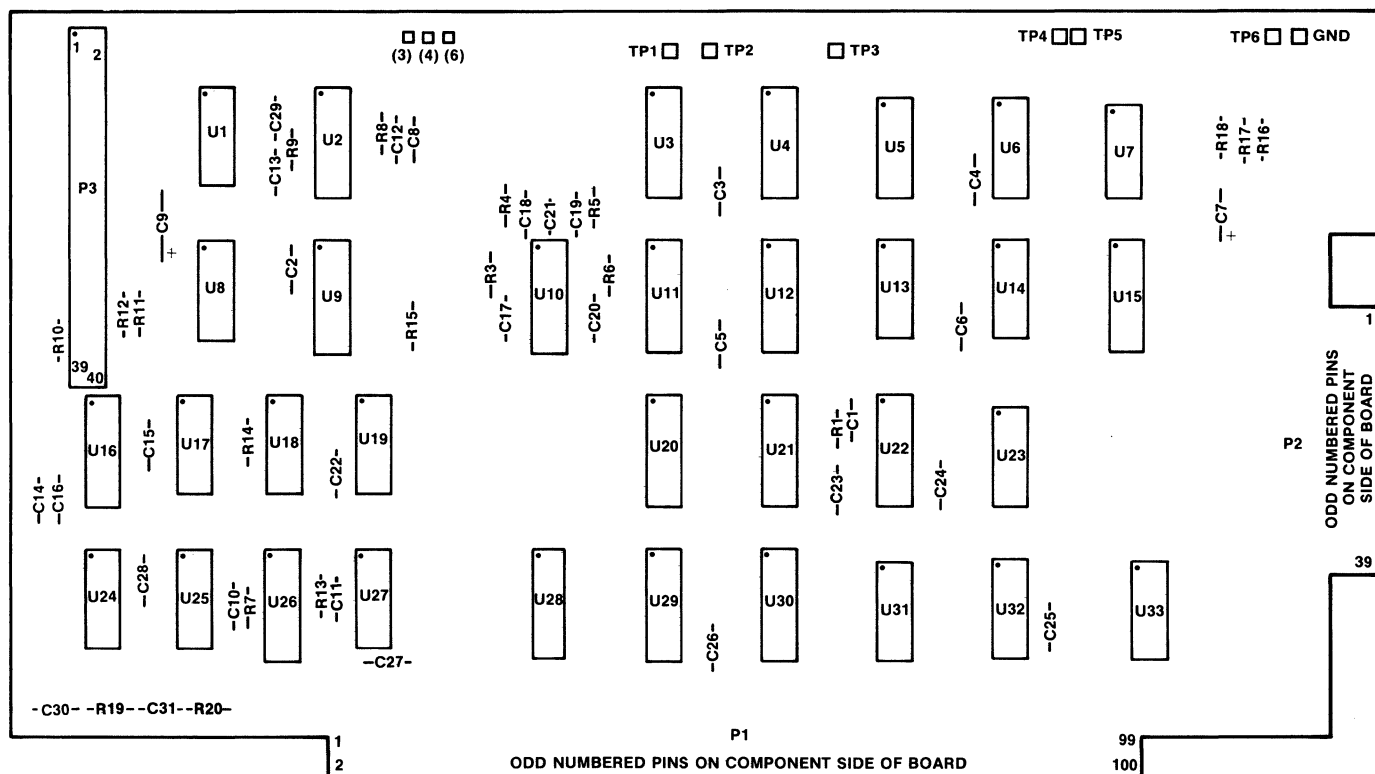


Figure 8-1.
Microprocessor Probe A13 (Sheet 3 of 3)
8-3



A9 Component Location (01611-66566)

Figure 8-2. Service Sheet 2, Personality Board A9 (Sheet 1 of 3)

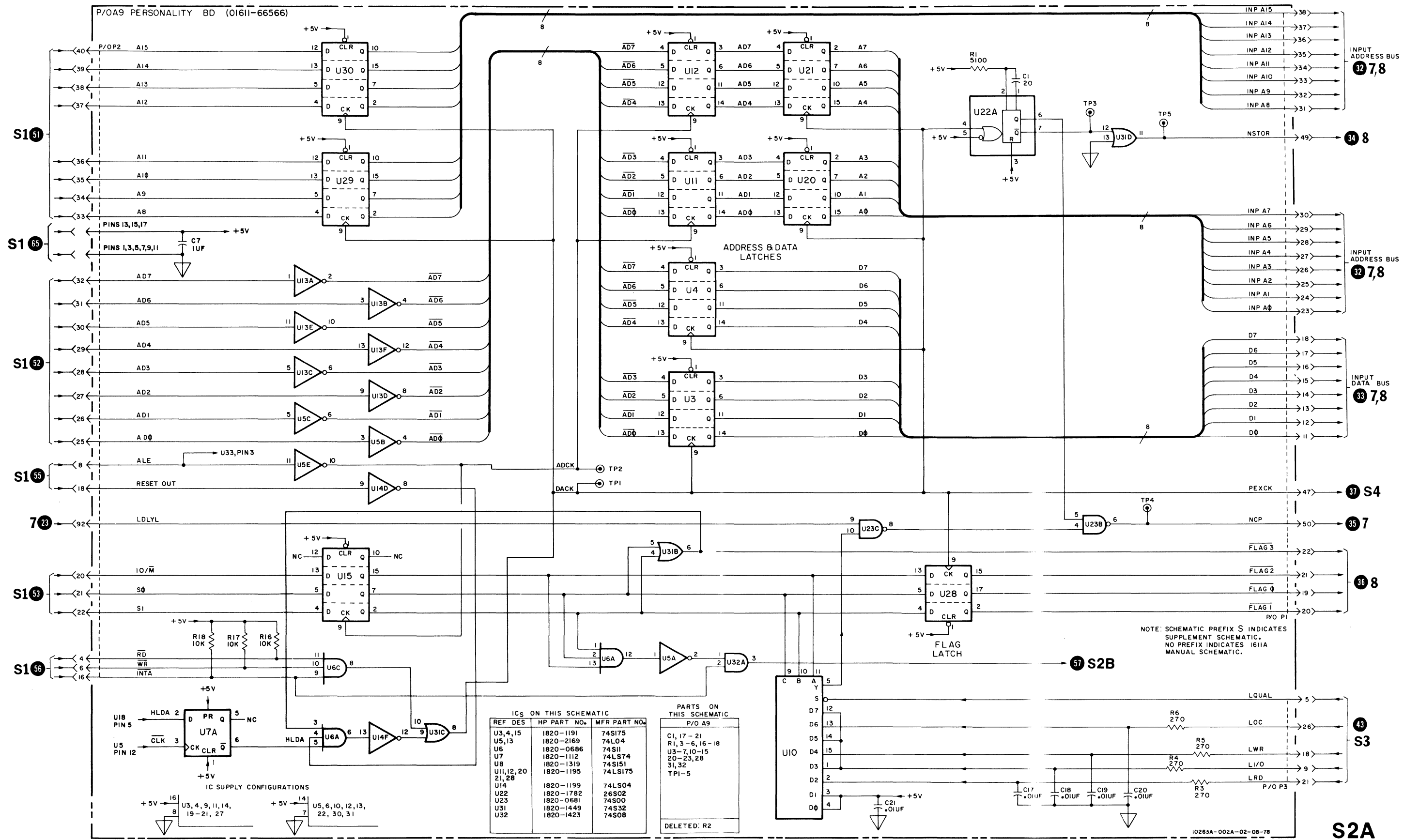


Figure 8-2.
Service Sheet 2, Personality Board A9 (Sheet 2 of 3)
8-5/(8-6 blank)

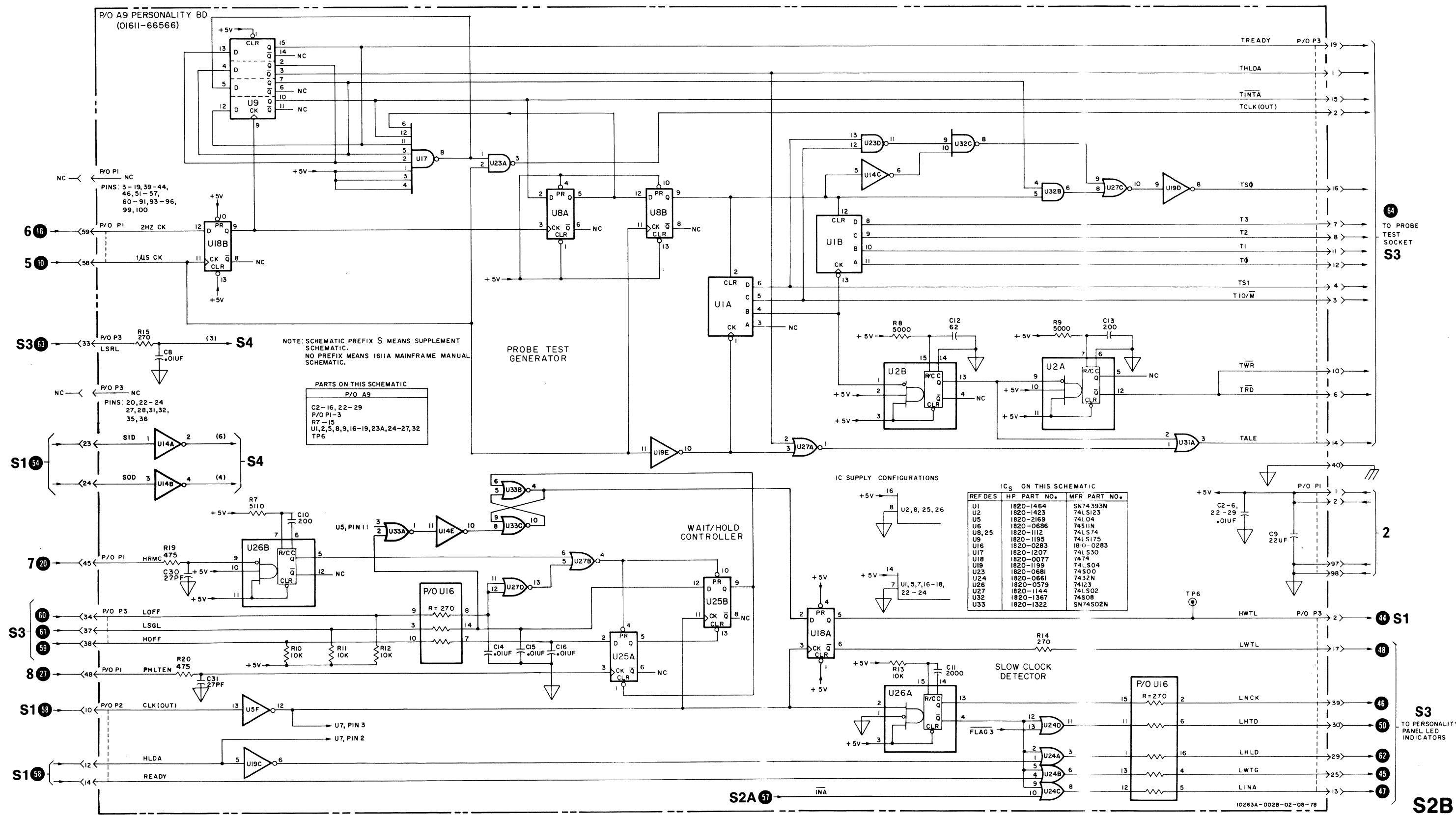
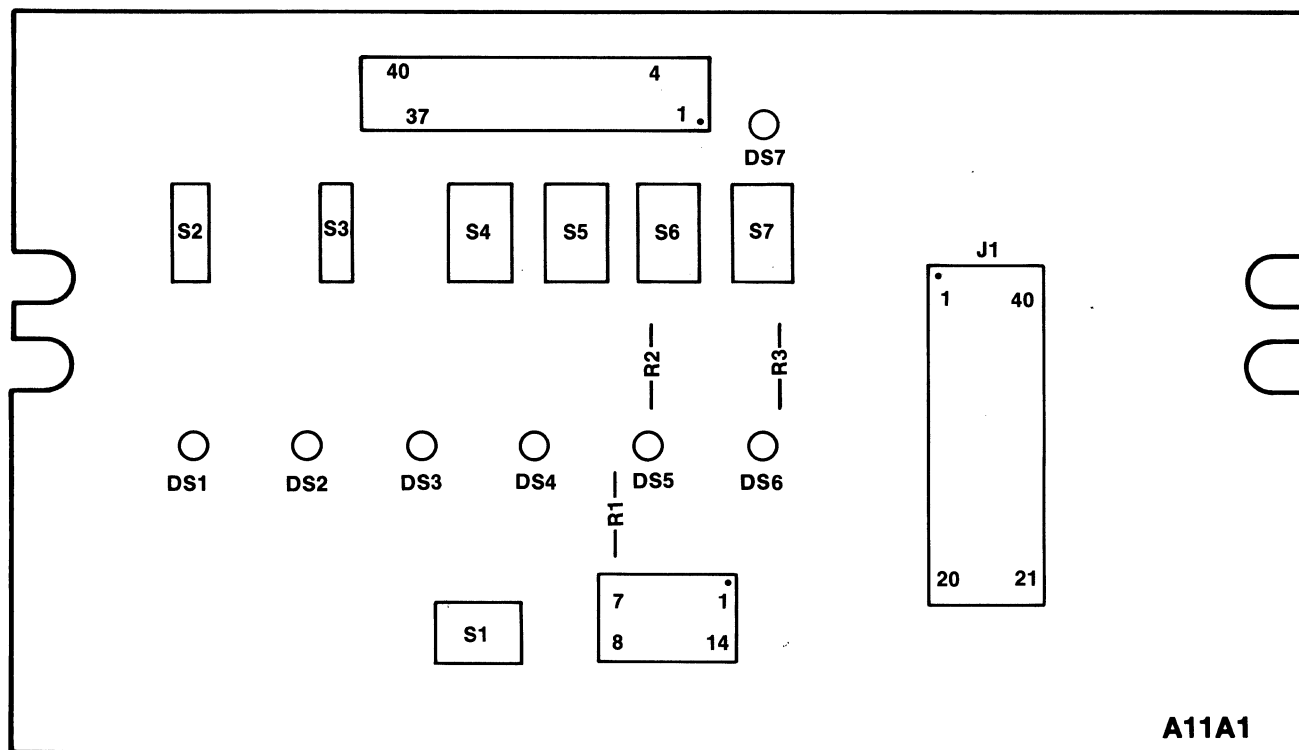


Figure 8-2.
Service Sheet 2, Personality Board A9 (Sheet 3 of 3)
8-7



A11A1 Component Location (01611-68710)

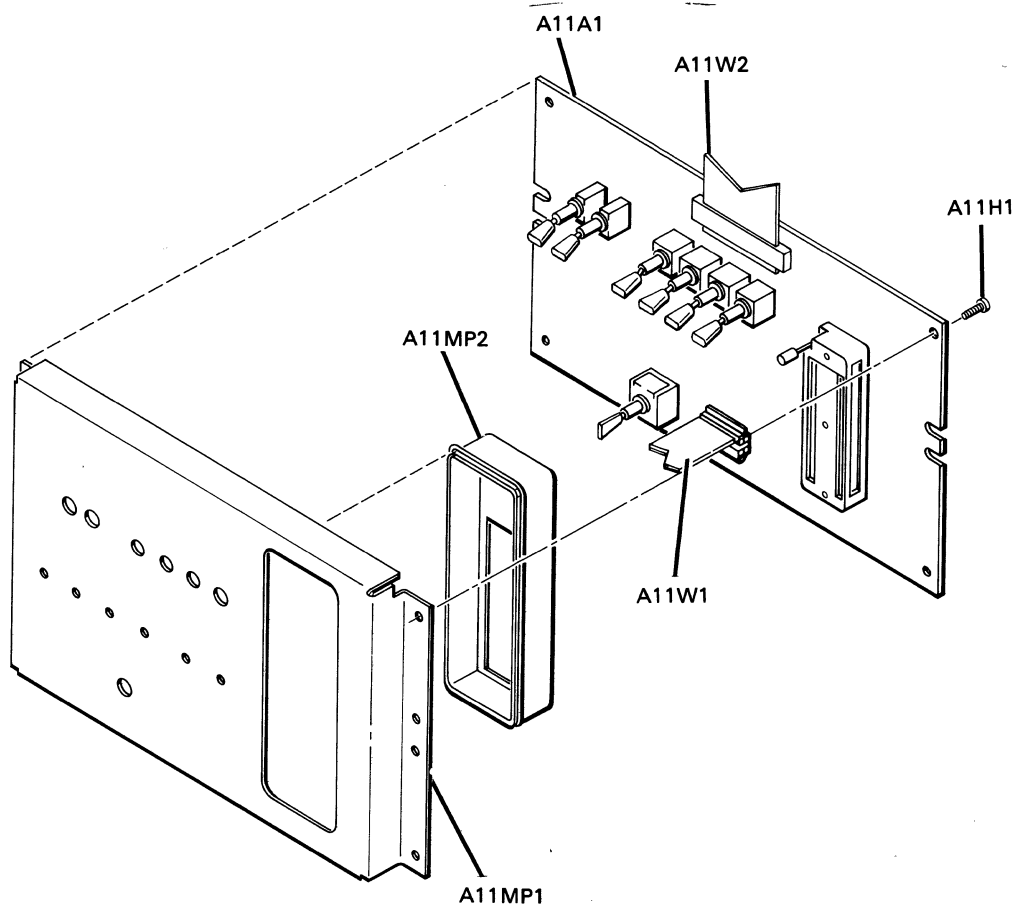


Figure 8-3. Service Sheet 3, Personality Panel A11 Parts Identification (01611-68710) (Sheet 1 of 2)

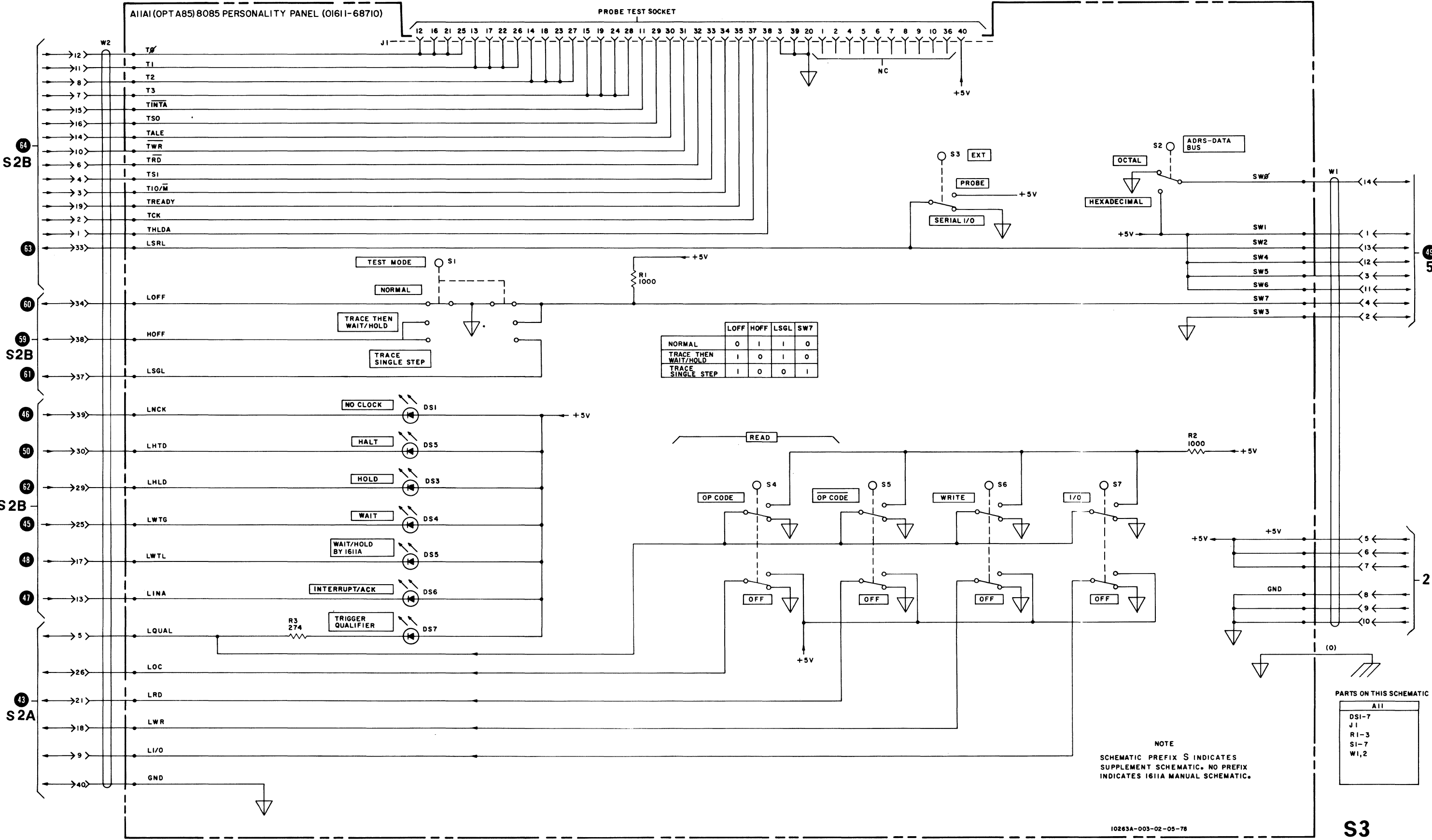


Figure 8-3.
Service Sheet 3, A85 Personality Board A11, (Sheet 2 of 2)
8-9

Table 8-1. Signature Analysis Procedure No. 1

- a. Set 1611A LINE switch to off position.

b. Remove A6, A7, A9, and A10 boards from 1611A.

c. Reinstall A10 Board on extender board A14.

d. Ground A5U3 pin 6 and A5U11 pin 6.

e. Connect signature analyzer probes to the following circuit points.
- CLOCK A5U11, Pin 17

GND A5TP (GND)

f. Set signature analyzer controls as follows:

START]

STOP]

CLOCK]

g. Set 1611A LINE switch to on position.

h. Verify that signatures called out in trouble-shooting tree match the following table.
- START A5U11, Pin 36

STOP A5U11, Pin 38

MEASUREMENT NUMBER	TEST POINTS	SIGNATURES	
	A10 boards 01611-66555 and 01611-66571	A10 board 01611-66555	A10 board 01611-66571
101	MEMORY BUS INPUTS V _H A5U8, pin 5 A5U8, pin 1 A5U8, pin 9 A5U8, pin 13 A5U9, pin 5 A5U9, pin 1 A5U9, pin 9 A5U9, pin 13	P254 A8PF UPAF AOUA 314F C583 600F A220 CH06	P254 U85F 131F 31CH C160 P5FP H35A 9HU9 5124
102	ROM 0 OUTPUT A10U10, pin 9 A10U10, pin 10 A10U10, pin 11 A10U10, pin 13 A10U10, pin 14 A10U10, pin 15 A10U10, pin 16 A10U10, pin 17	4AC8 1FU8 42AP H318 57H7 8258 4074 5U52	1A08 U148 H39P 5334 079A 310P 7UAH C370
	A10 boards 01611-66555 and 01611-66571	A10 board 01611-66555	A10 board 01611-66571
103	ROM ADDRESSING A10U10, pin 8 A10U10, pin 7 A10U10, pin 6 A10U10, pin 5 A10U10, pin 4 A10U10, pin 3 A10U10, pin 2 A10U10, pin 1 A10U10, pin 23 A10U10, pin 22 A10U11, pin 1 A10U11, pin 2 A10U11, pin 3	5P33 FA11 3HUA 12U0 C7A5 46HC 65CA 8AUC 9241 1U5P AAHU U665 826P	5P33 FA11 3HUA 12U0 C7A5 46HC 65CA 8AUC 9241 1U5P U665 826P 0000

Table 8-1. Signature Analysis Procedure No. 1 (Cont'd)

MEASUREMENT NUMBER	TEST POINTS	SIGNATURES	
	A10 ROM boards 01611-66555 and 01611-66571	A10 board 01611-66555	A10 board 01611-66571
104	DATA SELECTOR OUTPUTS A10U11, pin 15 A10U11, pin 14 A10U11, pin 13 A10U11, pin 12 A10U11, pin 11	879F 6CC3 931U 1UOA 4A78	OP7C 6P41 1A4A not used not used
	A10 boards 01611-66555 and 01611-66571	A10 ROM boards 01611-66555 and 01611-66571	
105	A5U11 ADDRESS OUTPUTS A5U21C, pin 7 A5U21C, pin 6 A5U21B, pin 5 A5U21B, pin 4 A5U21A, pin 3 A5U21A, pin 2 A5U21D, pin 9 A5U21D, pin 10 A5U21E, pin 11 A5U21E, pin 12 A5U12E, pin 11 A5U12E, pin 12 A5U12F, pin 13 A5U12F, pin 14 A5U12C, pin 7 A5U12C, pin 6 A5U12B, pin 5 A5U12B, pin 4 A5U2B, pin 6 A5U2B, pin 5 A5U2A, pin 3 A5U2A, pin 2 A5U12A, pin 3 A5U12A, pin 2 A5U2C, pin 8 A5U2C, pin 9	5P33 5P33 FA11 FA11 3HUA 3HUA 12UO 12UO C7A5 C7A5 65CA 65CA 8AUC 8AUC 9241 9241 1U5P 1U5P AAHU AAHU U665 U665 826P 826P VLP VLP	
VLP = 0000			

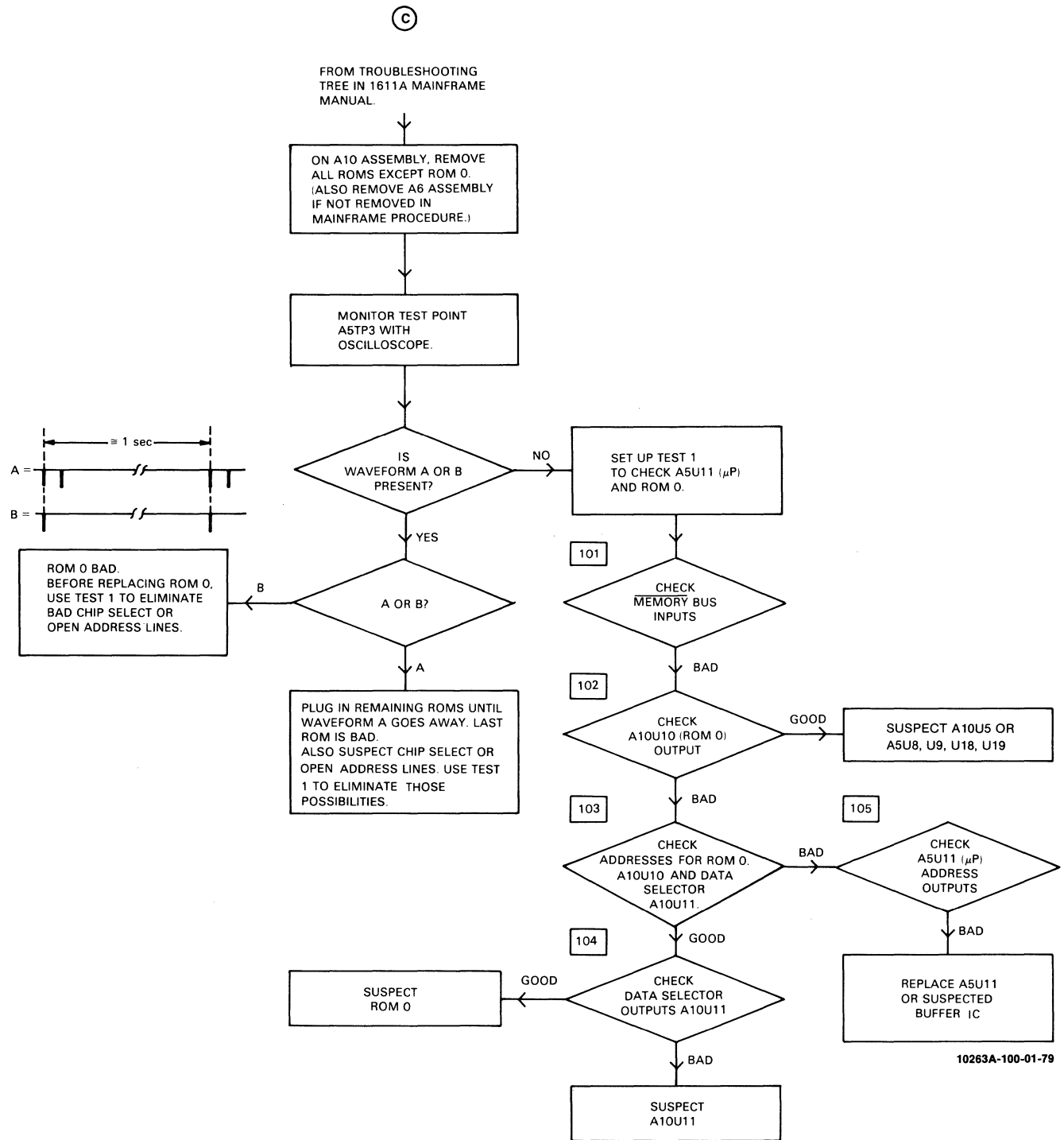
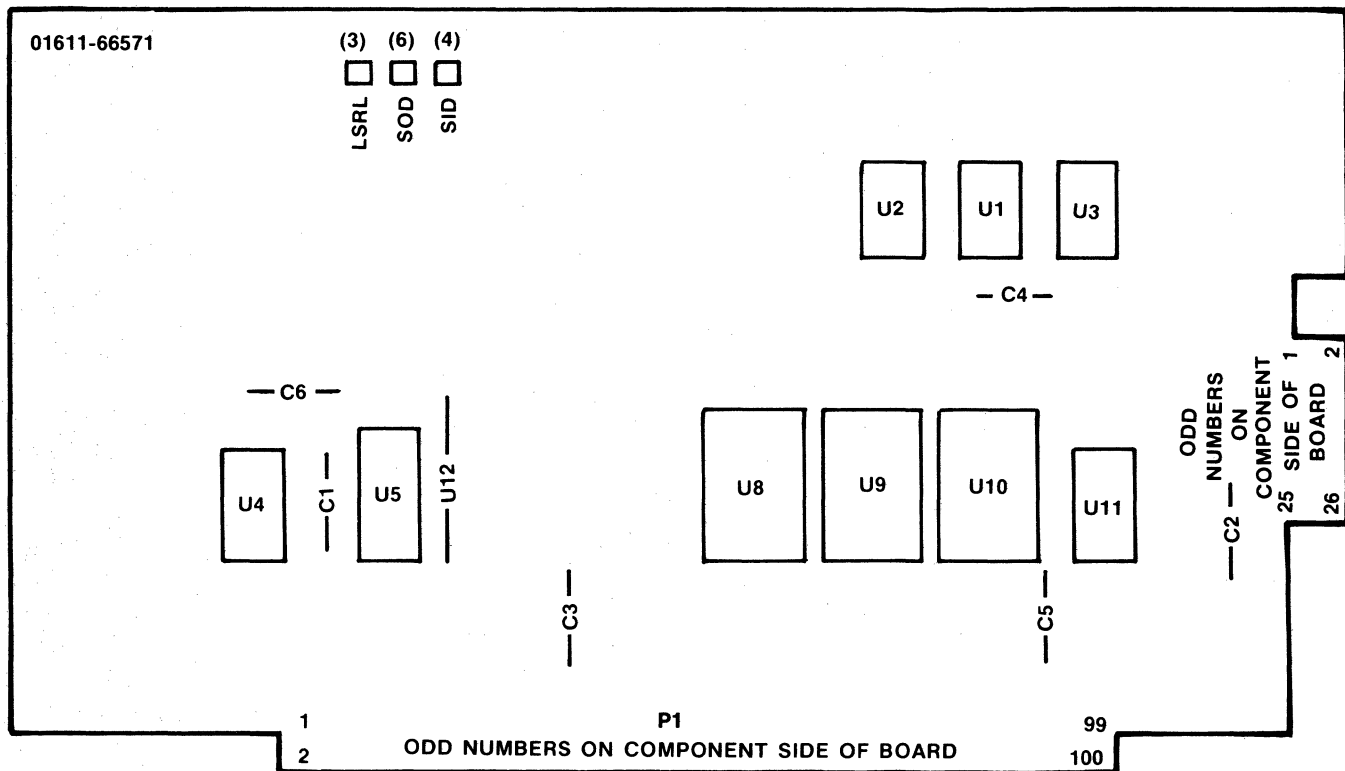


Figure 8-4. ROM Test Troubleshooting Tree



A10 Component Location (01611-66571)

Figure 8-5. Service Sheet 4, ROM Board A10 (Sheet 1 of 2)

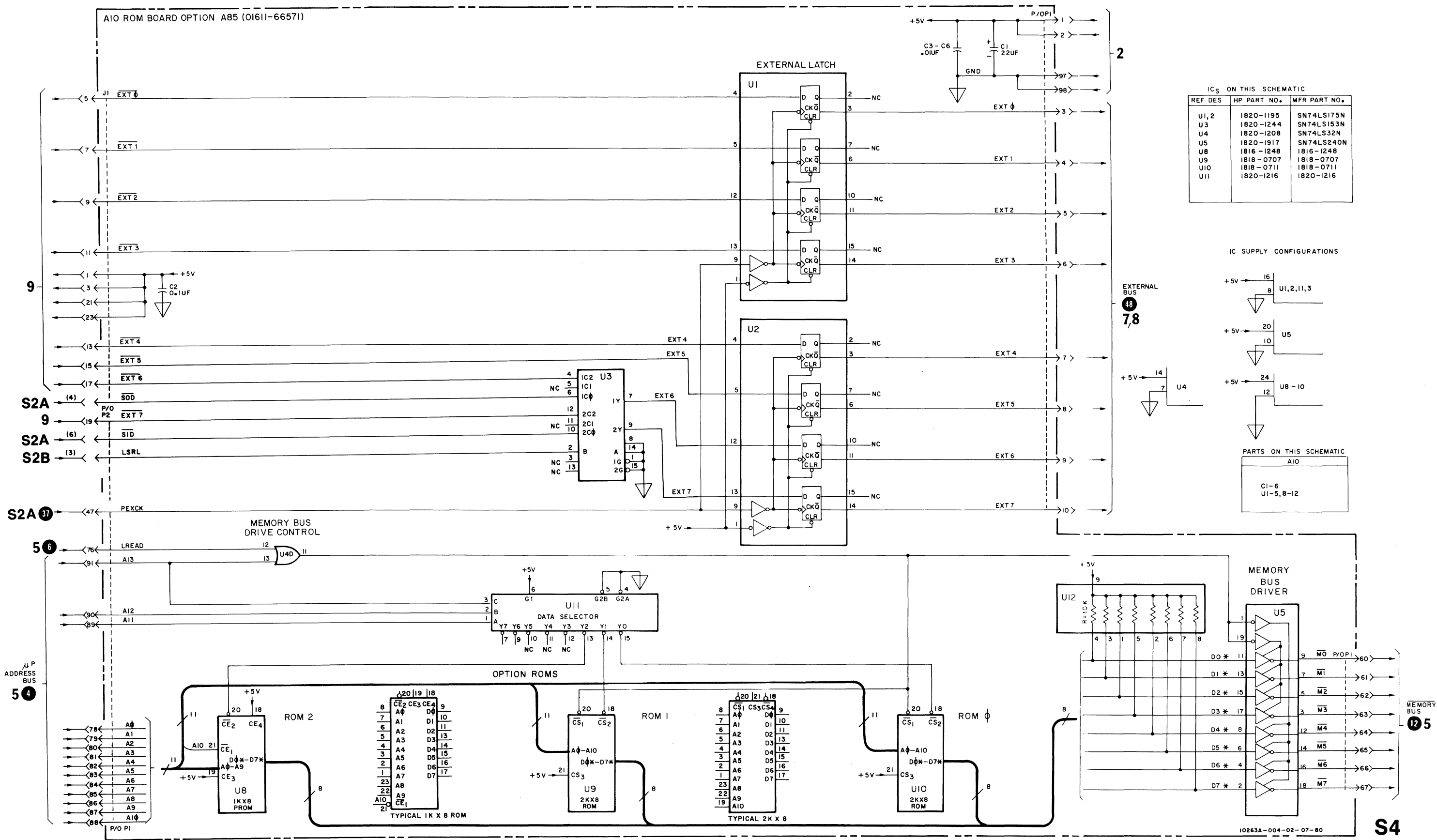


Figure 8-5.
Service Sheet 4, ROM Board A10 (Sheet 2 of 2)
8-13

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Arranged alphabetically by country

2



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Kilmore Road
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